

A Collection of RQMC Mean and Median Distributions

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This document reports detailed results and plots for numerical experiments with many quasi-Monte Carlo (QMC) and randomized QMC (RQMC) methods applied to a large set of integrands in various dimensions. For each integrand, method, and selected QMC/RQMC sample size (number of points), we provide an empirical distribution of the QMC/RQMC estimator X and a histogram representation. We also plot histograms of the empirical distributions of the average and median of r replicates of each estimator X , compare them, estimate their MSE, and plot these MSEs as functions of the number of points and as a function of r . The data, results, and code are given at

<https://github.com/pierrelecuyer/rqmc-mean-median>.

A subset of these results was used for the paper Seljak et al. (2026). The present document and the GitHub site provide a much larger set of results and plots than that paper. They are also meant to evolve in the future with the addition of new integrands and methods, and used for other projects.

1 Experimental Setting

$$\begin{aligned} \text{SmoothPerB4 : } f(\mathbf{u}) &= -1 + \prod_{j=1}^s [1 + (1/j)(30u_j^2(1 - u_j)^2 - 1)] \\ \text{SumUeU : } f(\mathbf{u}) &= -s + \sum_{j=1}^s u_j \exp(u_j) \\ \text{MC2 : } f(\mathbf{u}) &= -1 + (s - 1/2)^{-s} \prod_{j=1}^s (s - u_j) \\ \text{Polynomial : } f(\mathbf{u}) &= -1 + \prod_{j=1}^s [1 + (u_j - 1/2)a_j] \\ \text{Oscillatory : } f(\mathbf{u}) &= \cos\left(\sum_{j=1}^s u_j a_j\right) - \cos\left(\frac{1}{2} \sum_{j=1}^s a_j\right) \prod_{j=1}^s \frac{2 \sin(a_j/2)}{a_j} \\ \text{Gaussian : } f(\mathbf{u}) &= \exp\left(\sum_{j=1}^s u_j^2\right) - (1.462651745907181)^s \\ \text{SmoothGauss : } f(\mathbf{u}) &= -\Phi(1/\sqrt{2}) + \Phi\left(1 + s^{-1/2} \sum_{j=1}^s \Phi^{-1}(u_j)\right) \\ \text{PieceLinGauss : } f(\mathbf{u}) &= \max\left(s^{-1/2} \sum_{j=1}^s \Phi^{-1}(u_j) - \tau, 0\right) - \varphi(\tau) + \tau \Phi(-\tau) \end{aligned}$$

Figure 1: The selected integrand models

We consider the nine *integrands* (or *models*) defined in Figure 1, all parameterized by a dimension parameter s and re-centered so their exact integral is zero. These models are taken from Genz (1987), Goda and L’Ecuyer (2022), L’Ecuyer et al. (2024, 2023). Here, Φ is the cdf of the standard normal distribution and $a_j = j/s$ for all j . We consider all these models in $s = 2, 4, 8, 16, 32$ dimensions, with $n = 2^k$ QMC/RQMC points for $k = 10, 12, 14, 16$, and with the 13 different methods defined in Seljak et al. (2026) and listed in Figure 2. The methods in (a) and (b) are RQMC methods that provide unbiased estimators, while those in (c) are QMC methods with random parameters; they provide biased estimators. This gives $9 \times 5 \times 4 \times 13 = 2340$ cases.

- (a) RQMC methods with fixed choice of parameters, unbiased estimators:
 - Lat-RS: lattice rule with random shift (RS);
 - Lat-RSB: lattice rule with RS followed by baker transform;
 - Sob-RDS: Sobol’ net with random digital shift (RDS);
 - Sob-RDSB: Sobol’ net with RDS followed by baker transform;
 - Sob-NUS: Sobol’ net with nested uniform scrambling (NUS);
 - (b) RQMC methods with random parameters, unbiased estimators:
 - Lat-RvRS: lattice rule with random $\mathbf{a}$ and RS;
 - Lat-RvRSB: lattice rule with random $\mathbf{a}$ and RS followed by baker transform;
 - Lat-RpvRS: lattice rule with random number of points, random $\mathbf{a}$, and RS;
 - Lat-RpvRSB: lattice rule with random number of points, random $\mathbf{a}$, RS, and baker;
 - Sob-LMS-RDS: Sobol’ net with random LMS followed by RDS;
 - (c) QMC methods with random parameters, no shift, biased estimators:
 - Lat-Rv: lattice rule with random $\mathbf{a}$;
 - Lat-Rpv: lattice rule with random number of points and random $\mathbf{a}$;
 - Sob-LMS: Sobol’ net with random LMS, no shift;

Figure 2: The QMC/RQMC methods considered in the experiments; see Seljak et al. (2026).

The first stage of our experiments was to generate a data file of $N = 10^4$ independent observations of the estimator X for each of these 2340 cases. Each data file contains 10^4 real numbers, one per row, and nothing else. Its name indicates which case it is; the name is formed by the name of the integrand, followed by the value of s , the name of the method, the value of k , the number of observations which is usually 10000, and the extension `.dat`. These data files are all in the directory `datap1` on the GitHub site.

The lattice rules with fixed generating vector used $\mathbf{a} = (1, 103259, 73357, 46713, 58781, 112041, 32459, 50551, 40125, 128245, 18285, 124265, 98539, 130087, 113373, 22191, 120679, 98411, 94845, 33103, 47891, 15941, 30147, 43921, 81129, 3289, 50935, 63965, 55749, 38101, 70631, 116243)$. This vector was found by LatNet Builder (L’Ecuyer et al., 2022, L’Ecuyer and Munger, 2016) with the $\mathcal{P}_2$ criterion with product weights, with weight $\gamma_j = 2/(2 + j)$ for coordinate j , for $n = 2^k$ for all $k = 8, \dots, 18$, with a component-by-component construction in up to 32 dimensions. For the Sobol’ nets, we took the initial generating matrices from Joe and Kuo (2008).

To be able to visualize and compare the distributions of X quickly, we made histograms for these distributions, for each case. The histograms were made by pgfplots in L^AT_EX, with the L^AT_EX code generated in Java. There is one `.pdf` file for each integrand. It contains all the histograms for this integrand sorted first by the dimension s , for $s = 2, 4, 8, 16, 32$, with one row per method and four values of k per row, namely $k = 10, 12, 14, 16$. From the data files, we computed the empirical mean, variance, skewness, and excess kurtosis of the $N = 10^4$ realizations of X . Their values are displayed with each histogram. These `.pdf` files and their `.tex` source are in the directory `histograms` on the GitHub site. The single file `All-hist.pdf` contains all of them. These histograms are discussed

further in Section 2.

We also estimated the mean square error (MSE) and distributions of A_r and M_r , which are defined as the average and the median of r independent replicates of the given QMC/RQMC estimator. In our setting, the true value of the integral is always zero, although $\mathbb{E}[X]$ is nonzero in some cases, so we have $\text{MSE}[A_r] = \text{Var}[X]/r + \mathbb{E}^2[X]$, which we estimate by using the empirical mean and variance in the latter expression. We do not have such a direct formula for $\text{MSE}[M_r]$. To estimate it for each case, we took $m = 10000$ independent bootstrap sub-samples of size r from the corresponding data file, with replacement, computed the median M_r for each sample, and took the mean square of these m realizations of M_r . We did that for all cases, for selected values of r , and saved the MSE values in `.csv` files in the directory `mse`. These files are used by L^AT_EX to make the plots of Section 3.

These bootstrap subsamples of M_r and similar bootstrap subsamples of A_r were also used to construct empirical distributions and histograms of A_r and M_r and to estimate quantiles of these distributions and of the distributions of the absolute errors $|A_r - \mu|$ and $|M_r - \mu|$. These histograms are collected in the `histograms` directory. There is a separate `.pdf` file for each integrand and `All-ArMr-hist.pdf` collects all of them in a single file. Each plot contains two histograms on the same plot, one for A_r and one for M_r , and certain quantiles are also indicated for each histogram. This permits one to compare the distributions of A_r and M_r , and compare their quantiles in the tails. Plots of these quantiles and of the quantiles of the absolute errors as functions of k are provided in Section 3.5.

The Java code used for these experiments is in folder `src/main/docs/examples/rqmcexperiments` of the SSJ library available at <https://github.com/pierrelecuyer/ssj/tree/develop/>. For now, this code is only in the “`develop`” branch of this repository. It will eventually be integrated to the main repository at `umontreal-simul`. The integrand implementations are in that same directory, one class per integrand. To run the code, one must first install SSJ by following the instructions given in the README of the GitHub site of SSJ. The main programs are `Samo25SamplesMain` and `Samo25Samples` to generate the data files, `HistSamo25All`, `HistSamo25ArMrAll`, `HistCollectionLatex`, and `HistSamo25ArMr`, to generate the histograms, `MSESamo25` and `MeanMedianMSE` to compute all the data files on the MSEs and the moments of the distributions. This code was written by Pierre L’Ecuyer with some help from Otman Cherrati, Christiane Lemieux, and Gregory Seljak.

The total number of integrand evaluations required to generate all the `.dat` files for the 9 integrands, 6 values of s , and 5 values of k , was $9 \times 6 \times 13 \times 87296 \times 10000 = 612,817,920,000$ and required more than 7721 billion random numbers. That took about two days on a laptop computer and was the most computing time-consuming task. Producing all the histograms and all the MSE files and plots from these `.dat` files takes only a few minutes.

2 The Histograms

The histograms for the realizations of X and for the realizations of A_r and M_r are in the folder:

<https://github.com/pierrelecuyer/rqmc-mean-median/blob/main/histograms/>

All these histograms are made with a sample size of 10^4 . In either case, there is one `.pdf` file per integrand. This makes it easy to add new integrands in the future. In addition to that, all the histograms for all the integrands are also included in two large `.pdf` files, which are `All-hist.pdf` for the histograms of X and `All-ArMr-hist.pdf` for the histograms of A_r and M_r . We do not include these files in the present document to avoid making it too large, but they are easy to examine directly.

The readers should look at them to get an idea of what they look like. We comment them below.

2.1 Histograms of X

Some histograms of X resemble a normal distribution, but many do not, even when X is an average over a large number of points (n is large). Many of them have strange shapes. For the RQMC methods, for which $\mathbb{E}[X] = 0$, most of them appear very symmetric about their mean. Some are asymmetric in the tails because the two far tails do not contain the same (small) number of observations, but most often they become more symmetric when we increase the sample size. For the QMC methods with random parameters (Lat-Rv, Lat-Rpv, Sob-LMS), on the other hand, the distribution of X is sometimes very asymmetric. We give examples of these two situations in Figure 3.

Certain shapes of distributions favor using the median rather than the average, while other shapes favor the opposite. Example 2.7 in Seljak et al. (2026) shows that for symmetric distributions centered at 0, using the median is asymptotically better (for large r) if and only if $g(0)\sigma > 1/2$, where $g(0)$ is the density at 0 and σ^2 is the variance. Thus, the distributions with a large peak at 0 tend to favor the median, while the distributions with a small (or no) density around 0 favor the average.

The **Polynomial** integrand illustrates this. **Polynomial-2-Sob-RDSB** gives two very narrow peaks that are symmetric about zero and there is no density in the center, so in this case taking the median is clearly worse than taking the average. **Polynomial-4-Sob-RDSB** and **Polynomial-8-Sob-RDSB** also give strange histograms with two or more peaks, sometimes with no density in the center but not always. For the **Gaussian** and **MC2** integrands, **Sob-RDSB** gives distributions with several high peaks.

For **SmoothPerB4**, **Sob-RDSB** produces a strange behavior, with two or three or more narrow islands, while **Lat-RSB** gives a so-called bathtub distribution, with peaks at both ends.

The **PieceLinGauss** function may represent a typical computational finance integrand. With lattices, the random coefficients give a much larger variance (100 times larger) and a larger kurtosis for $s = 4$ and $k = 16$, but only slightly larger for larger s or smaller k . Since this function is not smooth, making it periodic with the baker transformation makes no significant difference.

For **IndSumNormal**, for which the function f can take only two values, we find histograms where only the bins at a certain regular spacing (e.g., one at every three bins or at every five bins) have observations and the other ones are empty.

For **SumUeU** with **Lat-Rv**, for any s and k , there is no visible histogram. The reason is that this function is a sum of one-dimensional functions, so the QMC estimator X with **Lat-Rv** does not depend on the choice of generating vector. It always takes the same value. This means that the variance is zero, and the skewness and kurtosis are undefined, because $\sigma = 0$. The empirical variance, skewness, and kurtosis values that we get for this case are just numerical noise. With **Lat-Rpv**, X is not always the same because it depends on the random p . The same behavior can also be observed for all other integrands when $s = 1$.

In many cases, mostly when we have random parameters, the true distribution of X has areas of very small but non-negligible density, in which we can find very few observations, so the histograms with $N = 10^4$ may give a poor representation of the true distribution in those areas. For example, we can have one or two outlier(s) far on one side, but not on the other side, which may suggest an

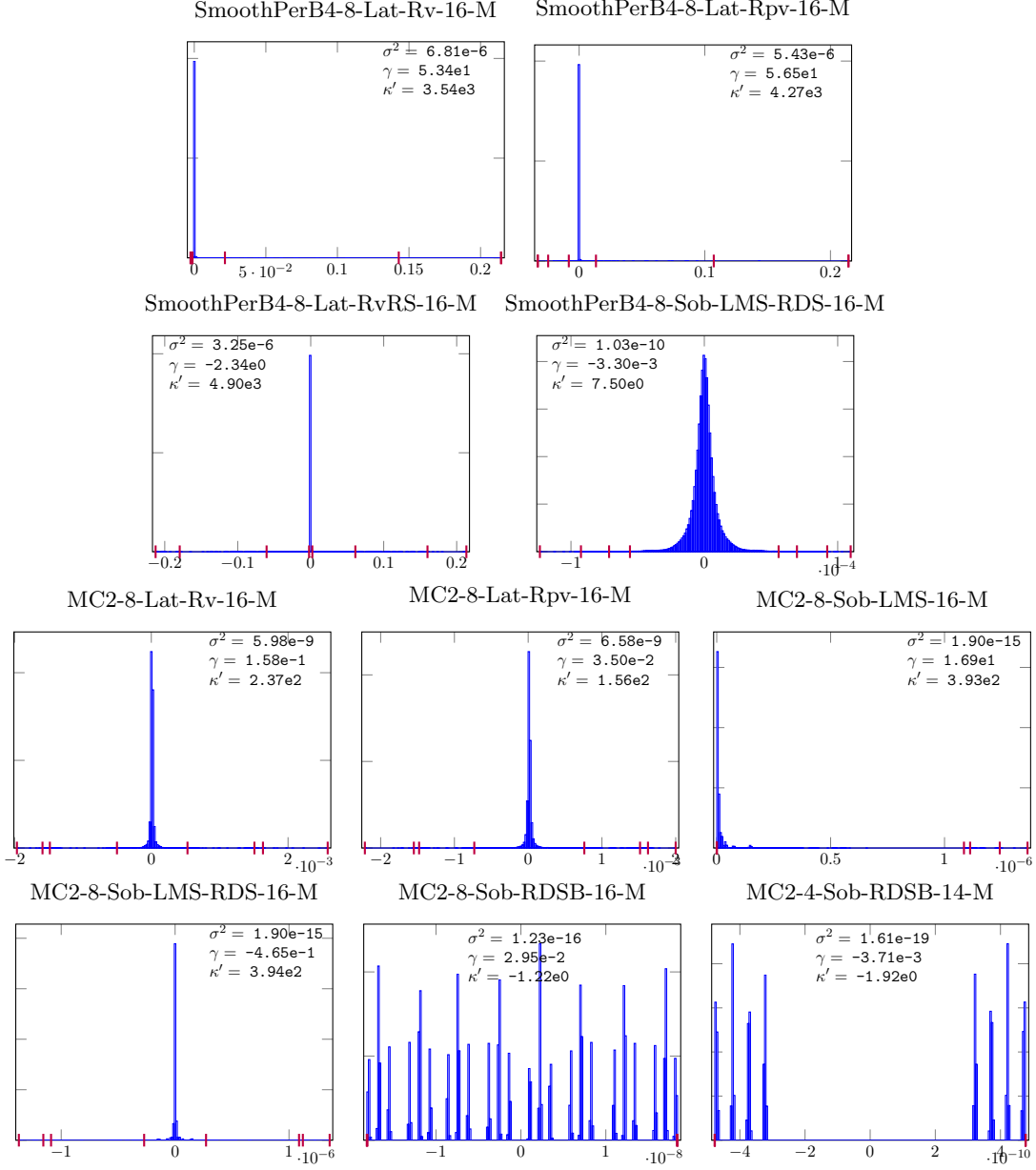


Figure 3: Histograms of 10^6 (one million) realizations of X . In these plots, the green marks on each side indicate the first, 10th, 100th, and 1000th smallest observations on the left and largest observations on the right. They give an idea of the symmetry/asymmetry of the tails.

asymmetric distribution even if the true distribution is symmetric. To better see what really goes on, we redid the experiment in some cases with the larger sample size of $N = 10^6$ (one million). This was done with the method `simulRepsSpecificCases` in class `Samo25Samples`. Figure 3 shows some of those histograms. They give a more accurate representation of the true distribution of X than those with $N = 10^4$. In general, they do not behave much differently than for $N = 10^4$, except that all the ones with a RS or RDS are more symmetrical, which suggests that the true distribution is symmetrical in those cases. The histograms for the QMC methods (Lat-Rv, Lat-Rpv, Sob-LMS) are all very asymmetric. This can be seen by looking at the green bars below the histograms. The only QMC case where the distribution may be symmetric is `MC2-8-Lat-Rv-16-M`, for which the smallest and largest observations are at a different distance from zero, but the 10th (and also the 100th) smallest and largest are at about the same distance.

2.2 Histograms of A_r and M_r

All the histograms of A_r and M_r are collected in `All-ArMr-hist.pdf`. There is also a separate `.pdf` file named `<Model>-ArMr-hist.pdf` for each model. Each plot contains two histograms on the same plot, one for A_r in green and one for M_r in red. There are also marks on the horizontal axis of each histogram, in the same color, to indicate the smallest and largest observation, as well as the 1%, 5%, 95%, and 99% quantiles. This makes six marks of each color.

The general tendency is that for the methods with random parameters, the tail quantiles of A_r are farther away from zero than those of M_r , so the distribution is more spread-out. For the methods with fixed parameters, we usually see the opposite.

For the QMC methods Lat-Rv and Lat-Rpv, which are lattice rules with random parameters and no RS, and even more for Sob-LMS, the distributions are sometimes highly asymmetric. For the other methods, they are usually symmetric or nearly so.

For Sob-RDSB, the histograms often have very bizarre shapes, usually symmetric but with separate high peaks, sometimes just a few of them and sometimes many.

For the `IndSumNormal` integrand, the histograms of M_r are made of high peaks with empty spaces in between, due to the discrete nature of that integrand and of the distribution of X .

3 The MSE Plots

3.1 The files used to make the plots

The program `MSESamo25` was used to compute estimates of $\text{MSE}[M_r]$ and $\text{MSE}[A_r]$ for all cases, by using static methods from the class `MeanMedianMSE`. This program produces several `.csv` files and puts them in the `mse` directory.

For each model, value of s , and value of r , the method `quantilesAndMSEArMrOneModel` produces a `.csv` file with one row per value of k and 14 columns per RQMC method. These columns give estimates of $\text{MSE}[M_r]$ and $\text{MSE}[A_r]$, the 1%, 5%, 95%, and 99% quantiles of the empirical distributions of A_r and M_r , and the 95% and 99% quantiles of the empirical distributions of the absolute errors $|A_r - \mu|$ and $|M_r - \mu|$.

The method `estimateMSEArMrManyr` produces similar `.csv` files, but whose rows are for different values of r instead of values of k . These `.csv` files are used by $\text{\LaTeX}$ to plot the MSEs as functions of k or r .

The method `estimateMSEOneCategory` computes the moments and the MSE estimates for all methods in a given category, for all models, s , and k . There are three categories of methods, as defined in Figure 2: (a) RQMC with fixed parameters, (b) RQMC with random parameters, and (c) QMC with random parameters. The method `estimateMSEOneCategory` constructs a single `.csv` file per category that has one row for each case. These files are used to produce scatter plots like those in Figure 4 and in Figure 3 of Seljak et al. (2026).

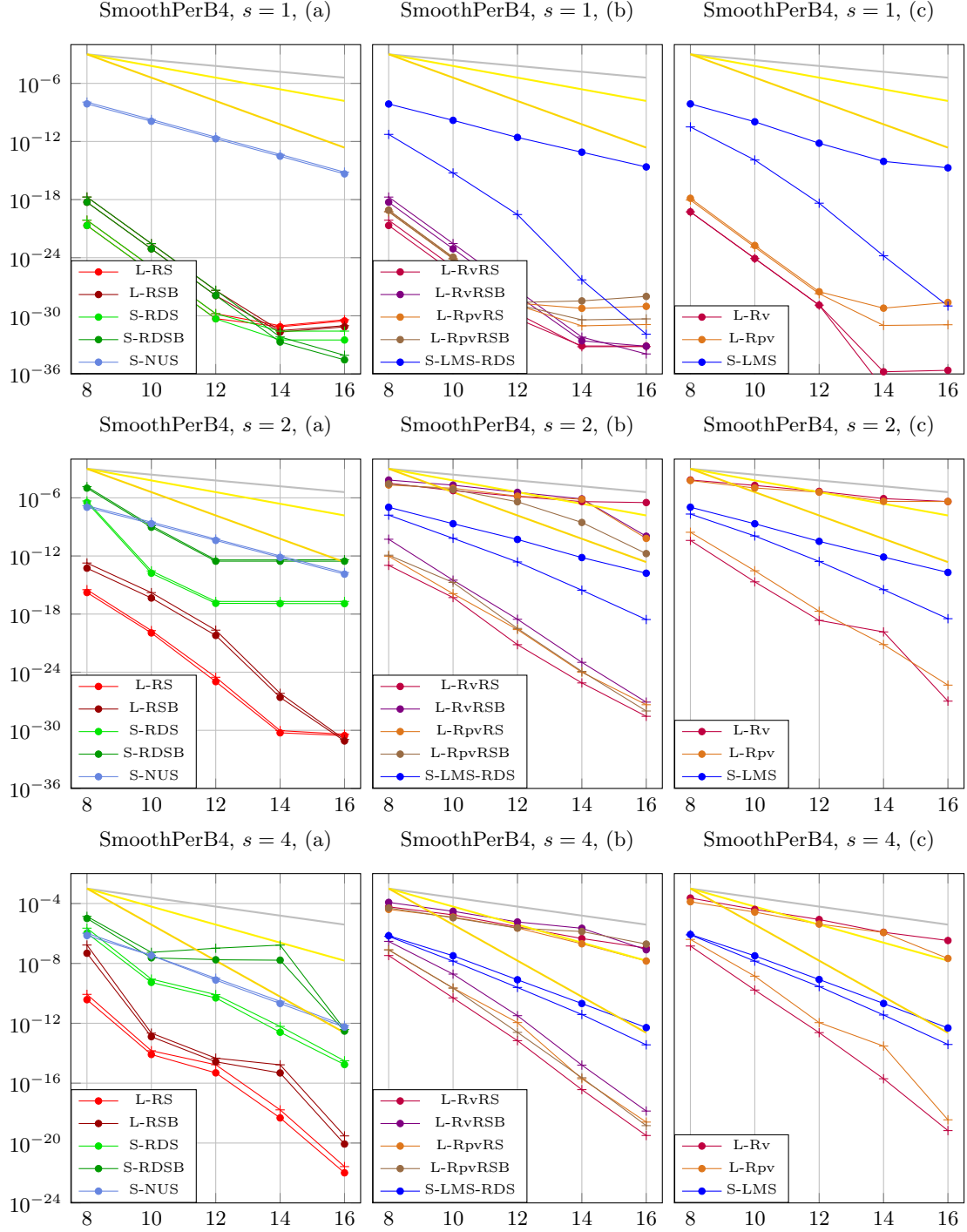
In the following subsections, for each combination of integrand and dimension s , we show three plots per row. These three plots correspond to these three categories of methods. All three use the same horizontal and vertical scale, so the values can be compared between them. Each plot shows $\text{MSE}[A_r]$ and $\text{MSE}[M_r]$ as functions of k or as functions of r , in log scale, for each method in the category. The legend in the plot indicates the color for each method. We abbreviate Lat to L and Sob to S. The lines with a “•” mark are for A_r and those with a “+” mark are for M_r . The light grey, yellow, dark yellow, and orange straight lines, when they appear, show the convergence rates of n^{-1} , n^{-2} , n^{-3} , and n^{-4} , respectively. In Sections 3.2 and 3.3, the plots are made as functions of k for fixed r , while in Section 3.4 they are as functions of r for fixed k . These MSE plots show various types of behavior.

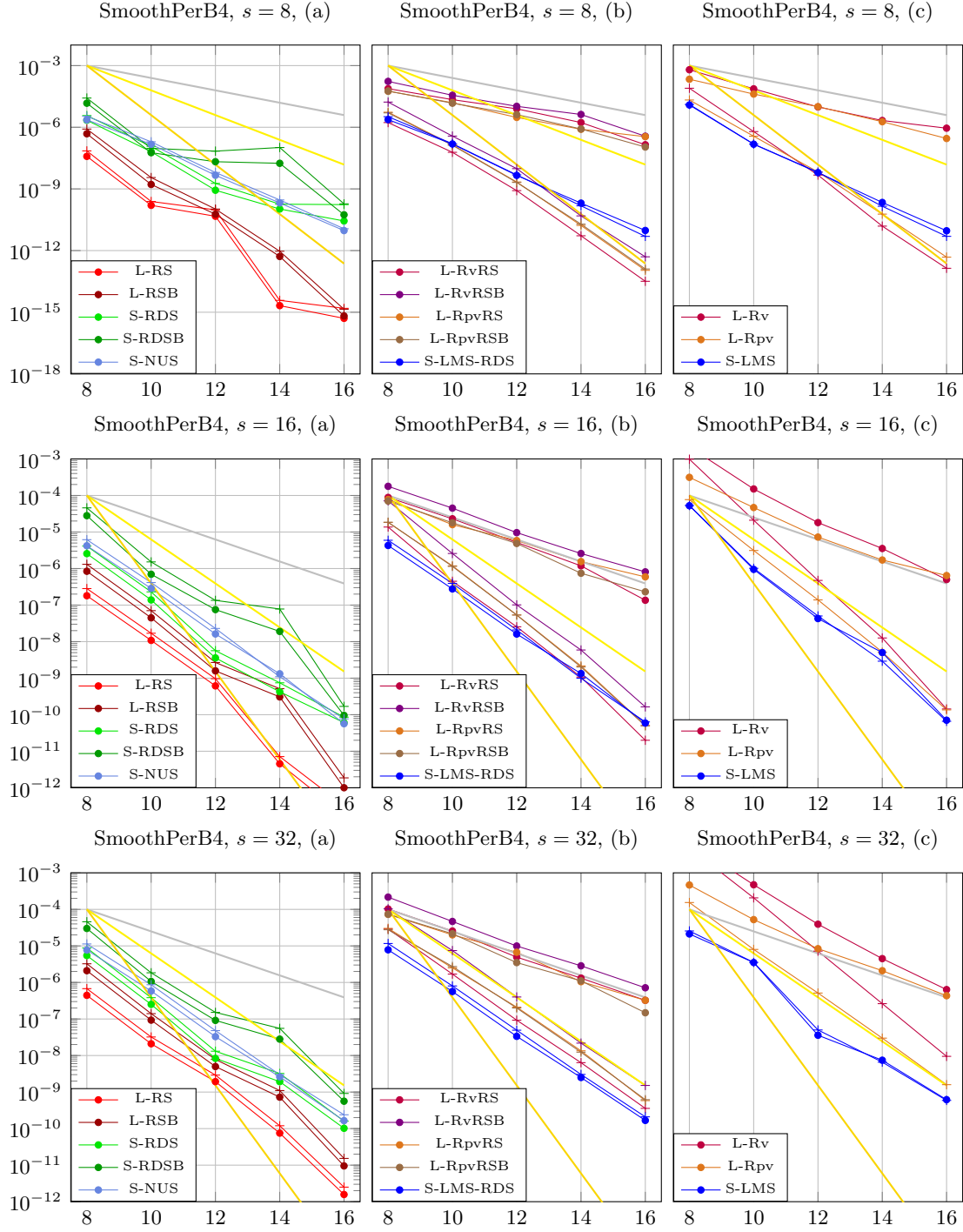
For the RQMC methods with fixed parameters, Category (a), we can see that taking the average A_r is almost always slightly better than taking the median M_r , for both values of r . The MSE ratio is usually not far from 1, however, except for a few cases, mostly for the larger r . For `SmoothPerB4`, the lattice rules are the best performers as expected, since they are designed for this type of smooth periodic function. For the other smooth and bounded functions that are not periodic, the best performers are `Lat-RSB` and `Sob-RDSB`, which outperform `Sob-NUS`. This behavior is explained in L’Ecuyer and Lemieux (2026).

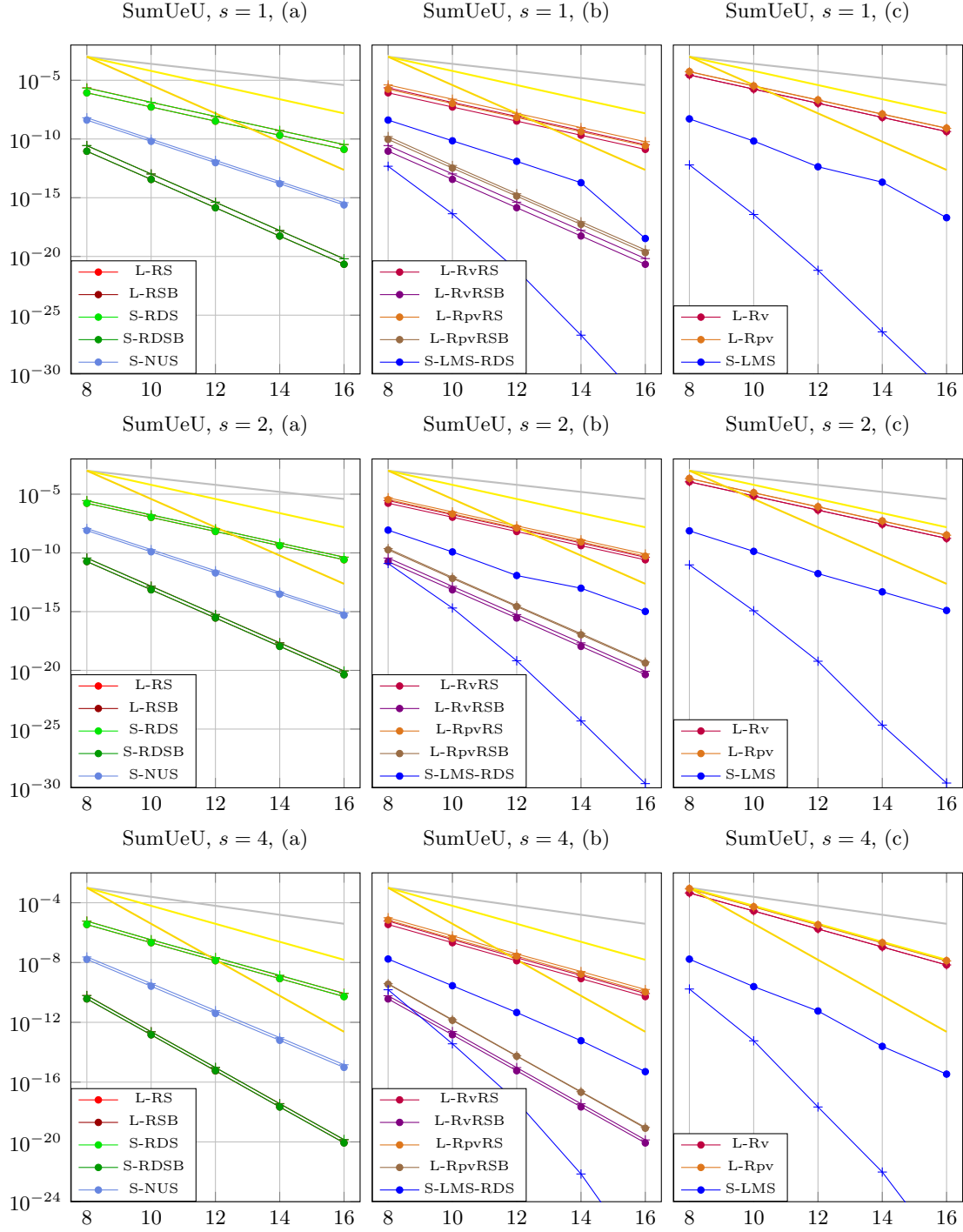
For the random-parameter categories, (b) and (c), `Sob-LMS` and `Sob-LMS-RDS` are often much better than the other methods, and taking the median with them is usually better and often provides a better convergence rate for the MSE than taking the average. For `SmoothPerB4`, `Lat-Rv` combined with the median wins, but only in small dimensions. The random-parameter methods rarely give a better MSE than the best fixed-parameter one. This occurs sometimes with `Sob-LMS-RDS` combined with the median.

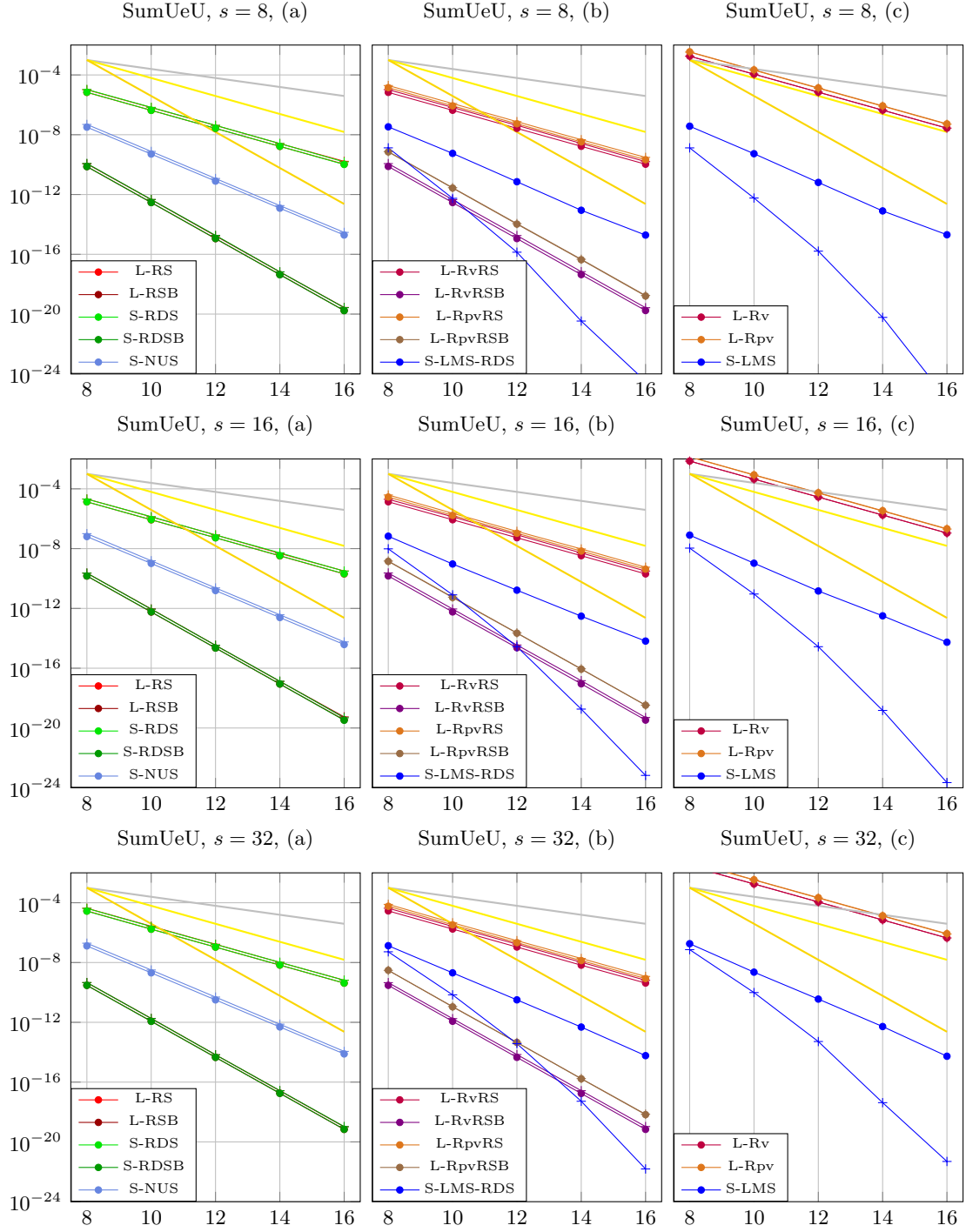
In the MSE plots as functions of r for fixed k , in Section 3.4, we find that the MSE decreases only very slowly with r . When the estimator is unbiased (RQMC), we know that $\text{MSE}[A_r]$ decreases as $1/r$ as a function of r , while for a biased QMC estimator, it does not converge to zero when $r \rightarrow \infty$ for fixed k . This is why for several cases in Category (c), the MSE function is pretty flat, practically constant in r .

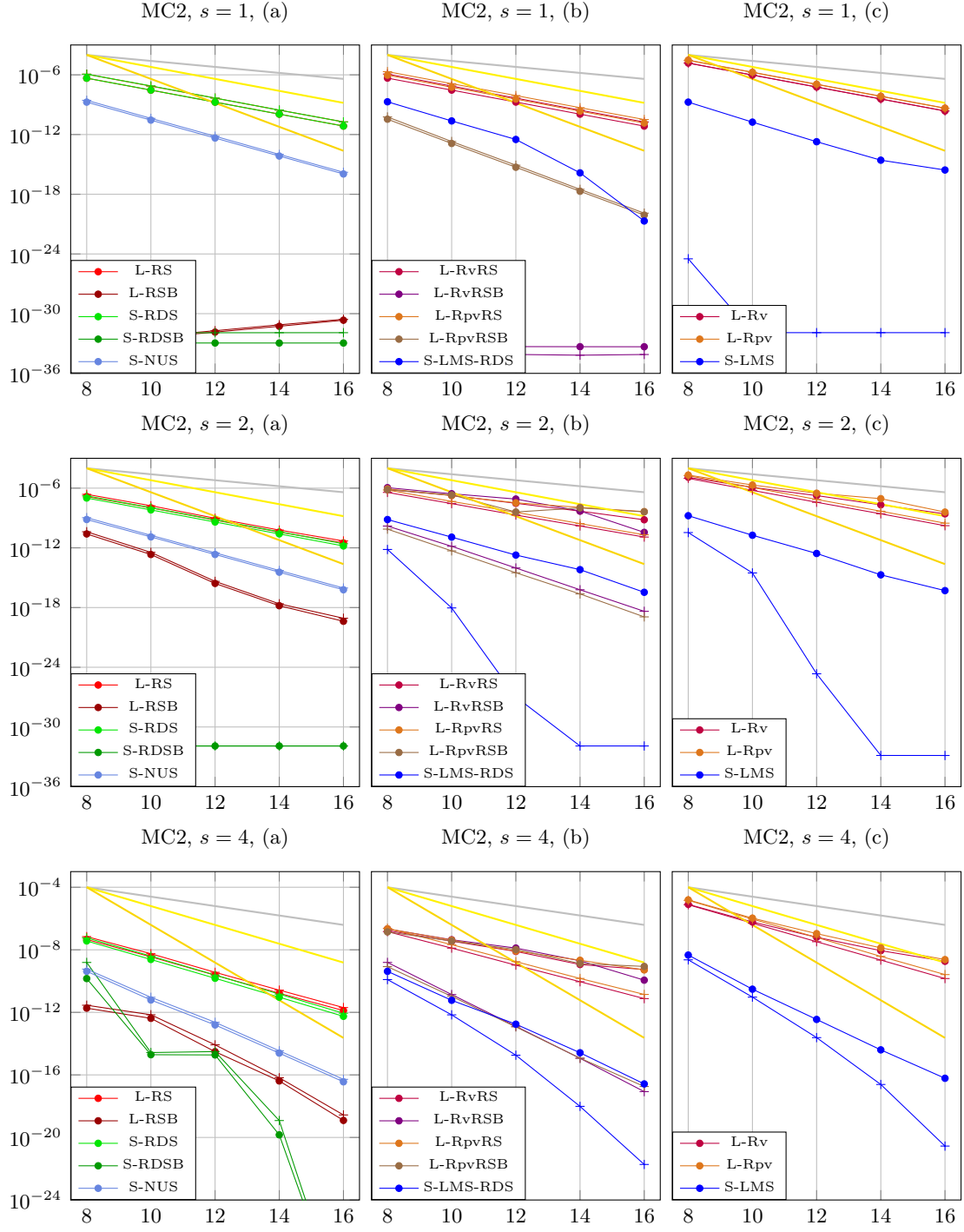
3.2 MSE plots as functions of k for $r = 11$

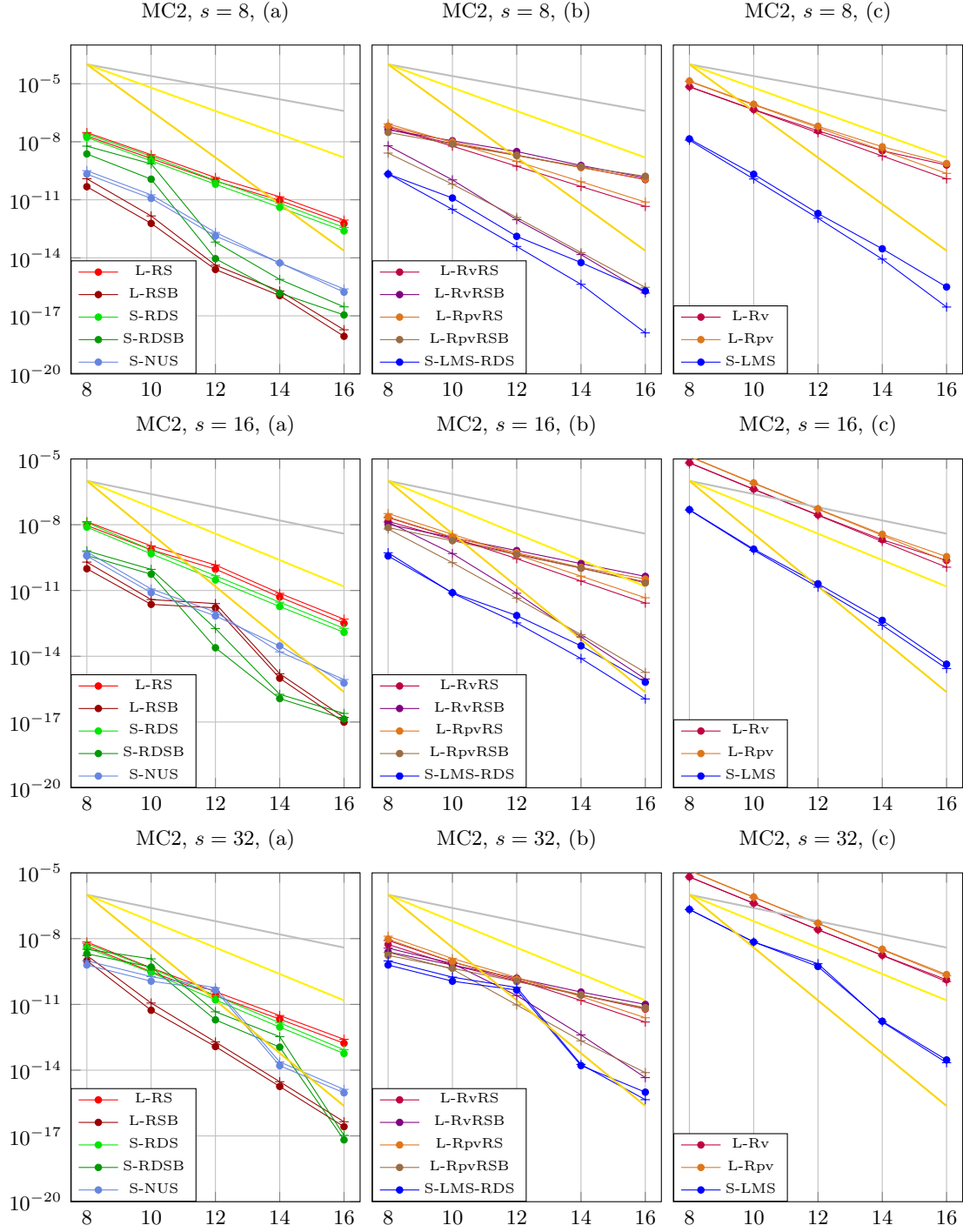


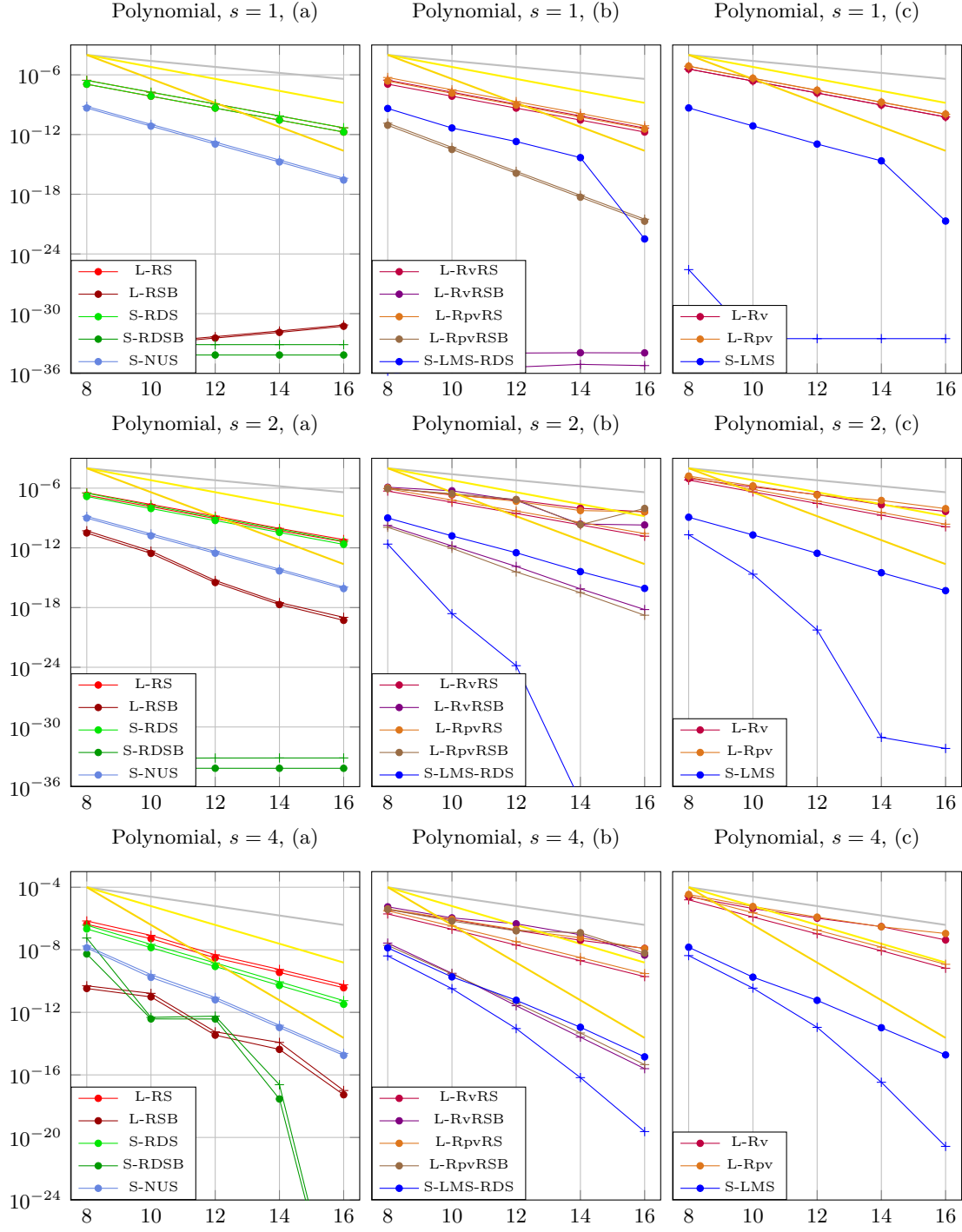


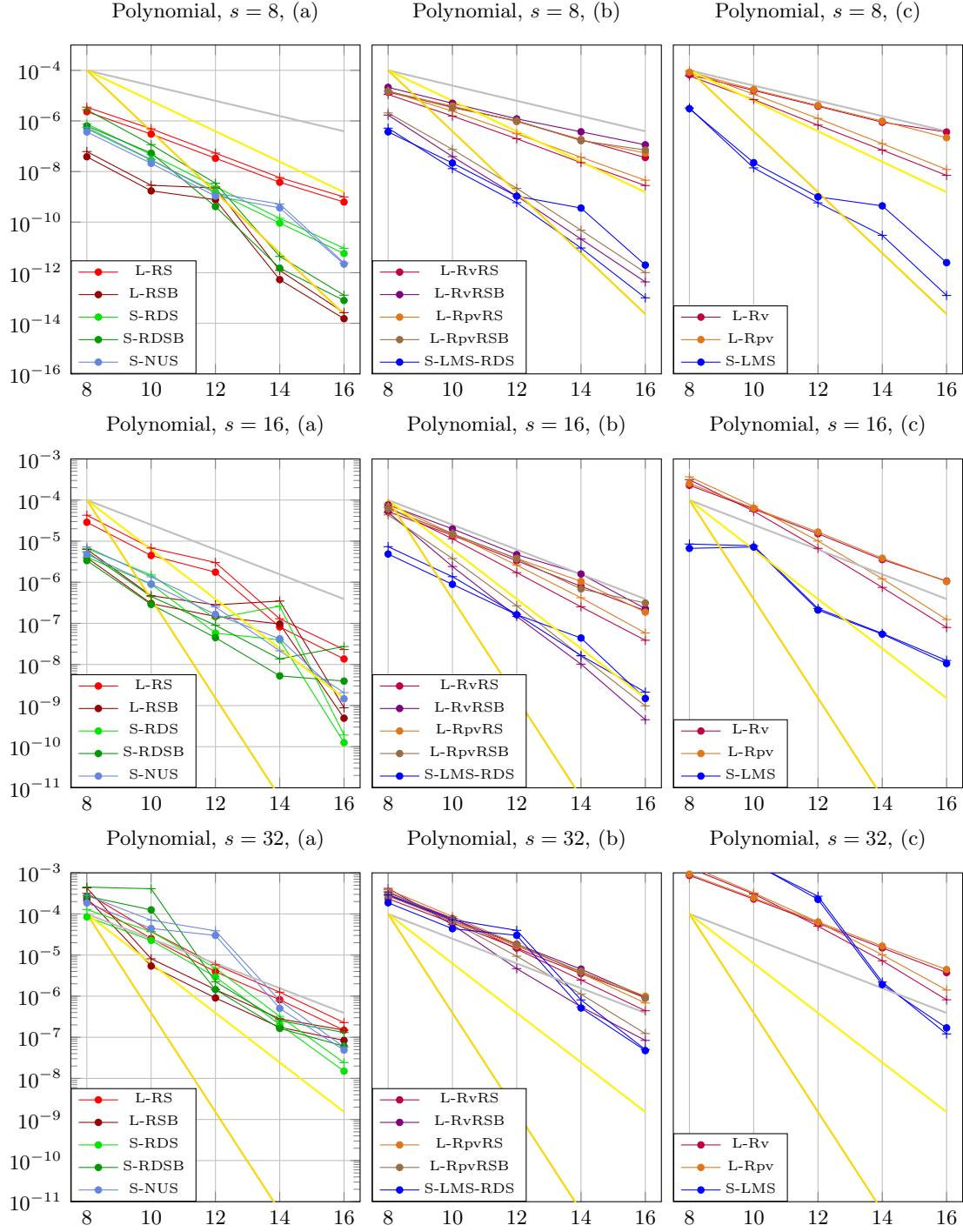


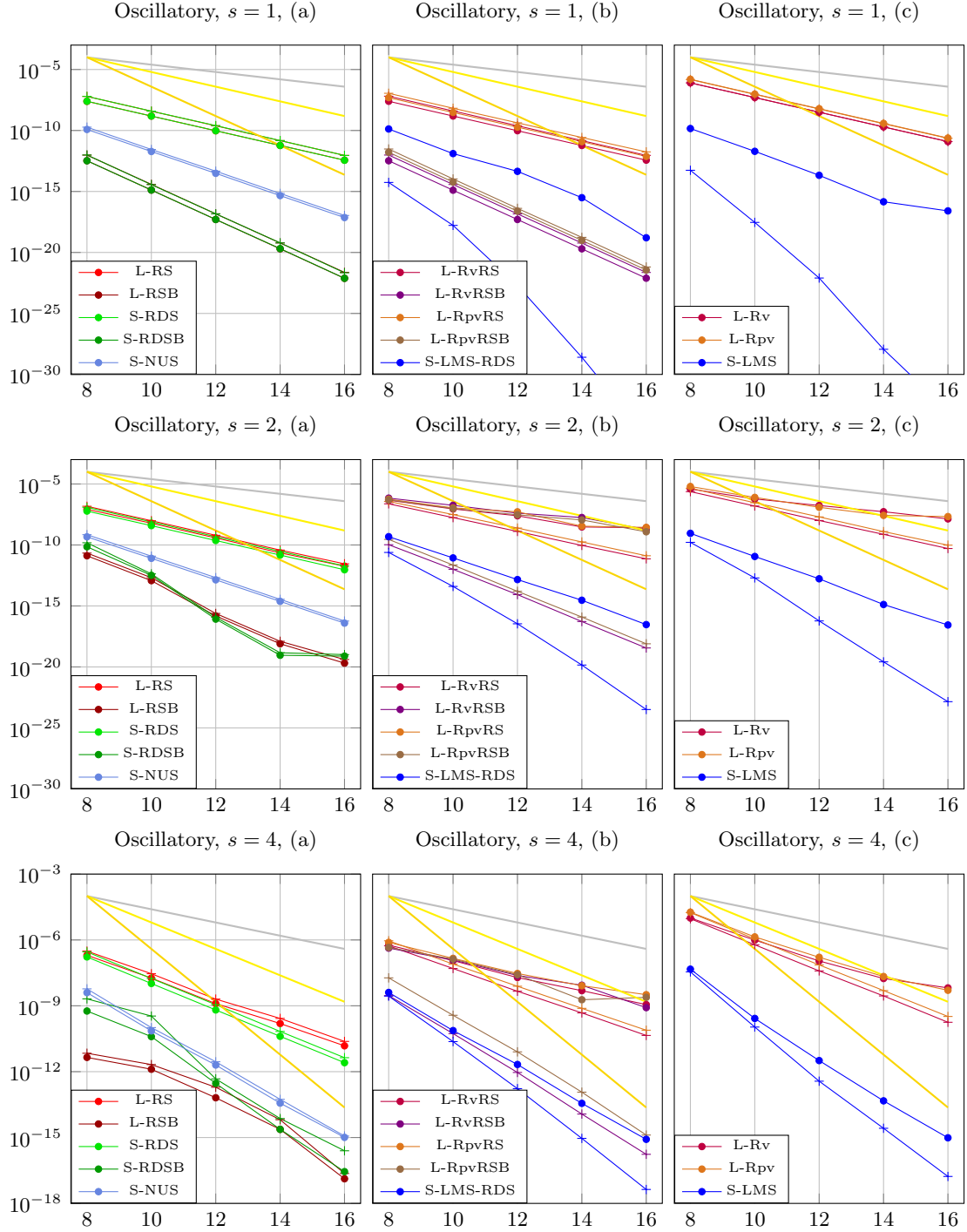


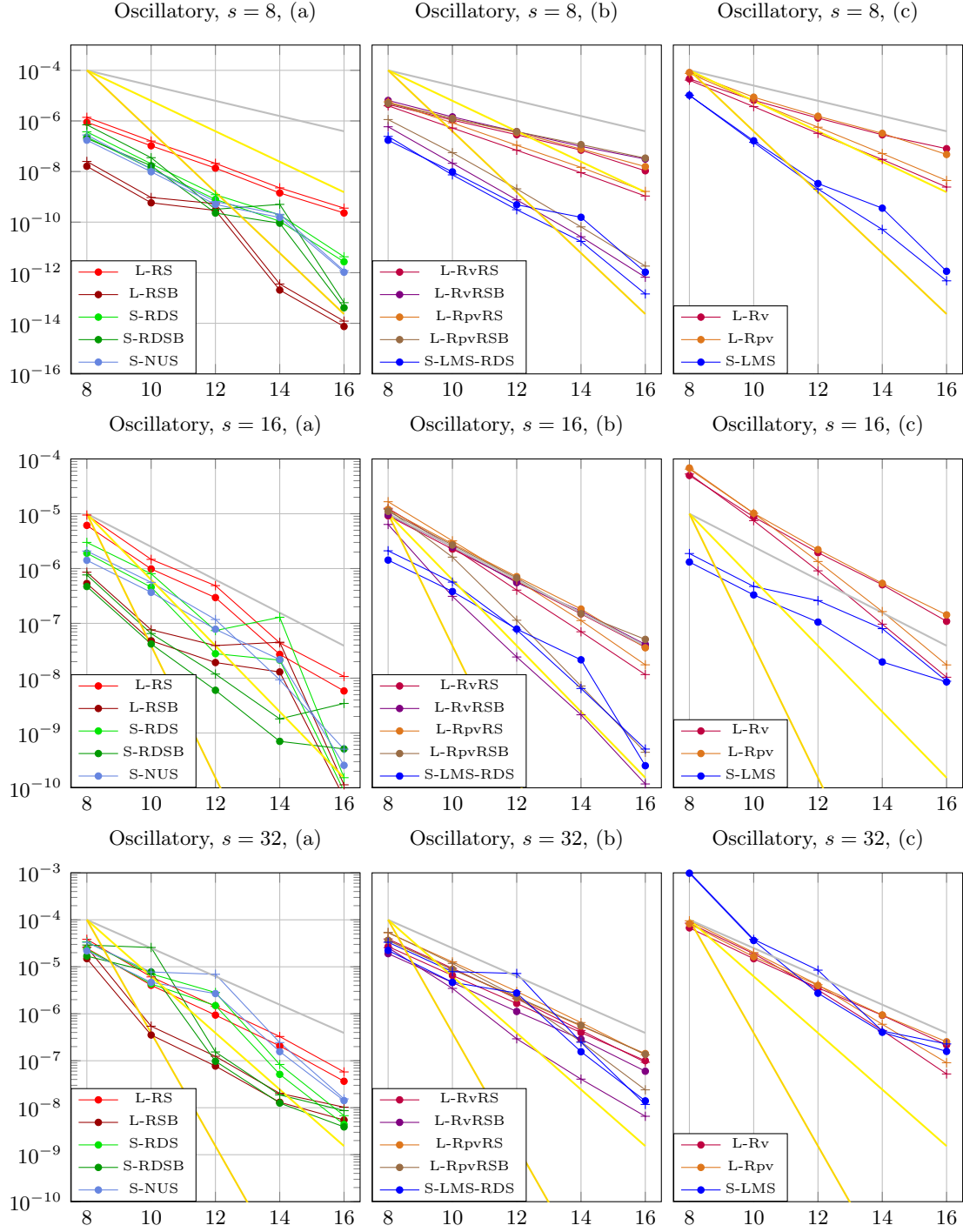


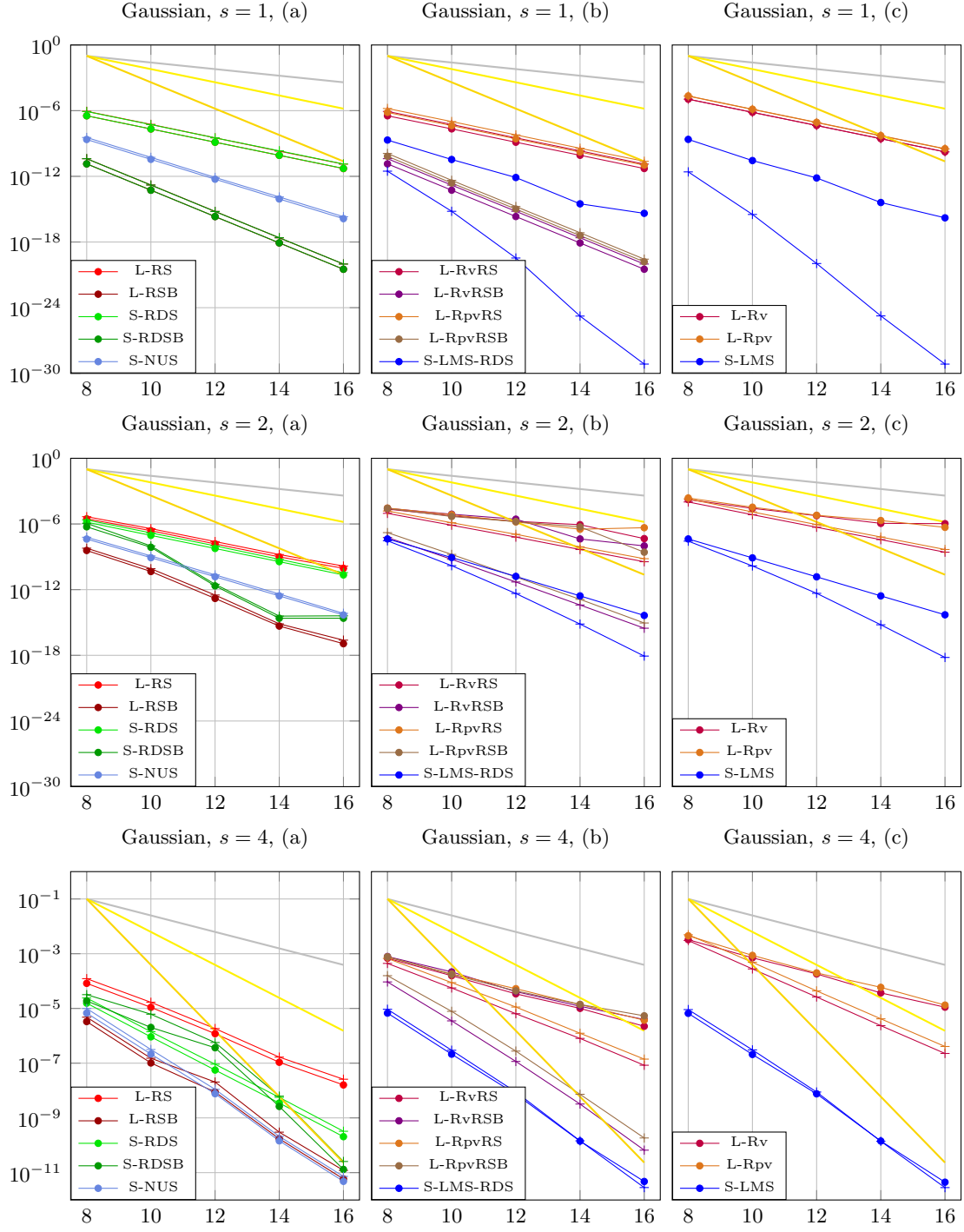


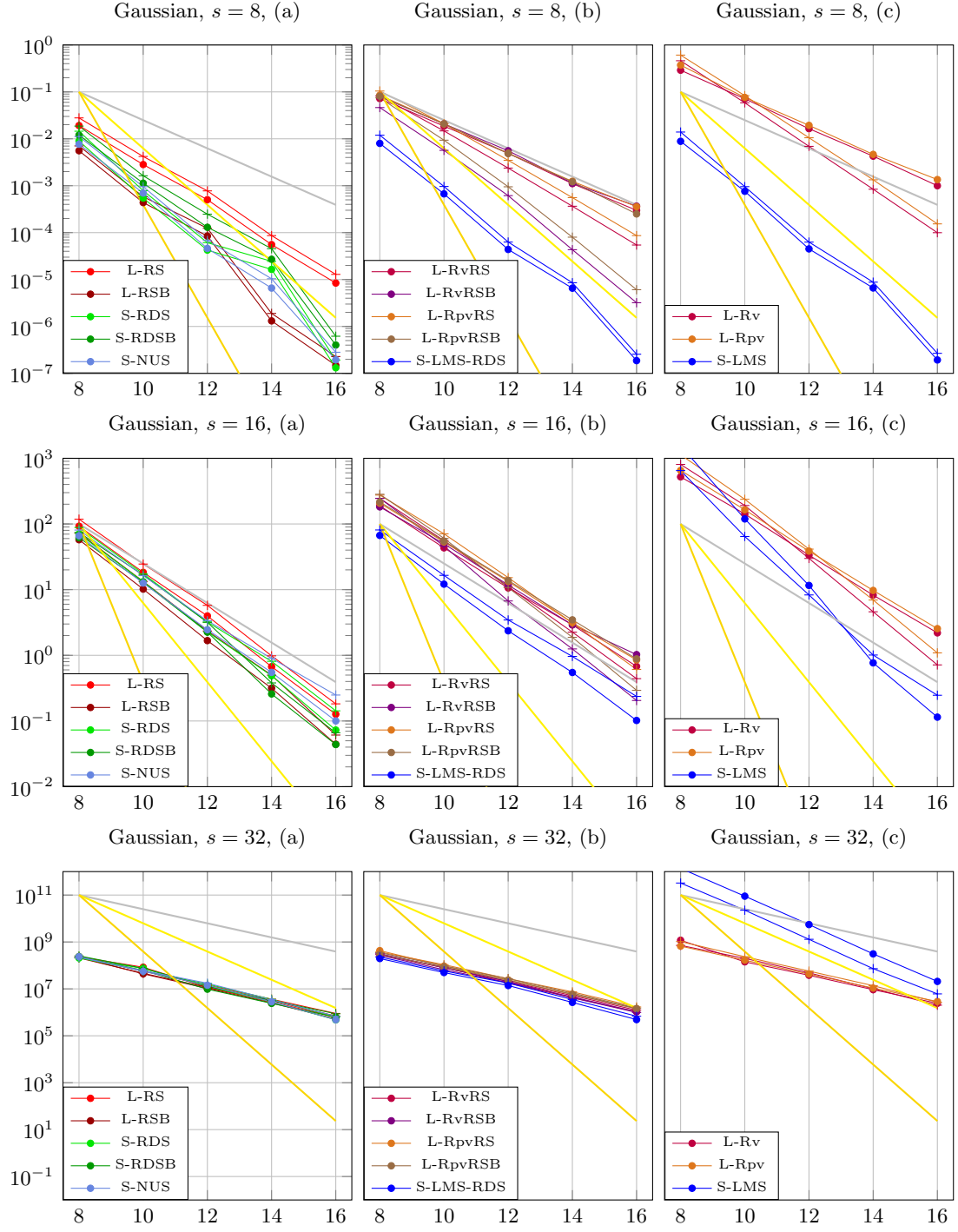


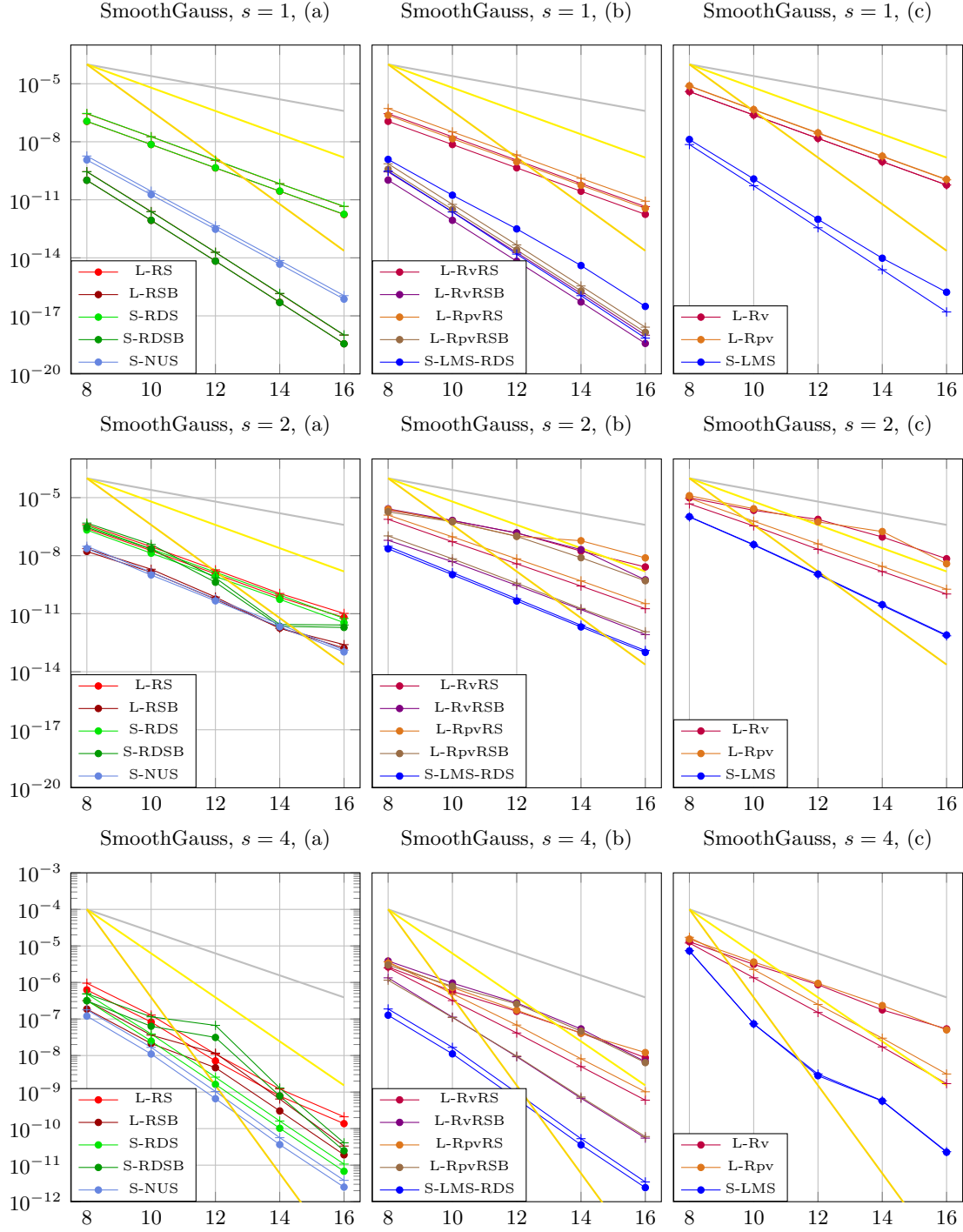


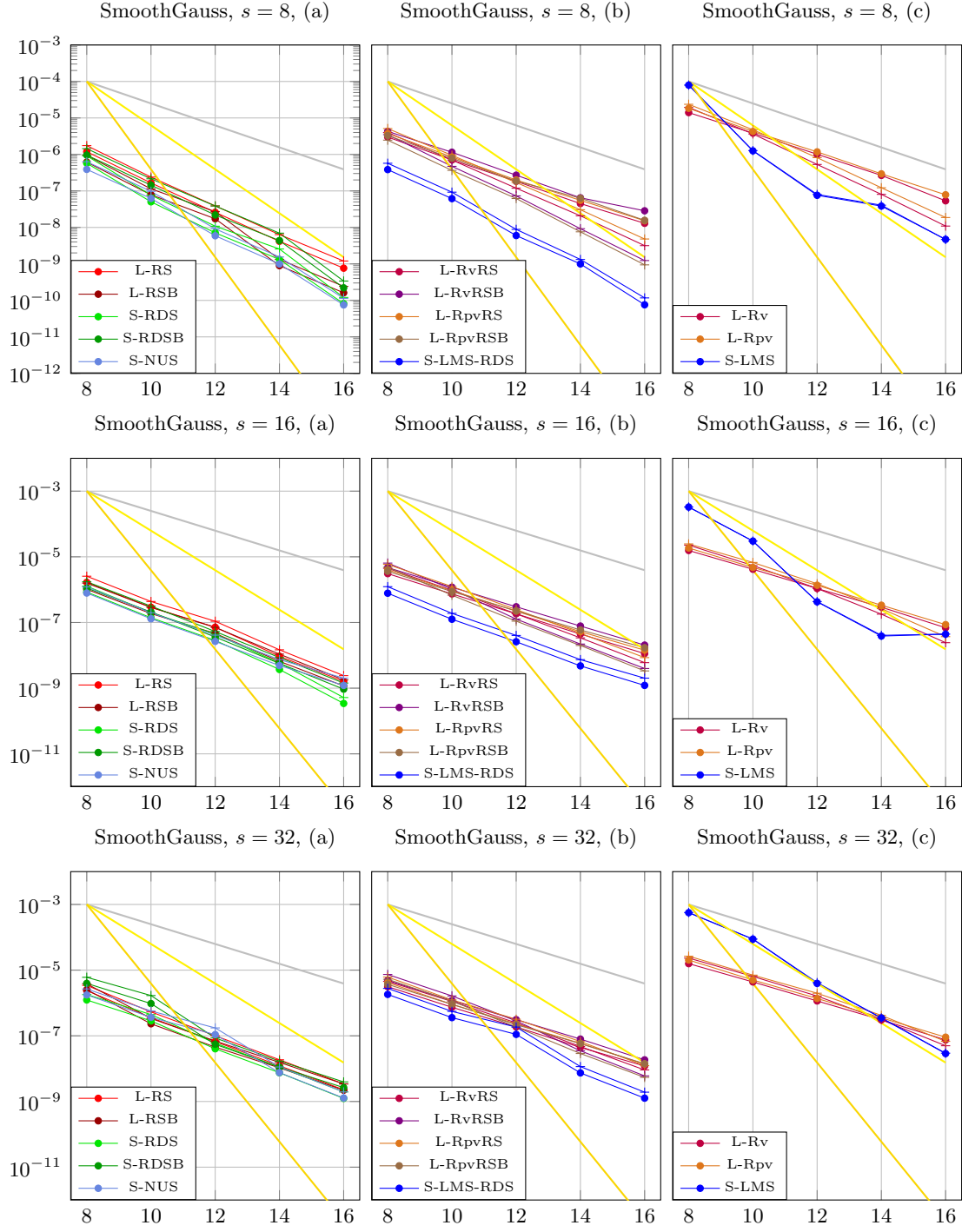


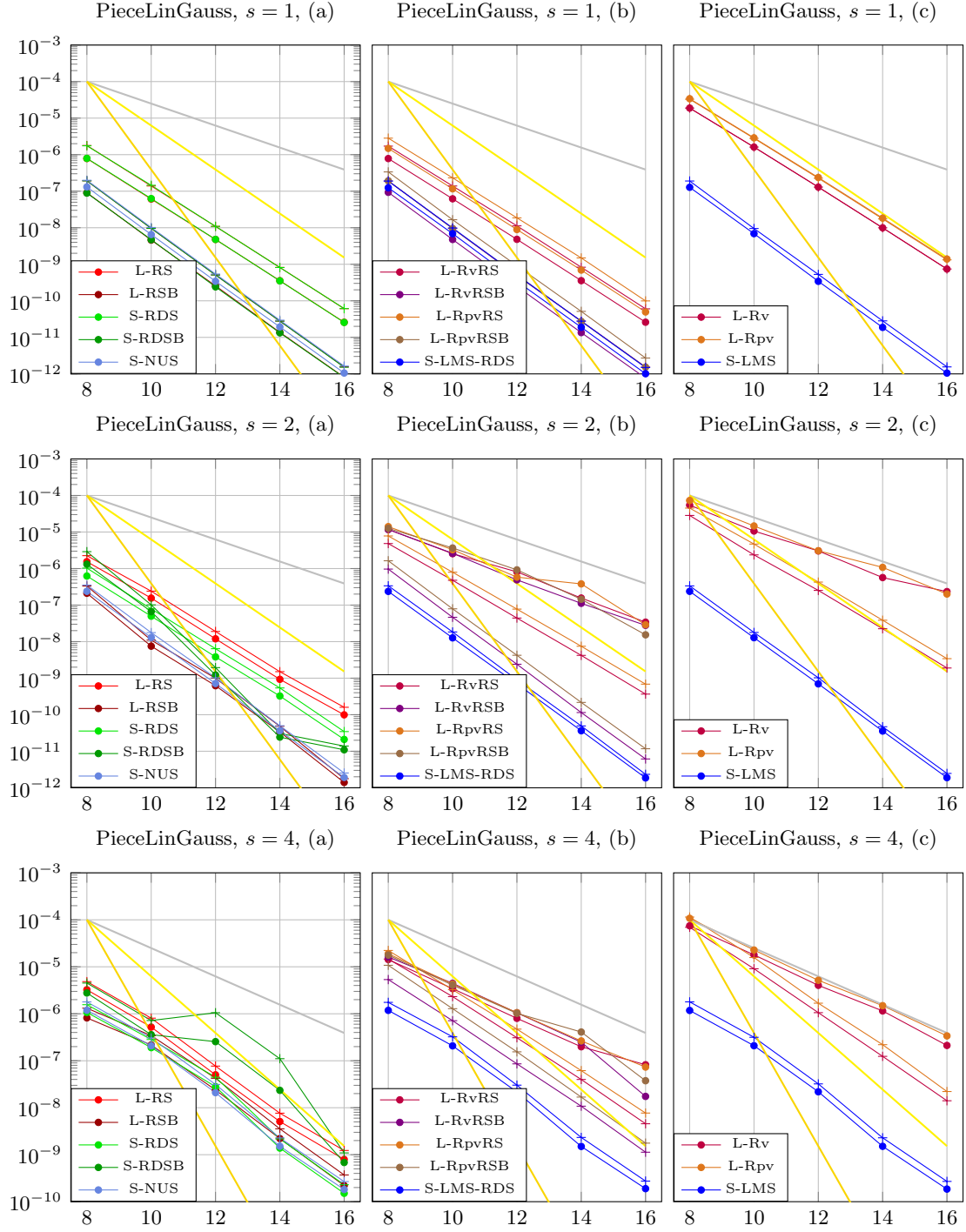


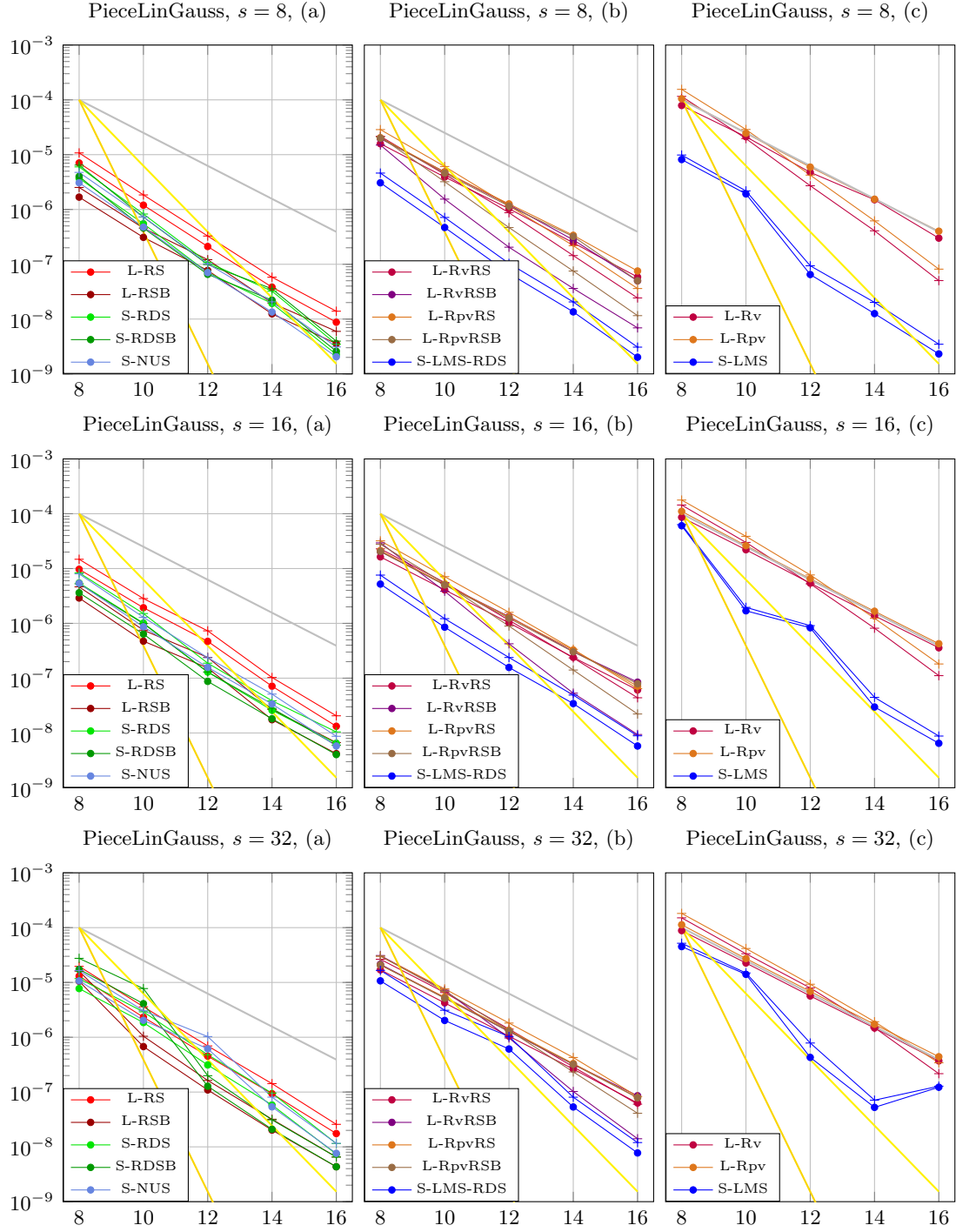


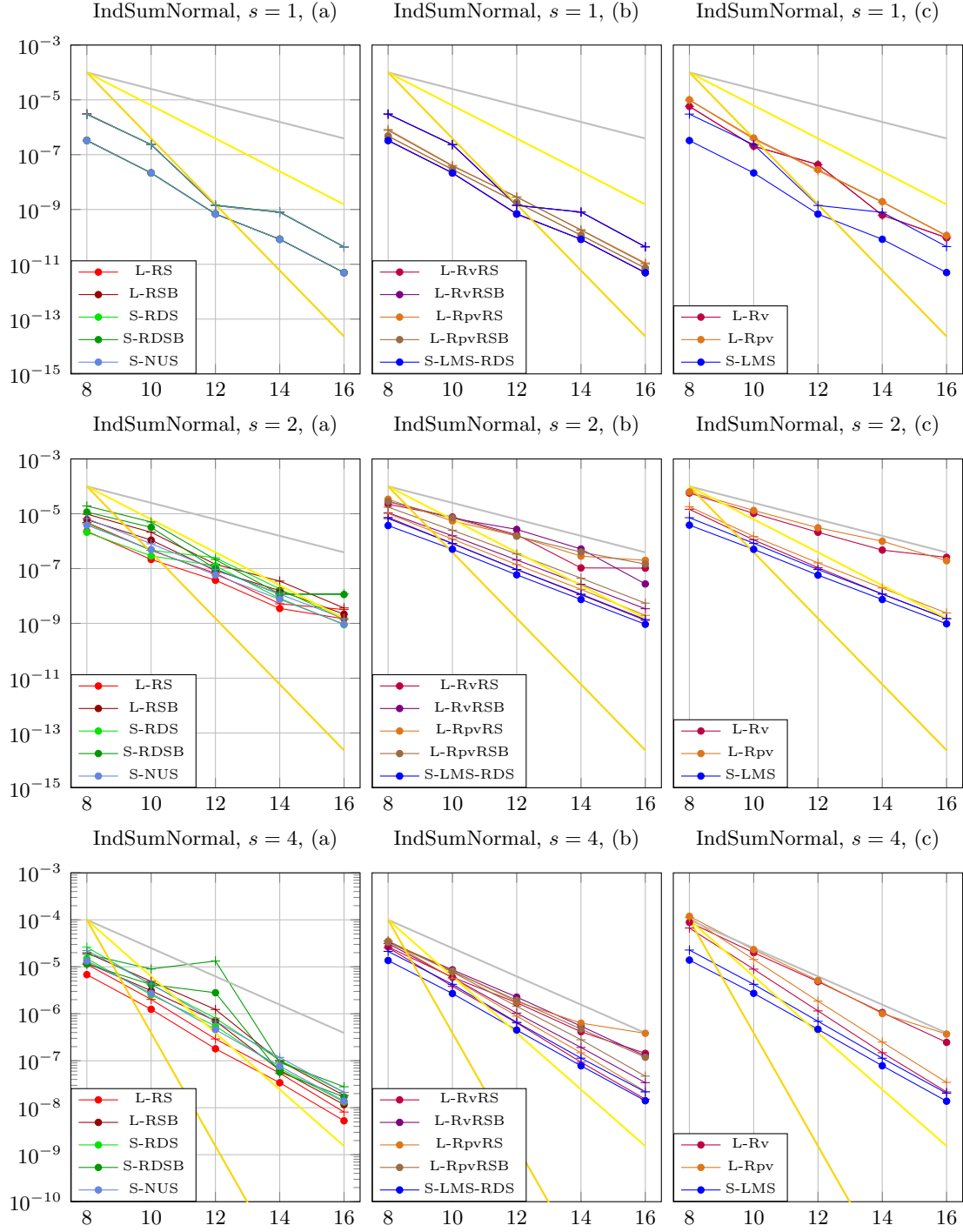


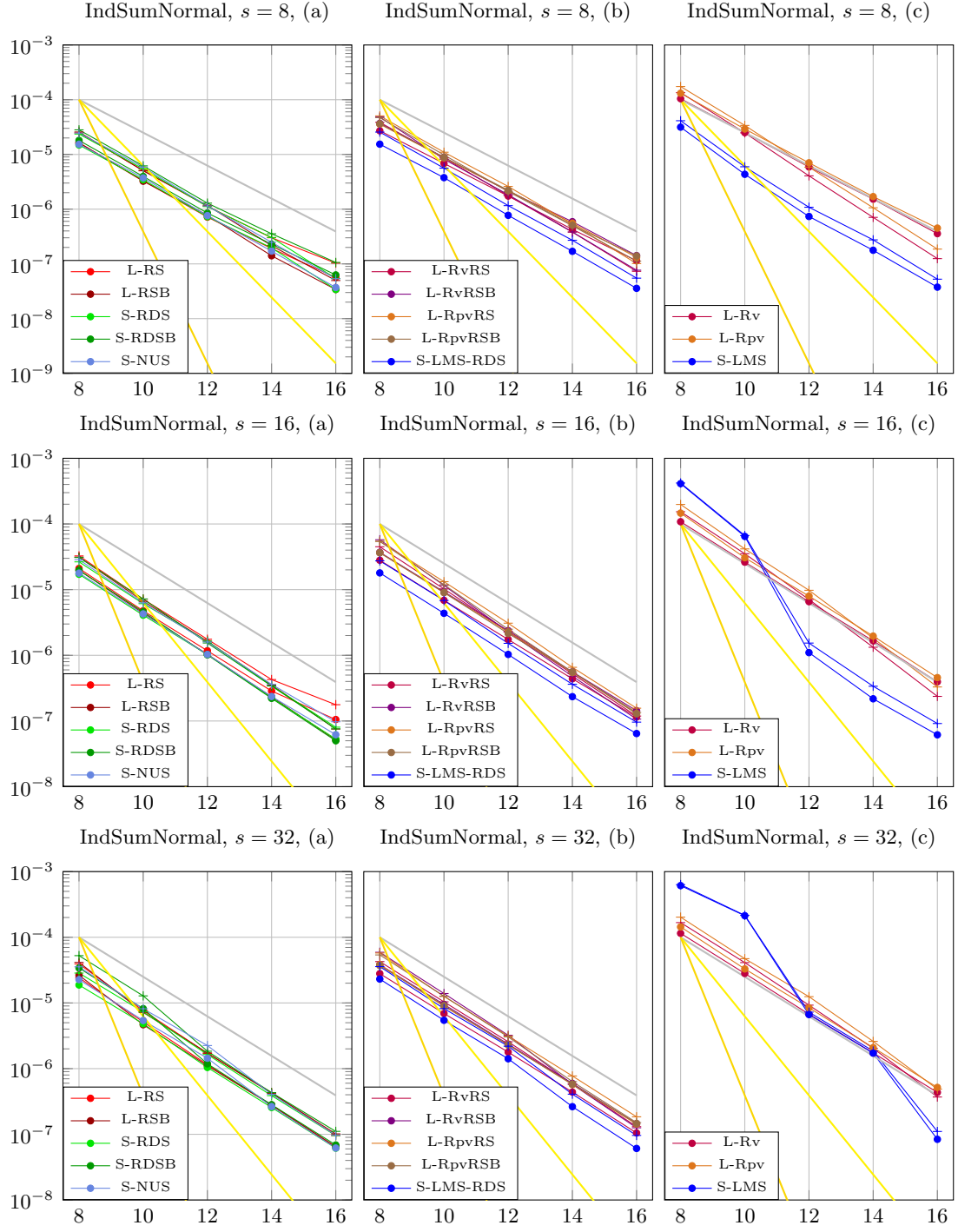




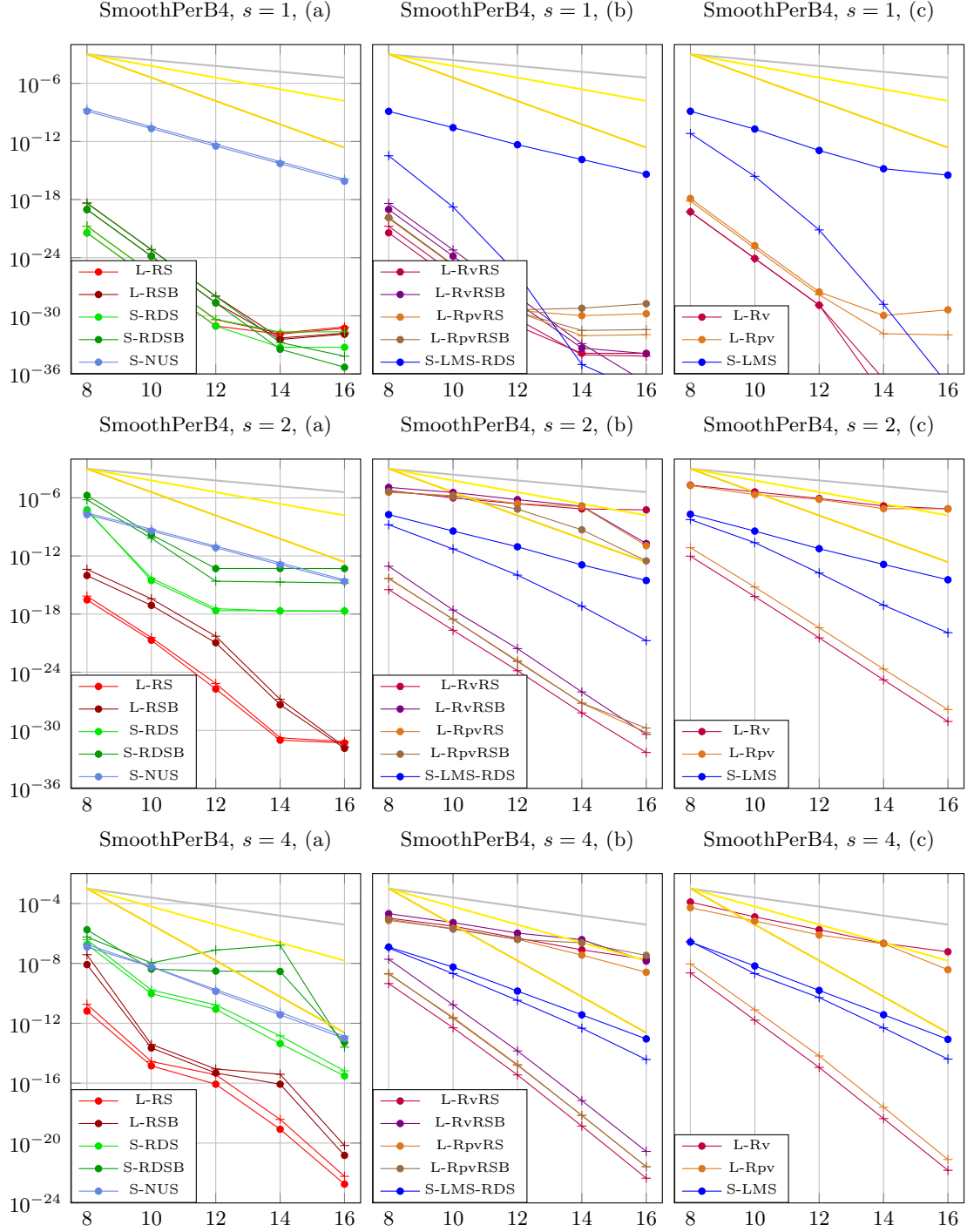


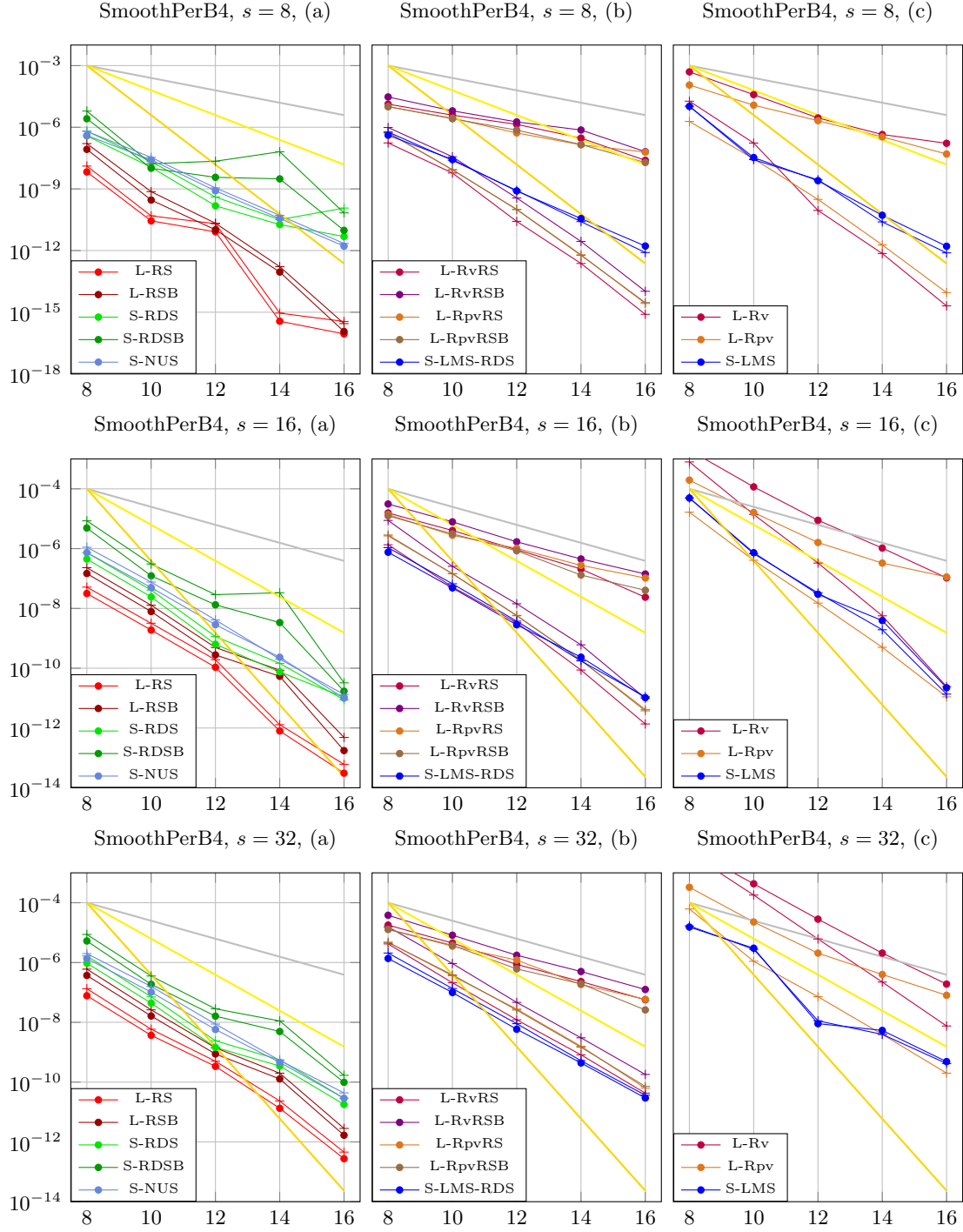


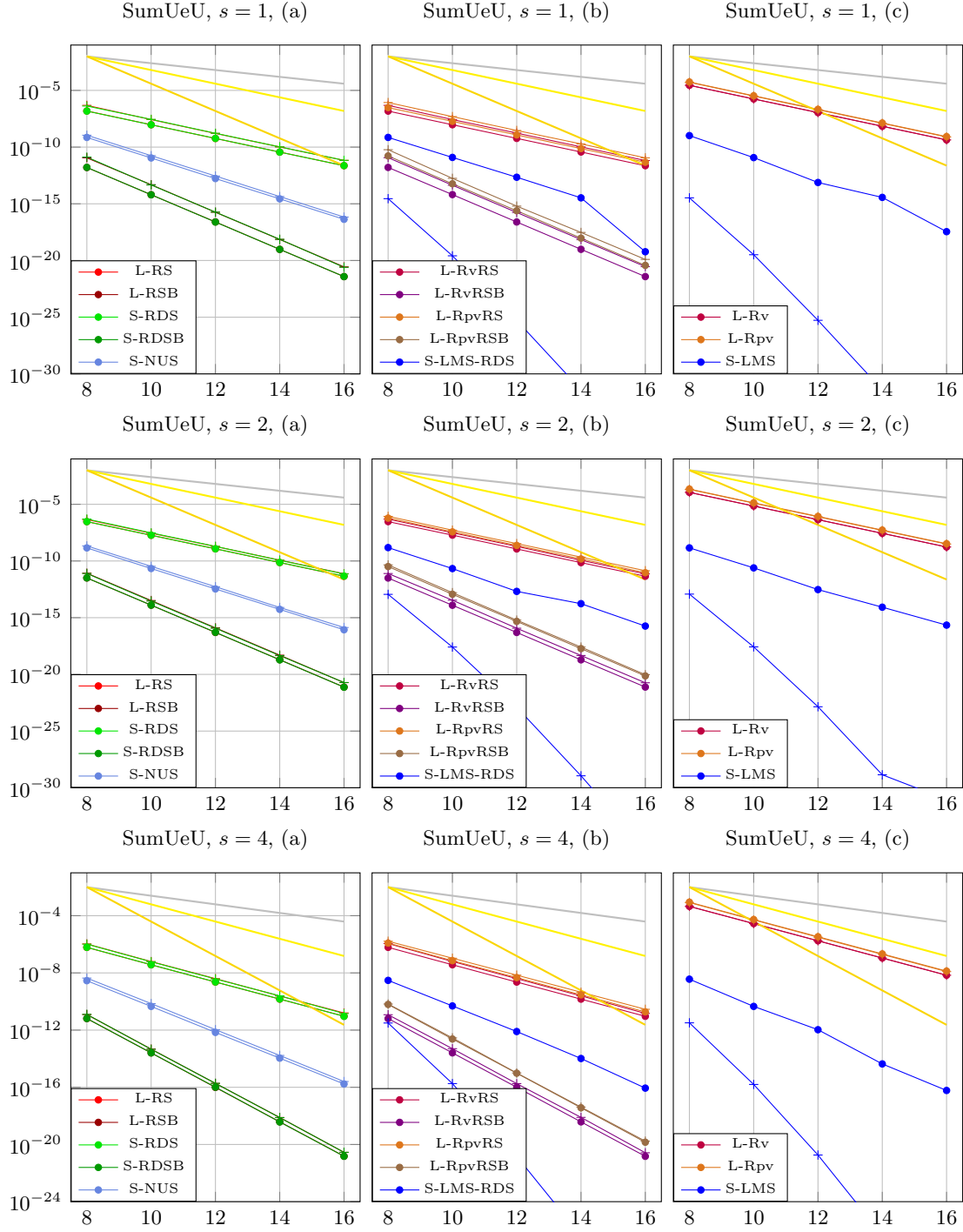


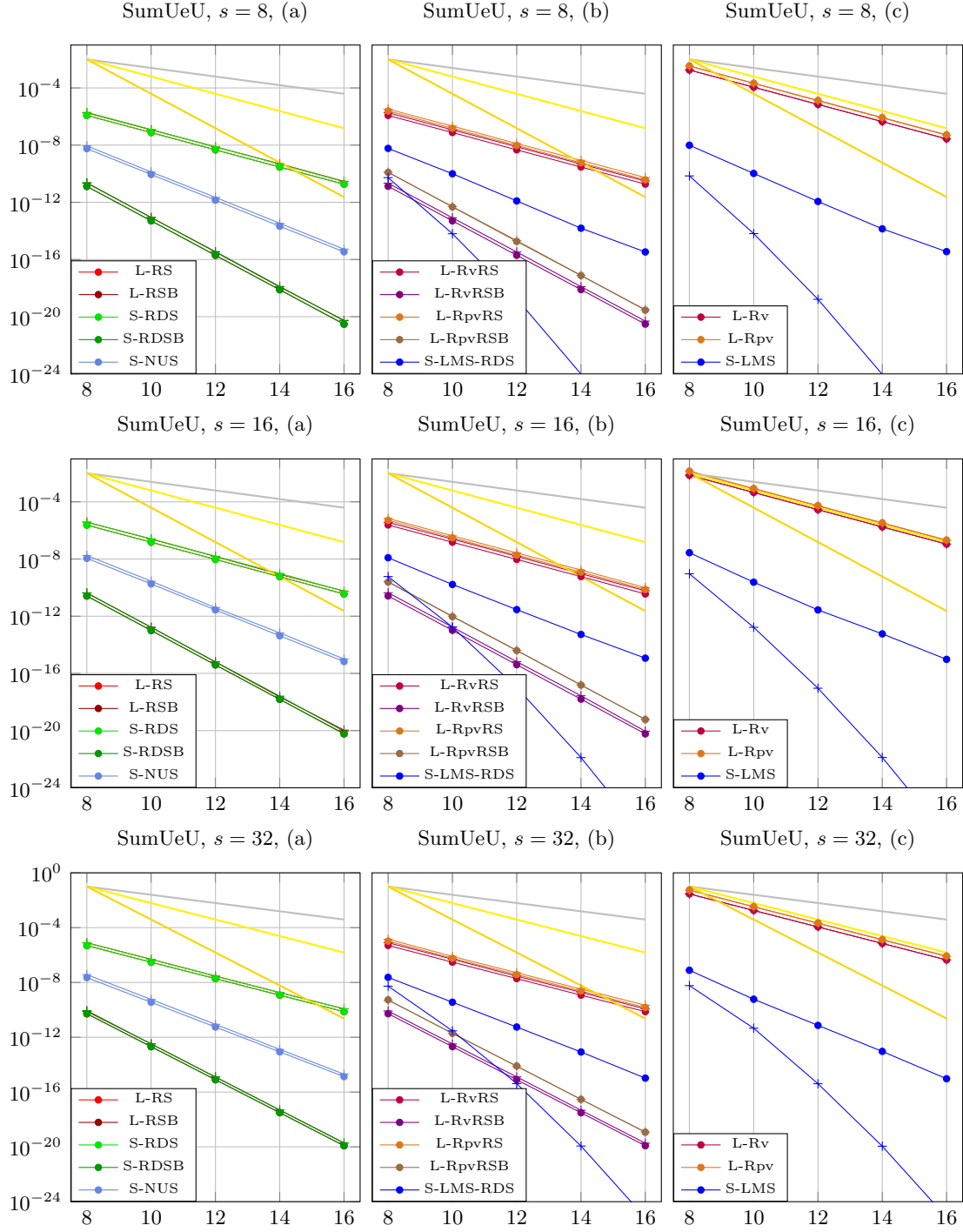


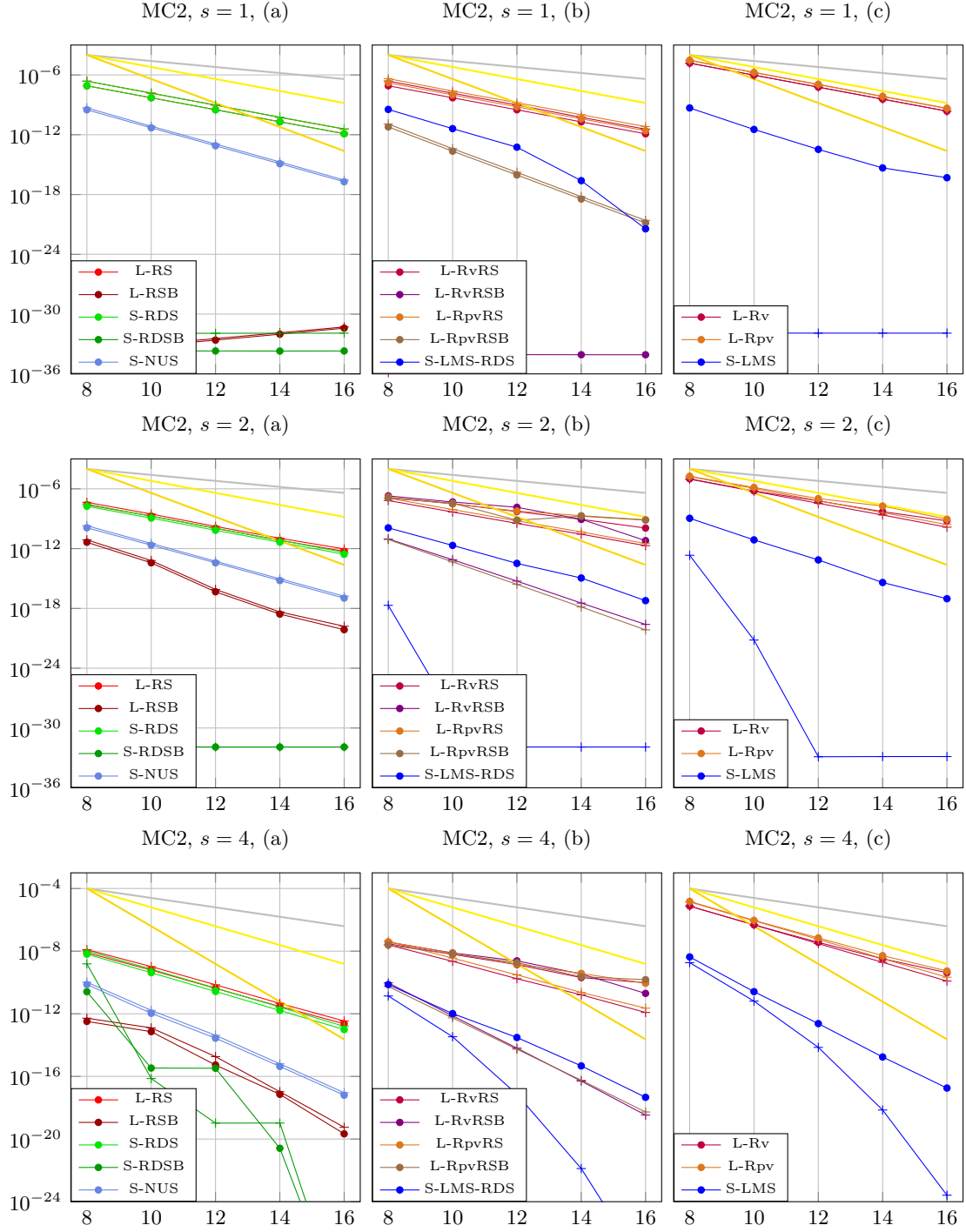
3.3 MSE plots as functions of k for $r = 63$

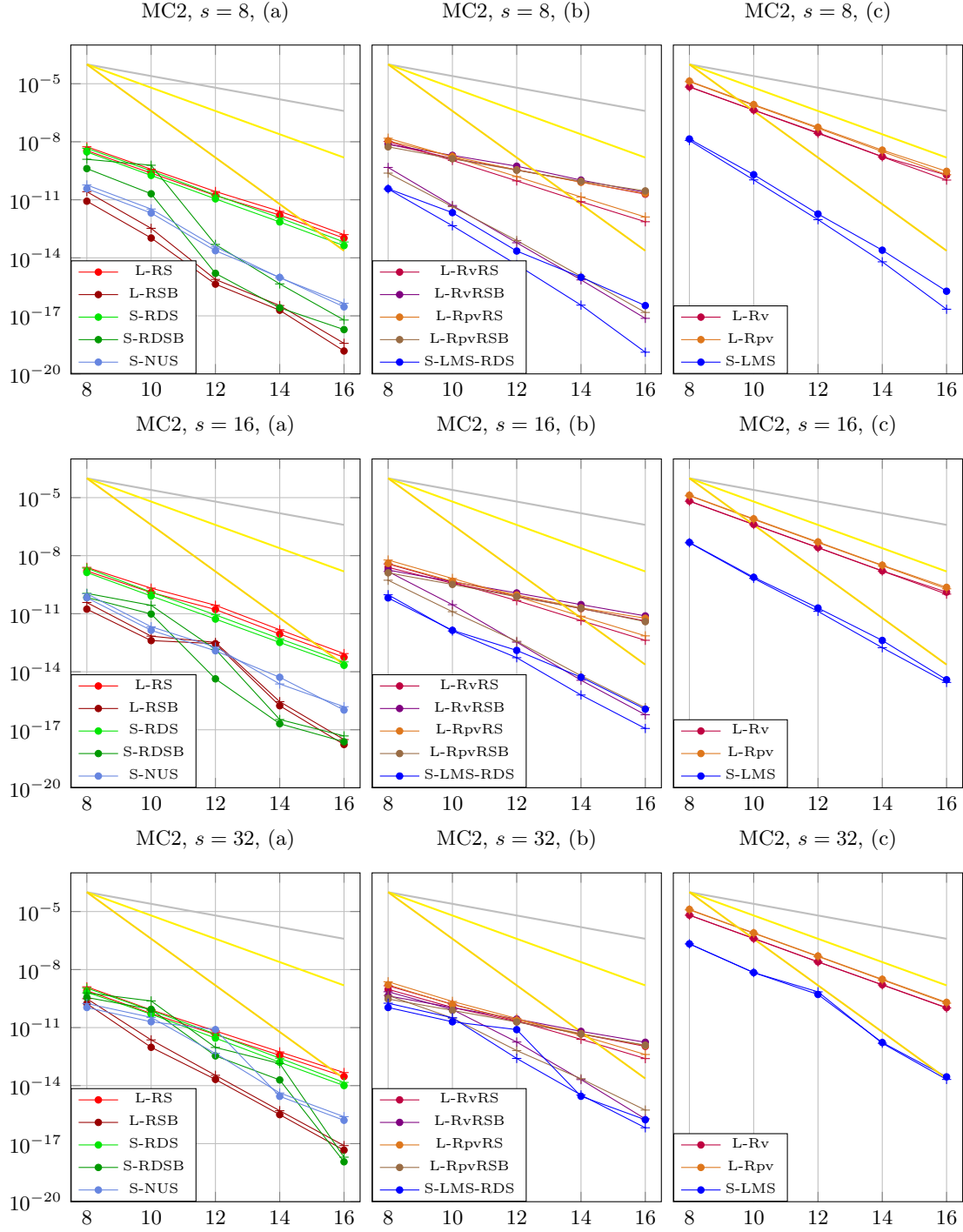


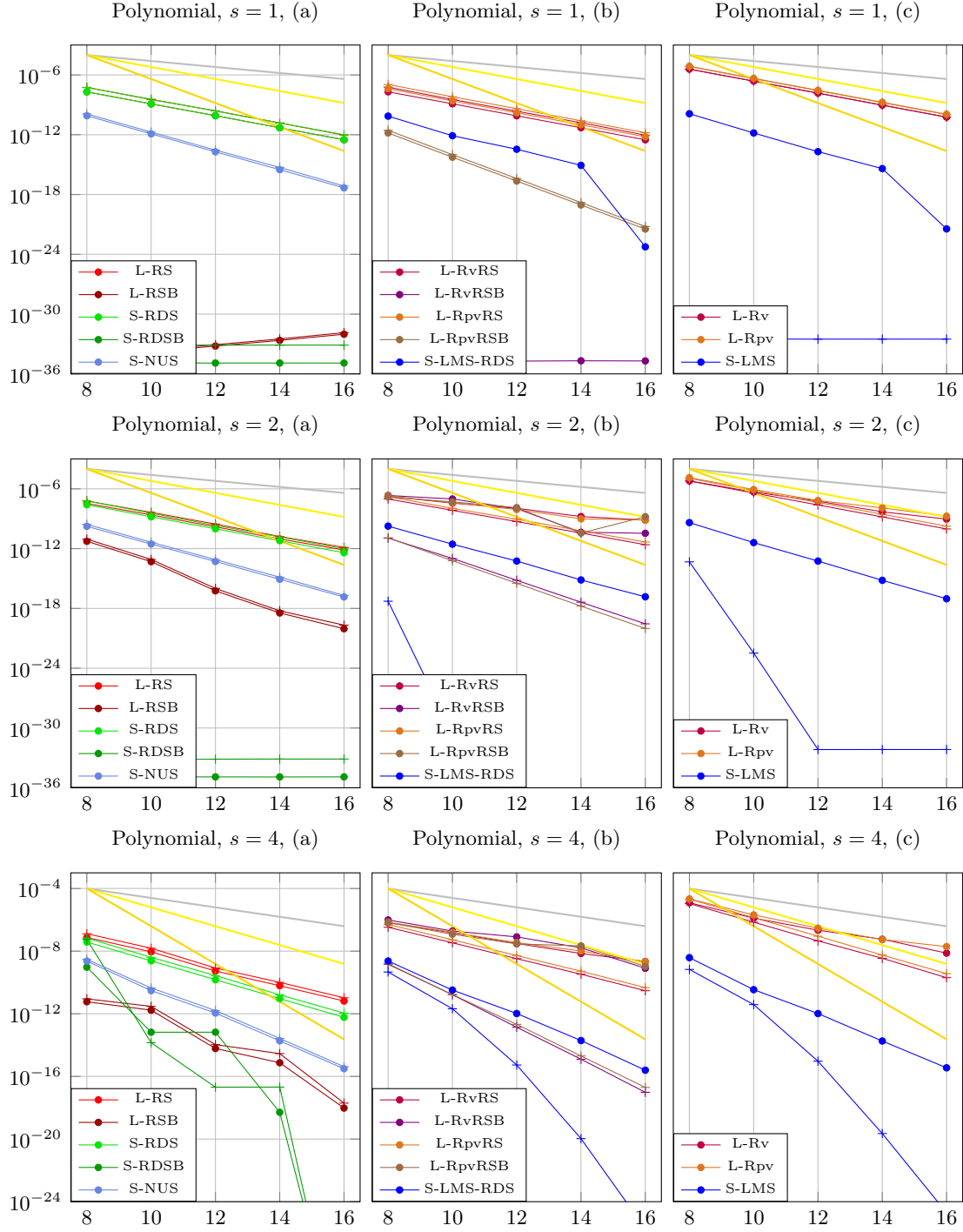


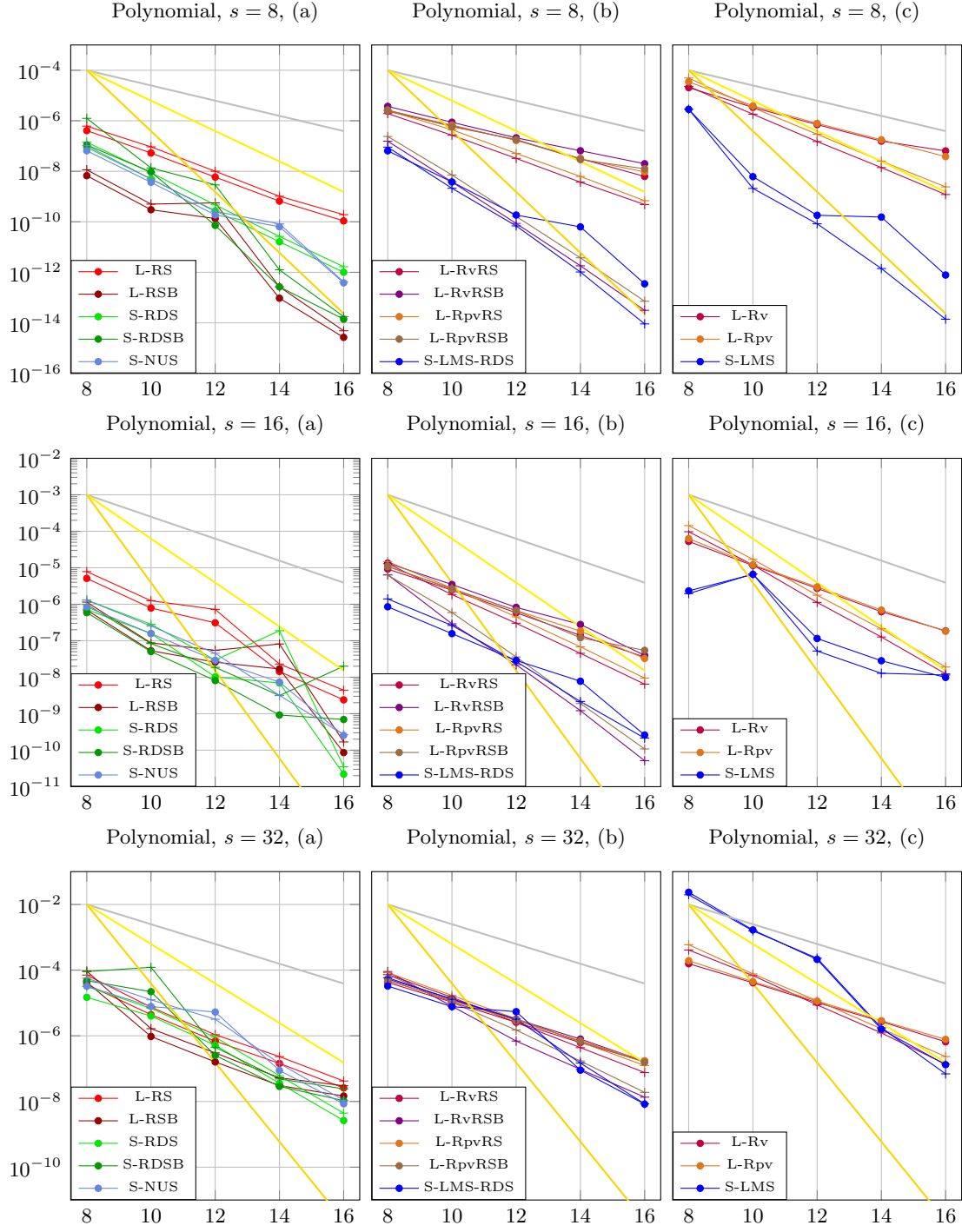


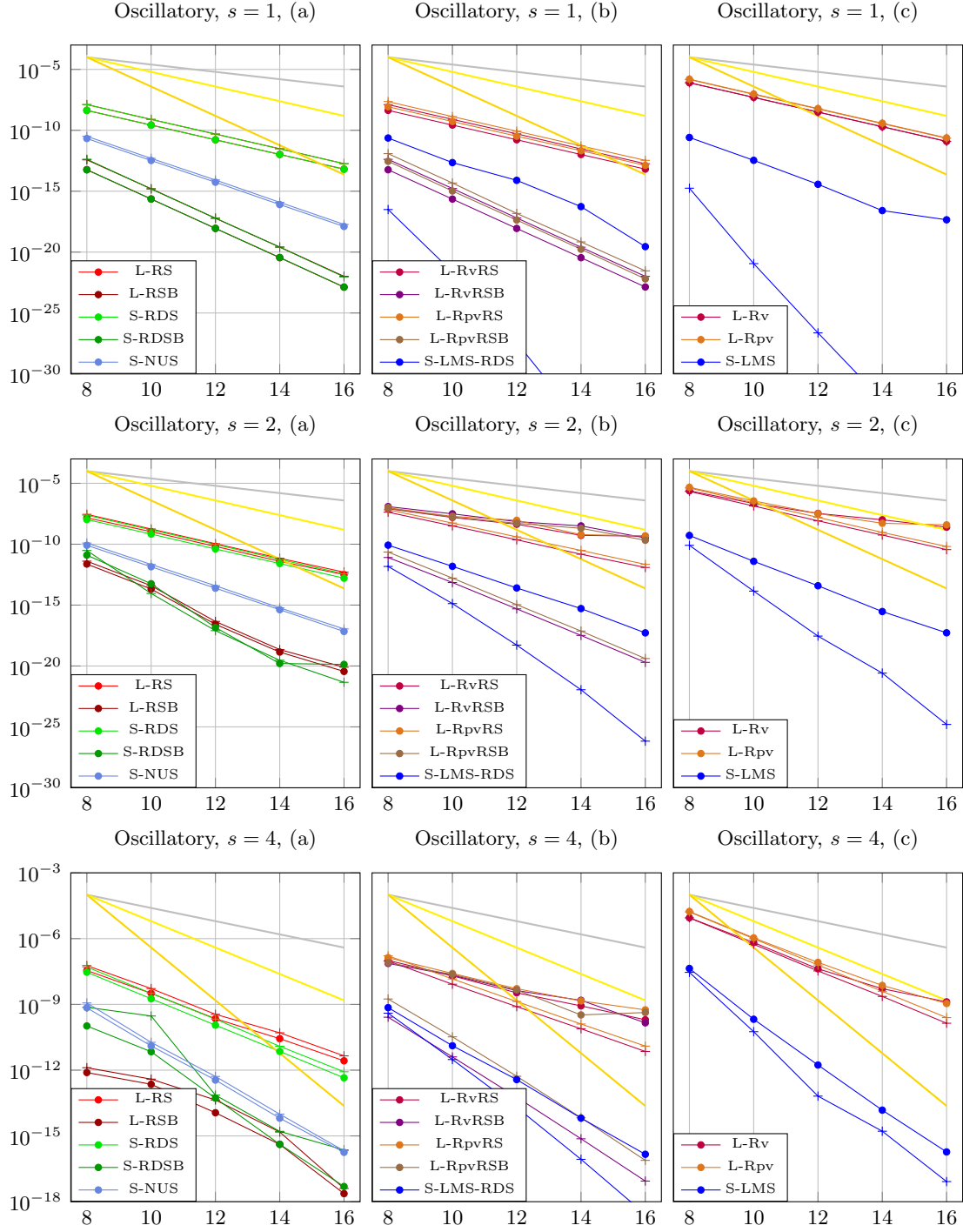


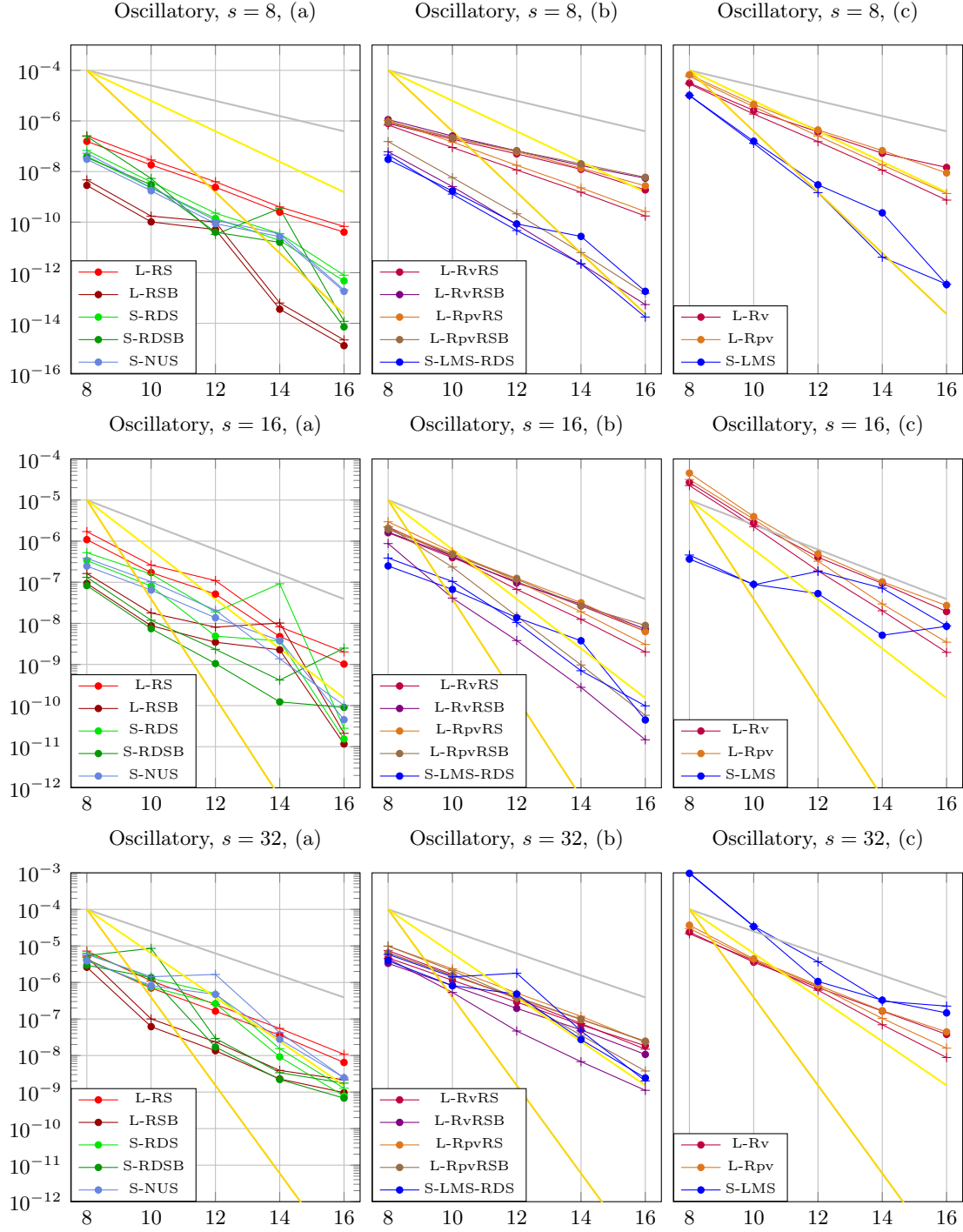


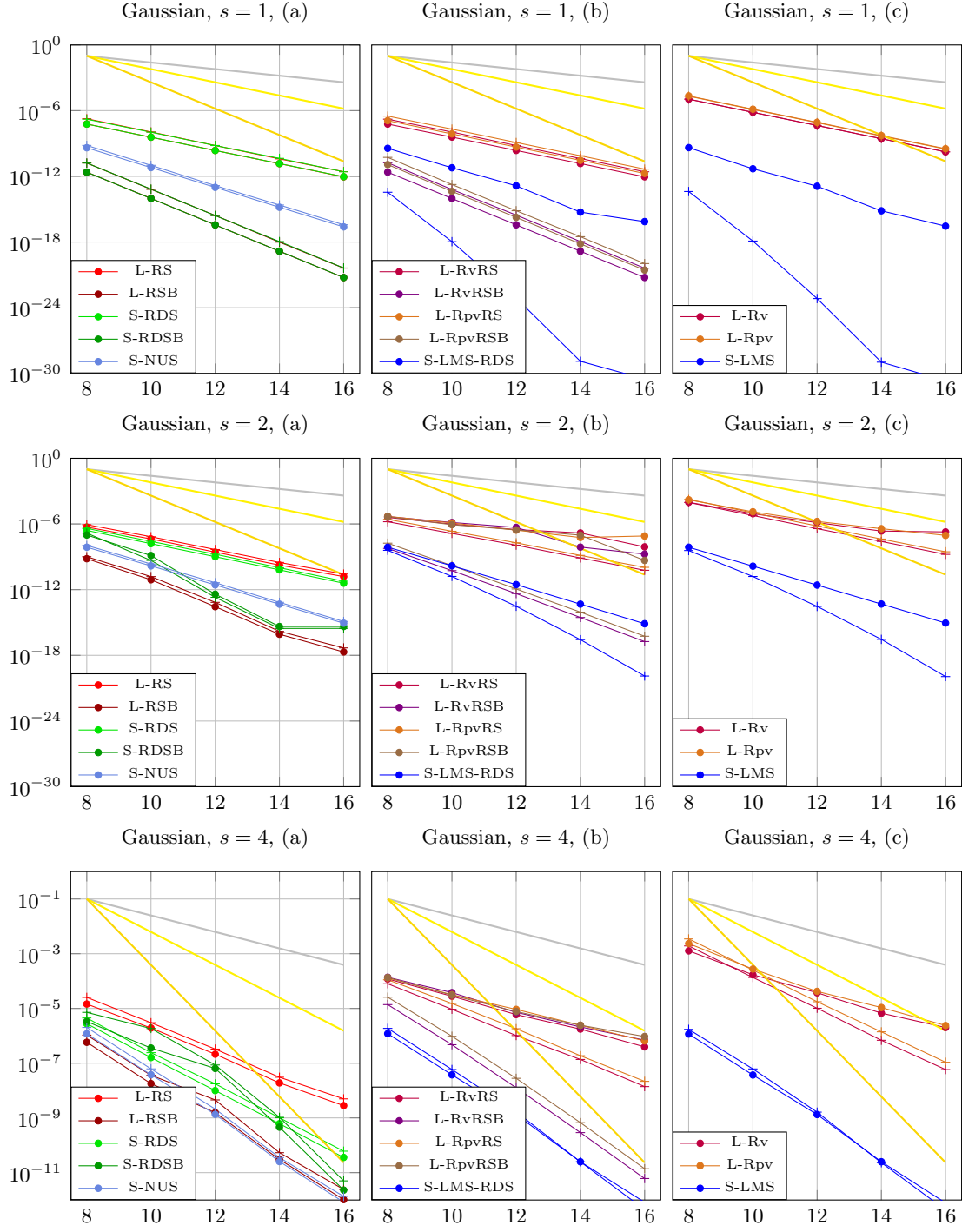


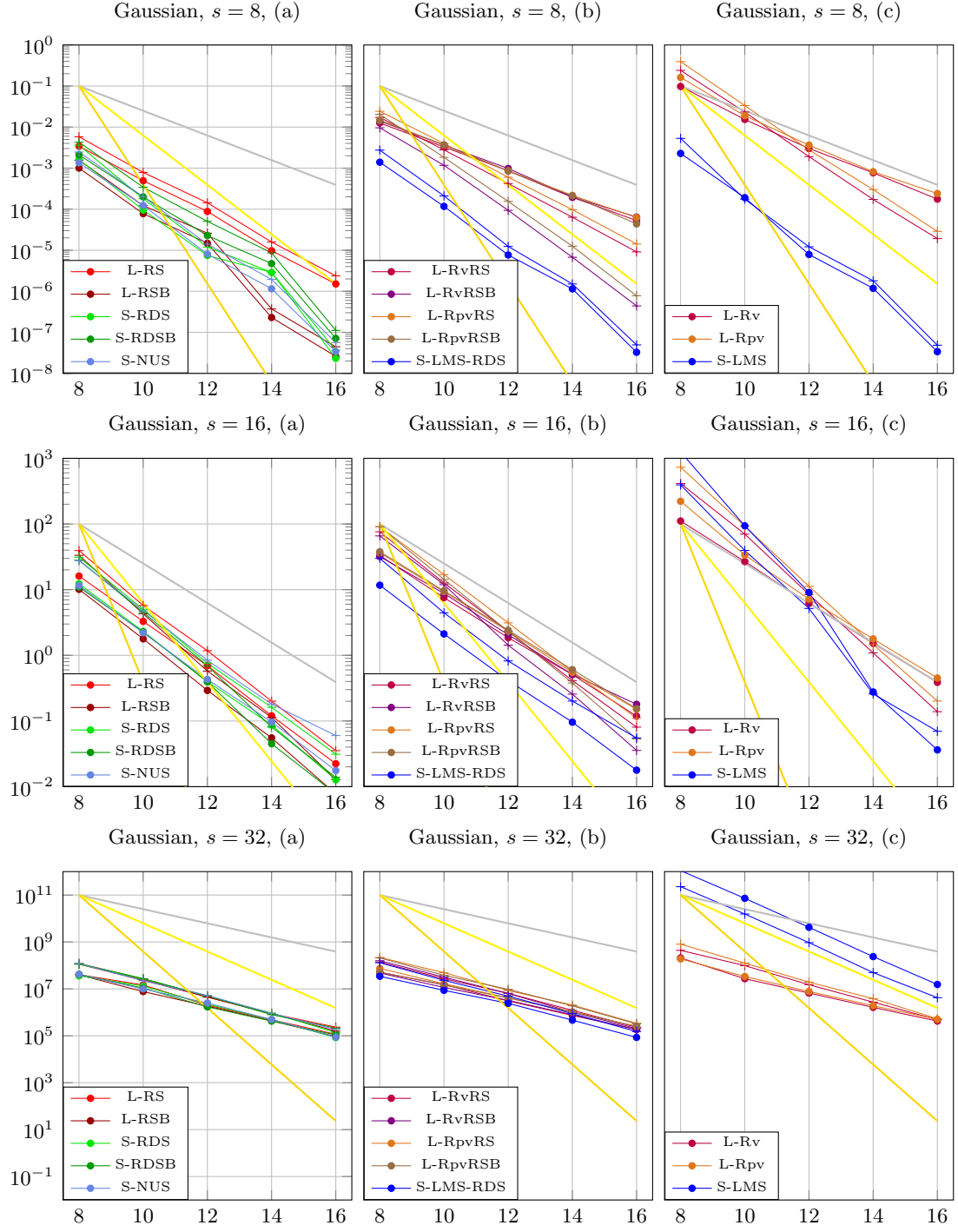


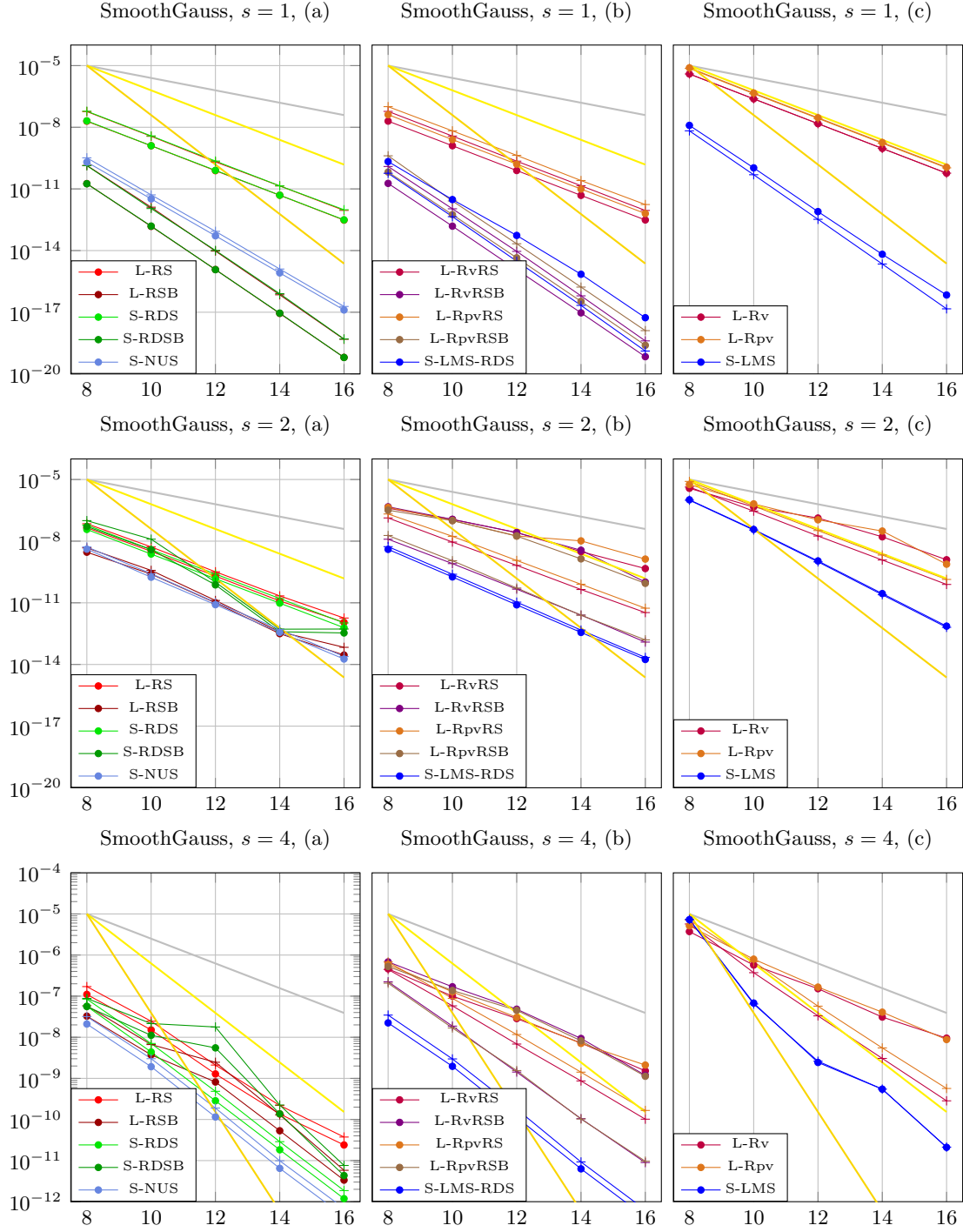


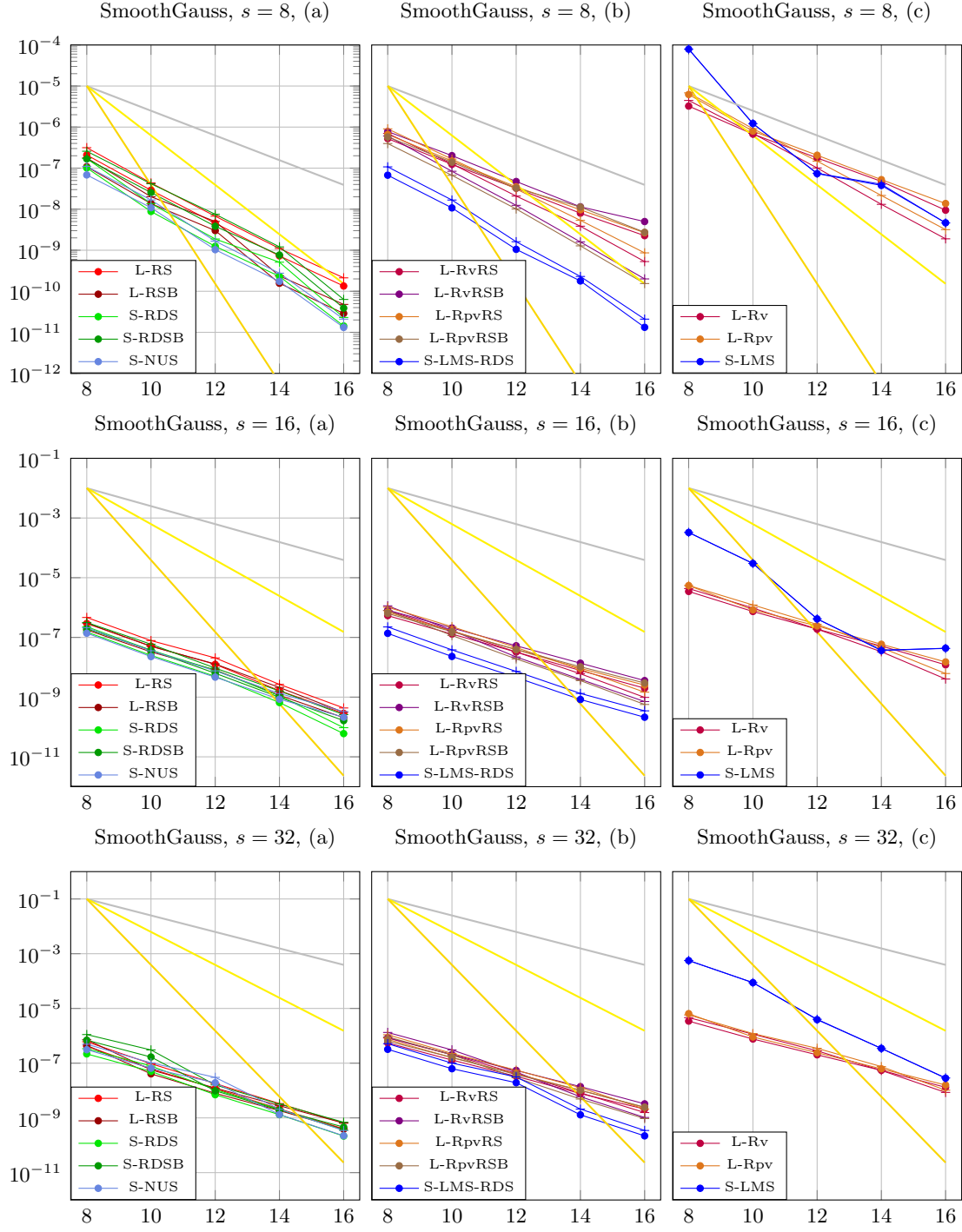


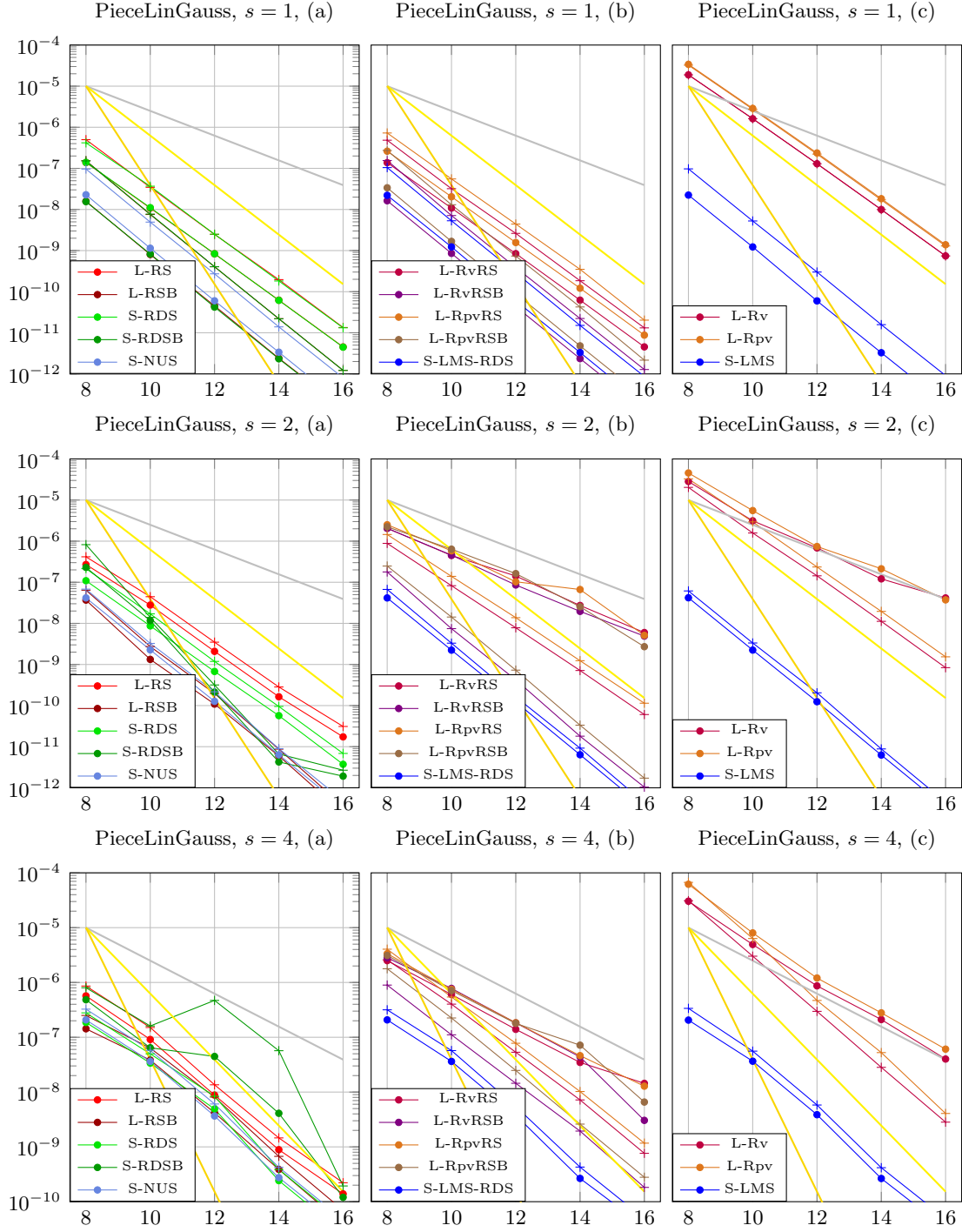


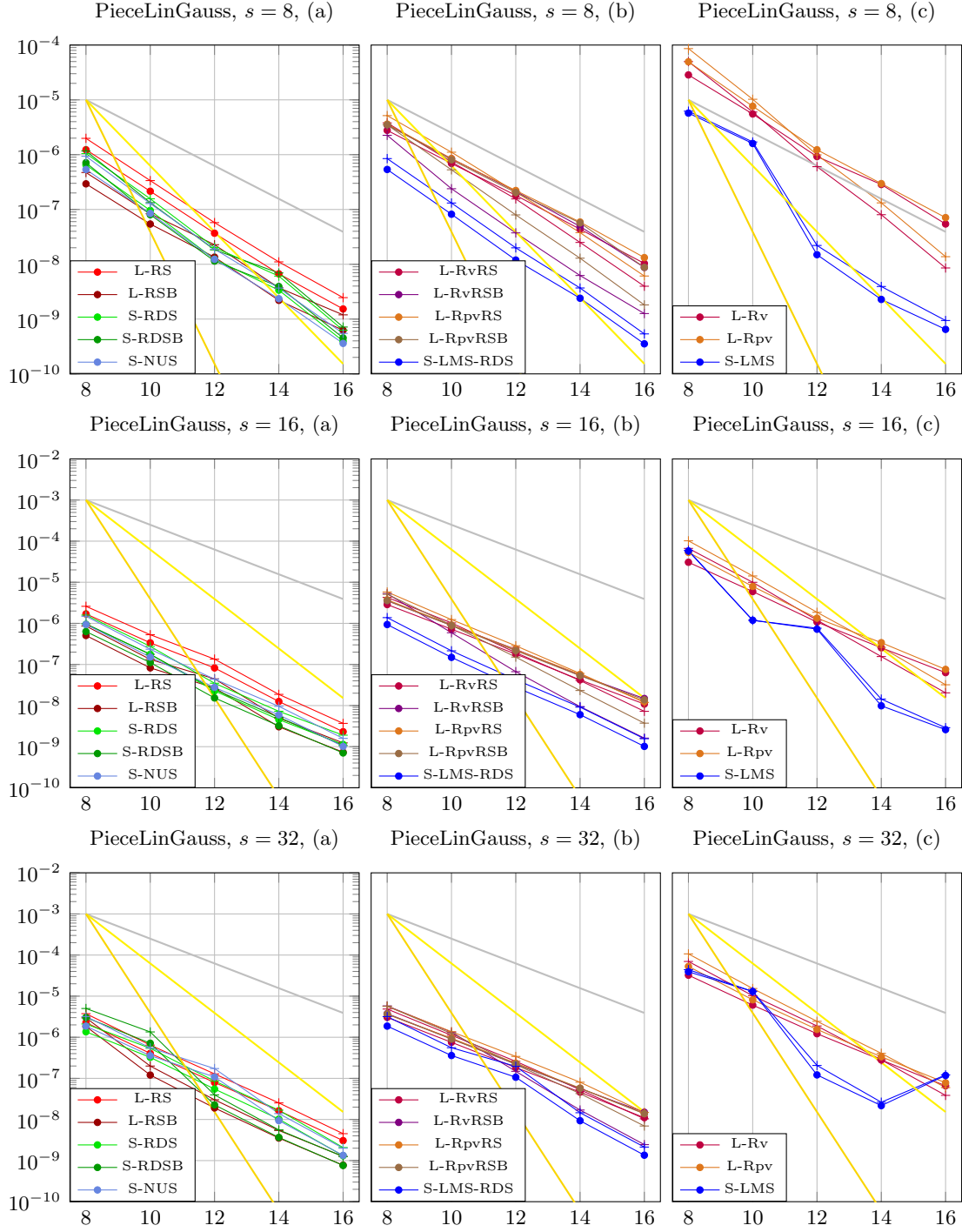


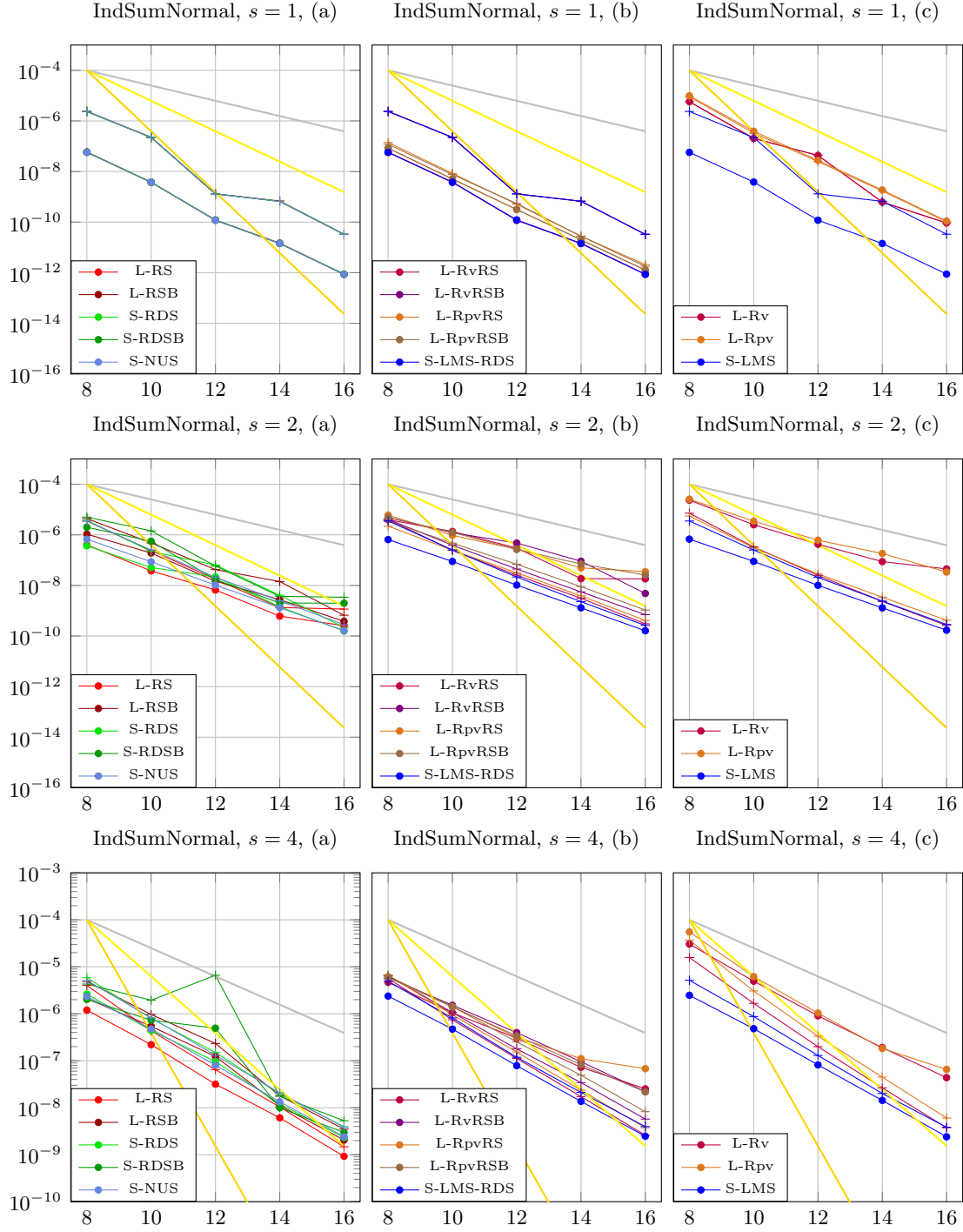


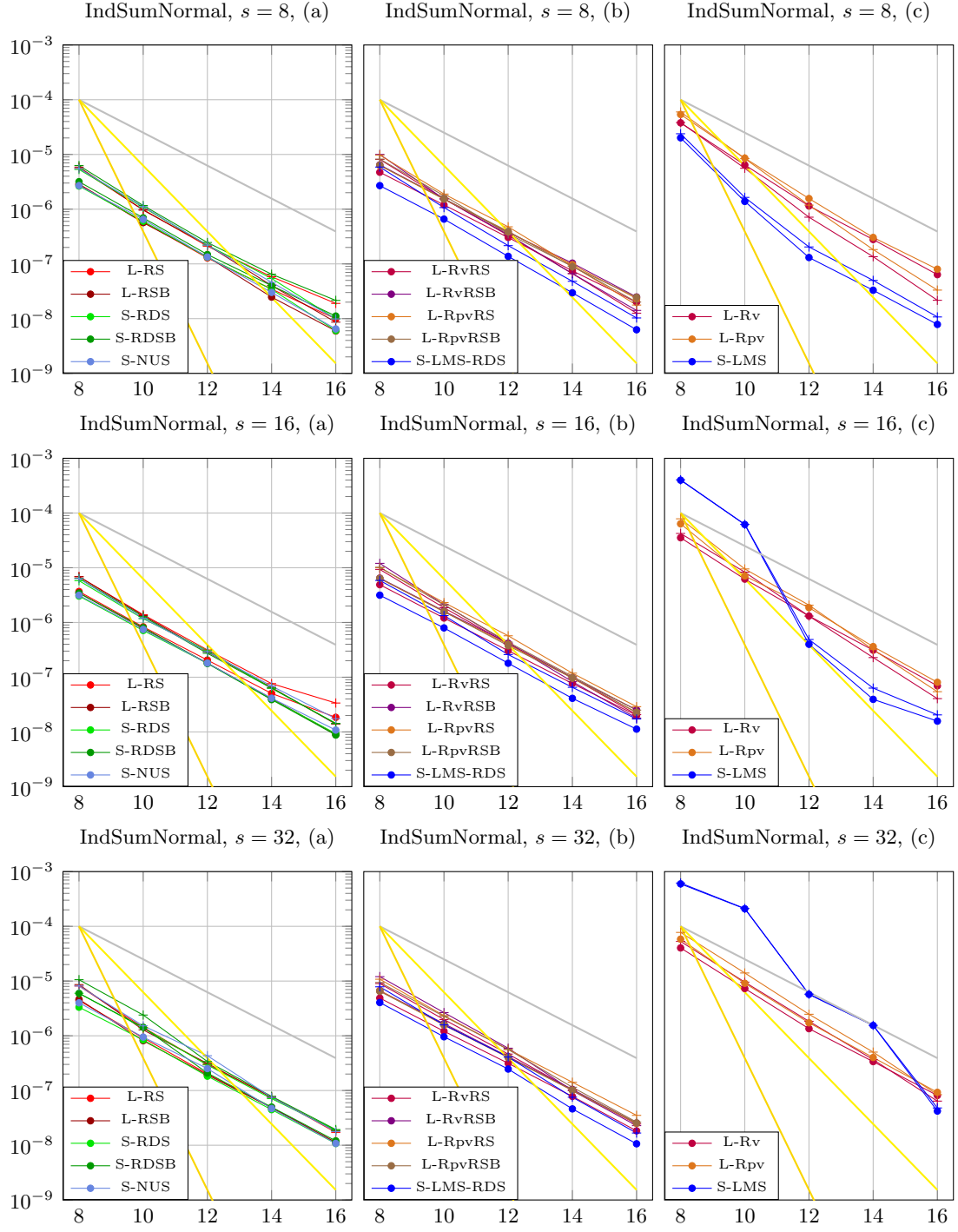




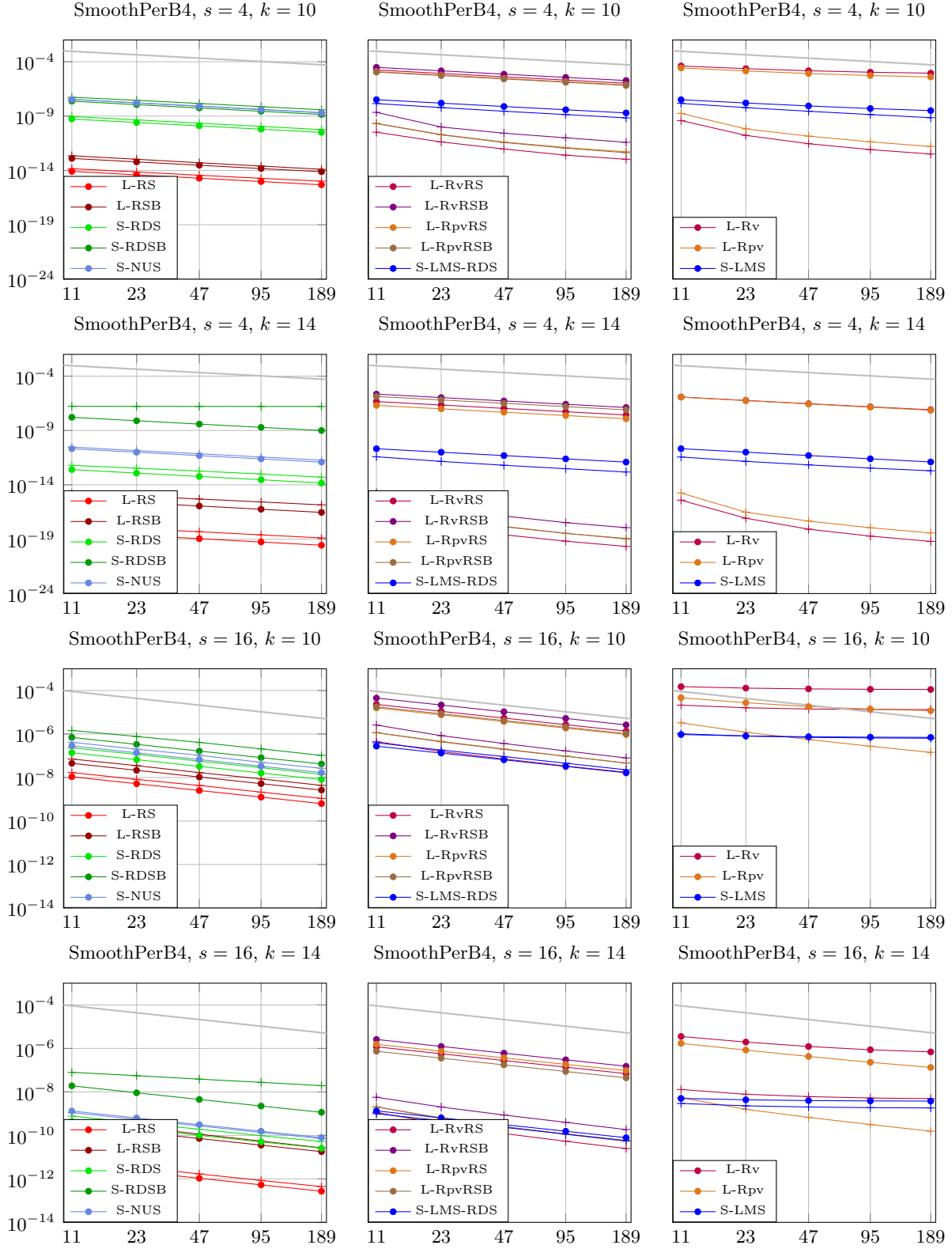


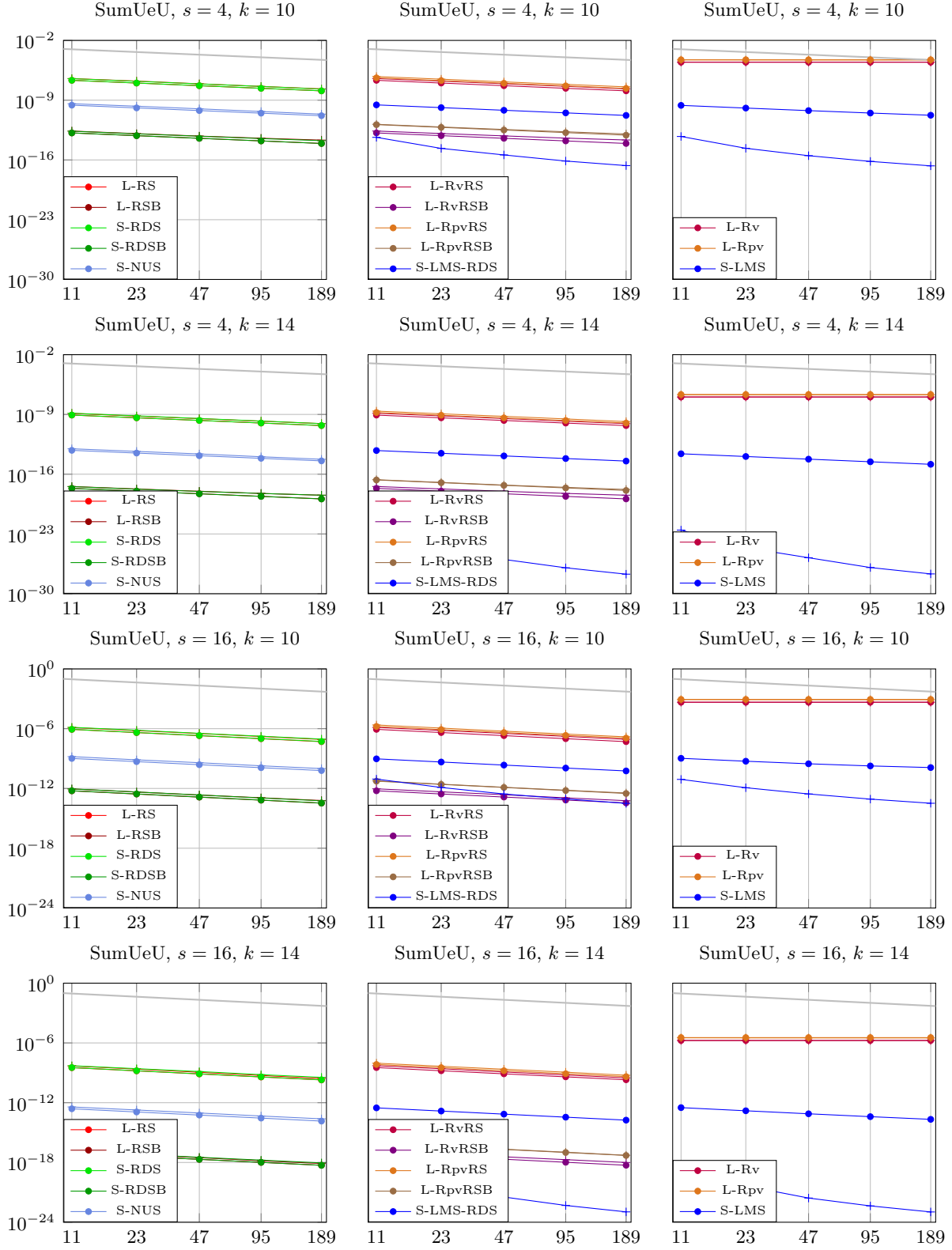


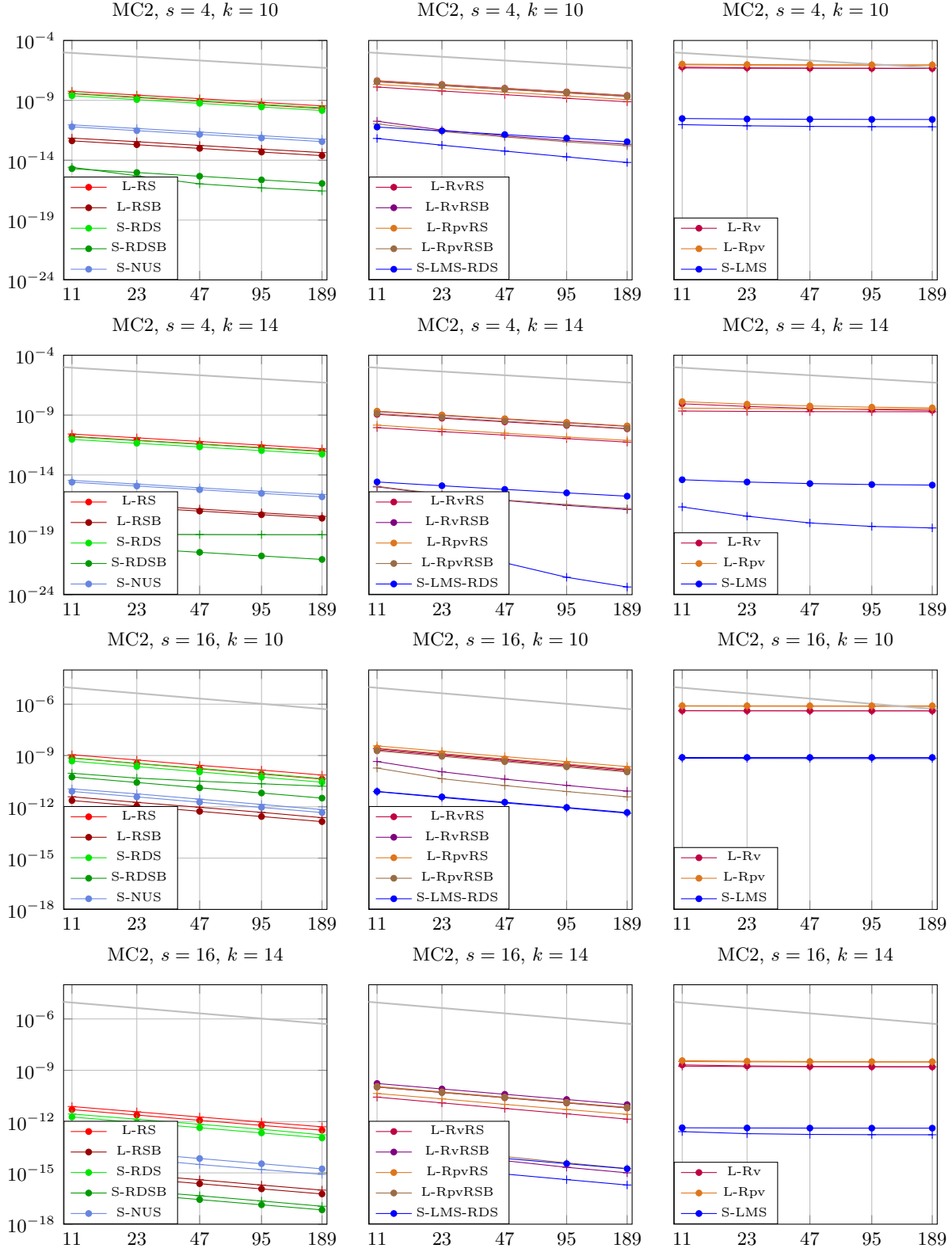


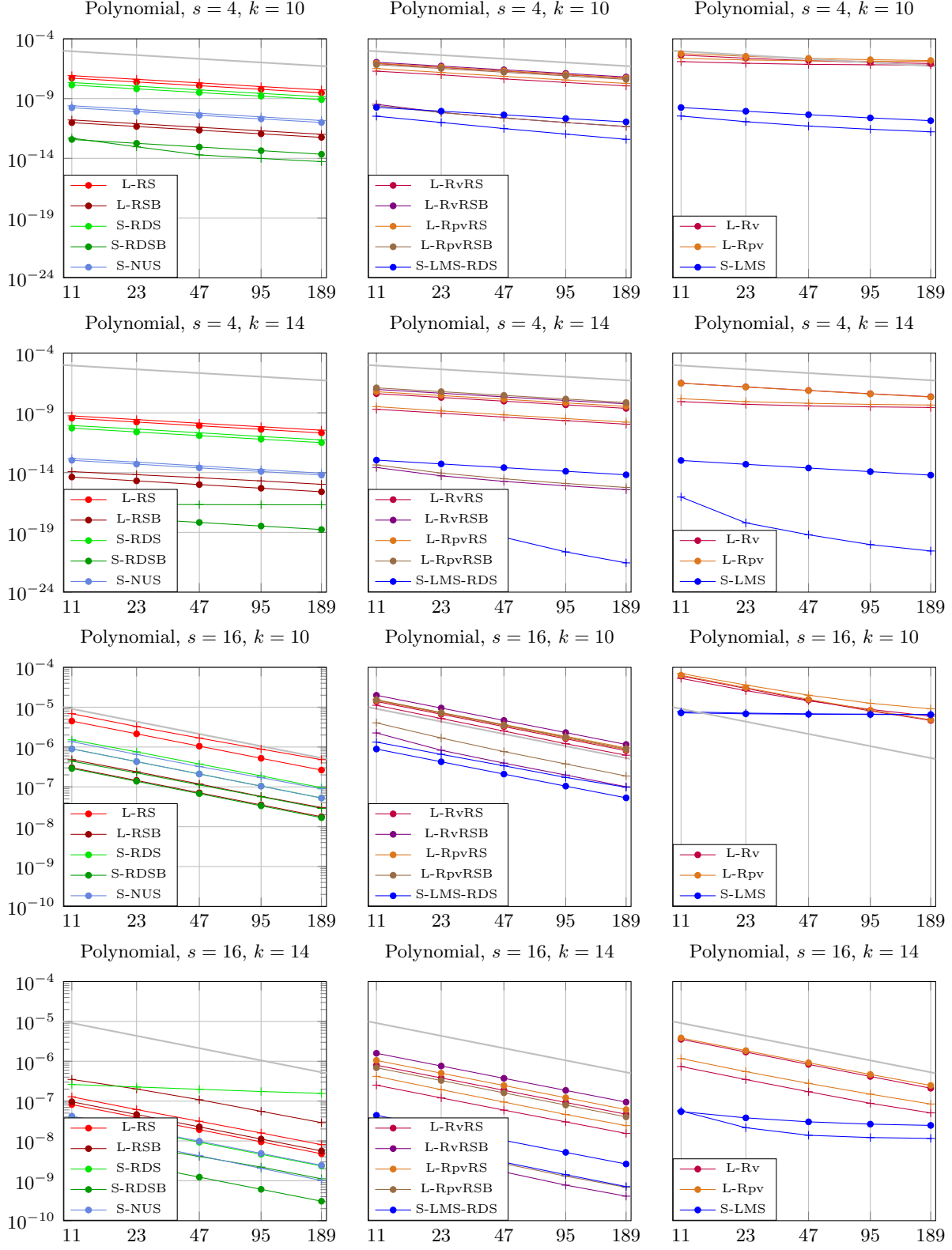


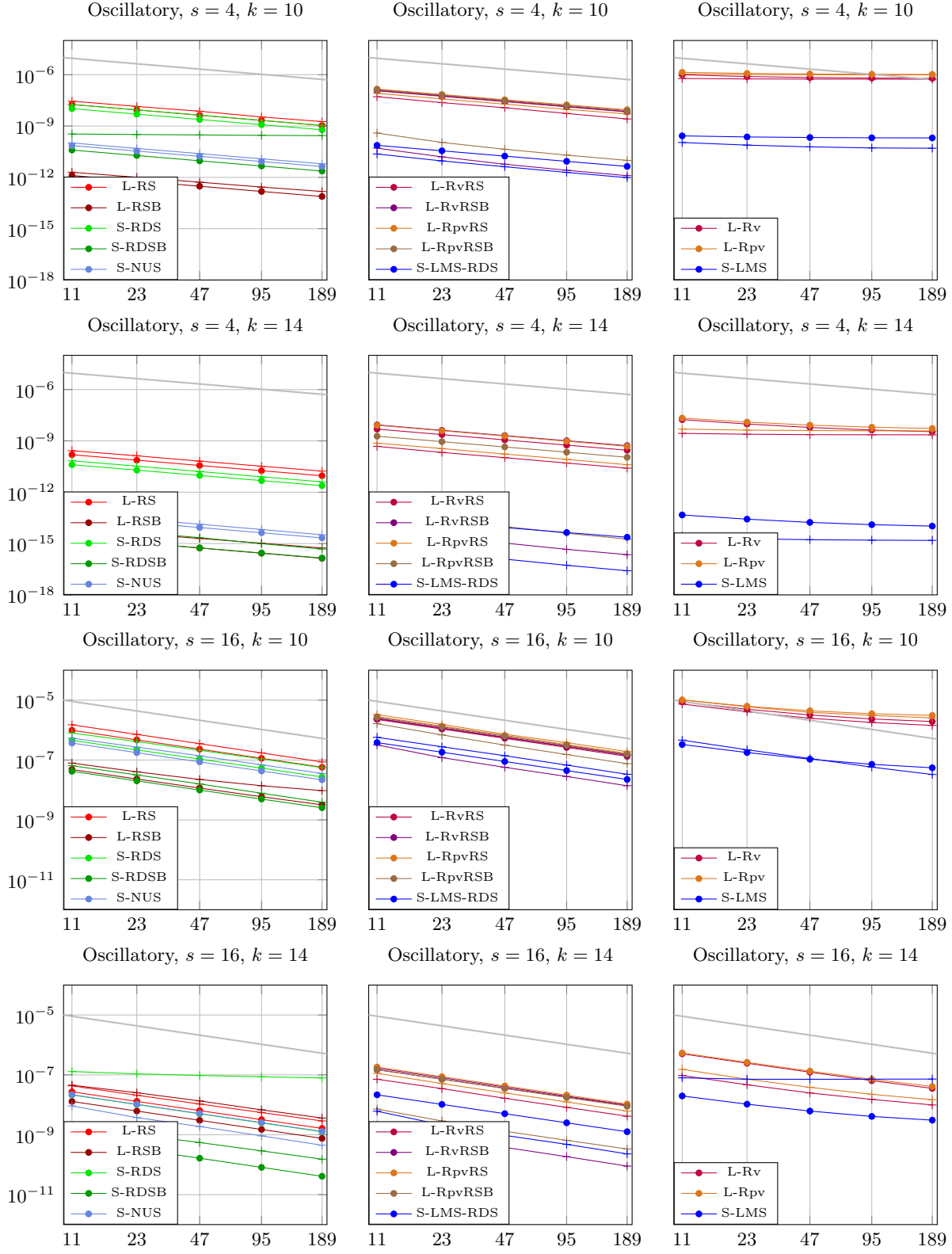
3.4 MSE plots as functions of r

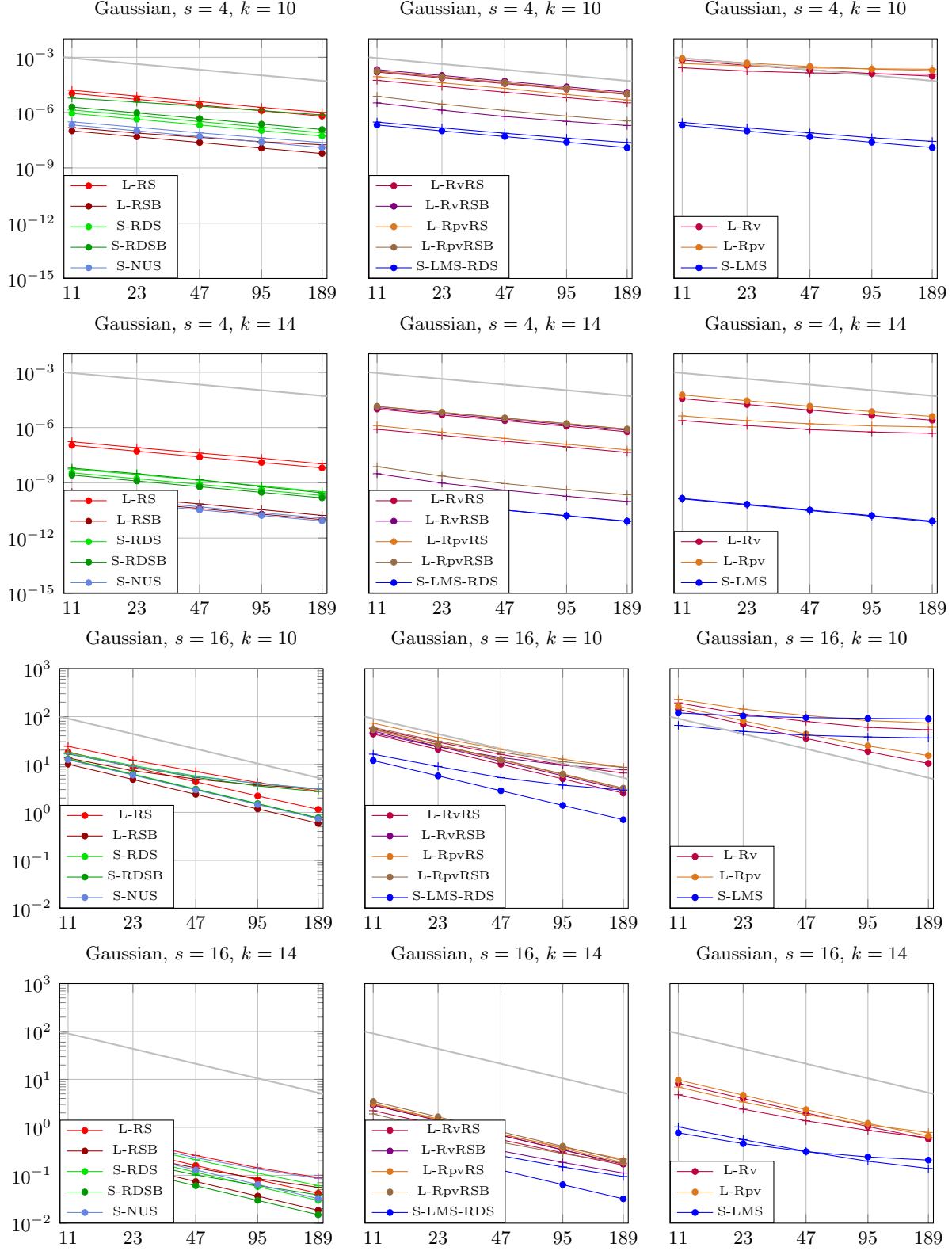


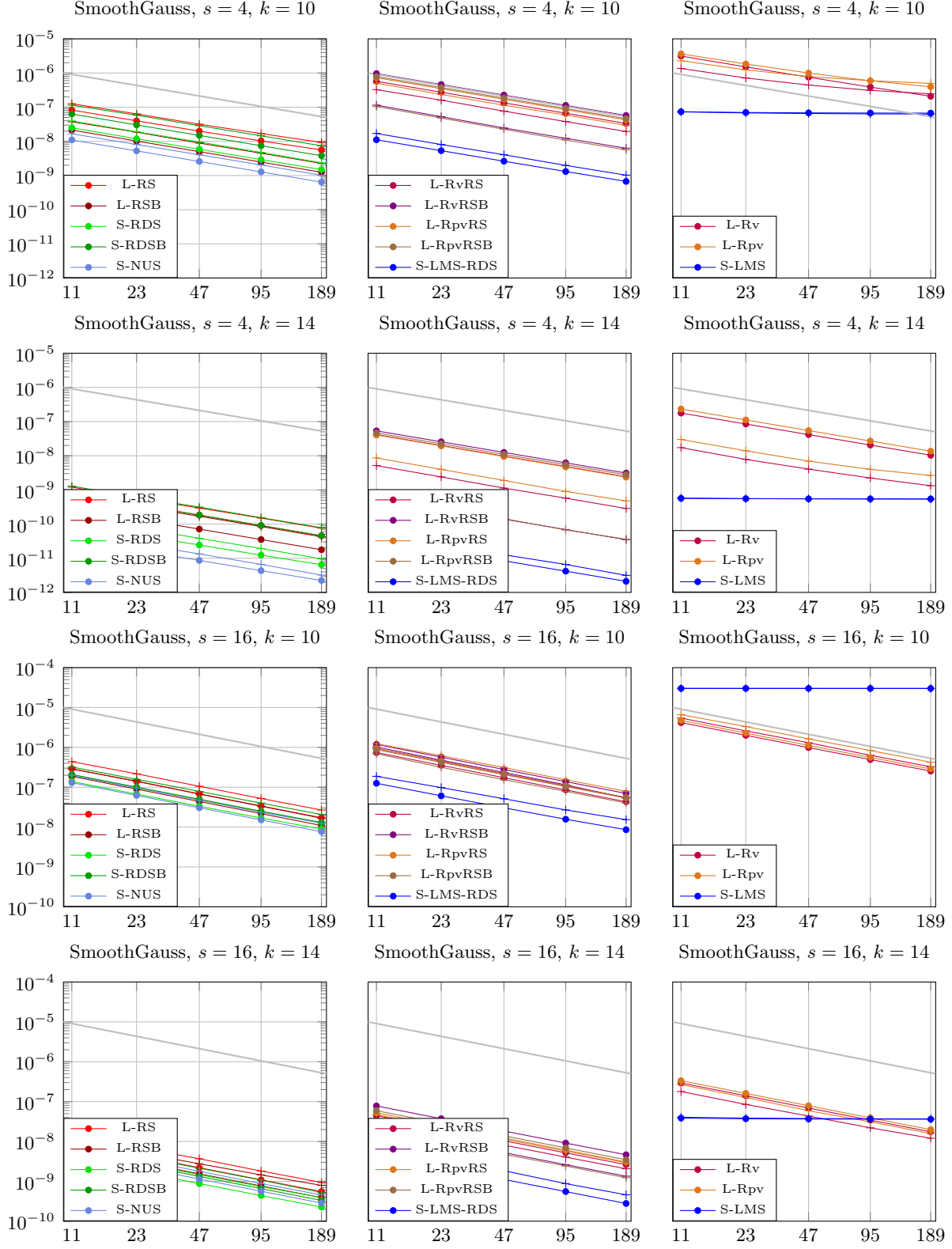


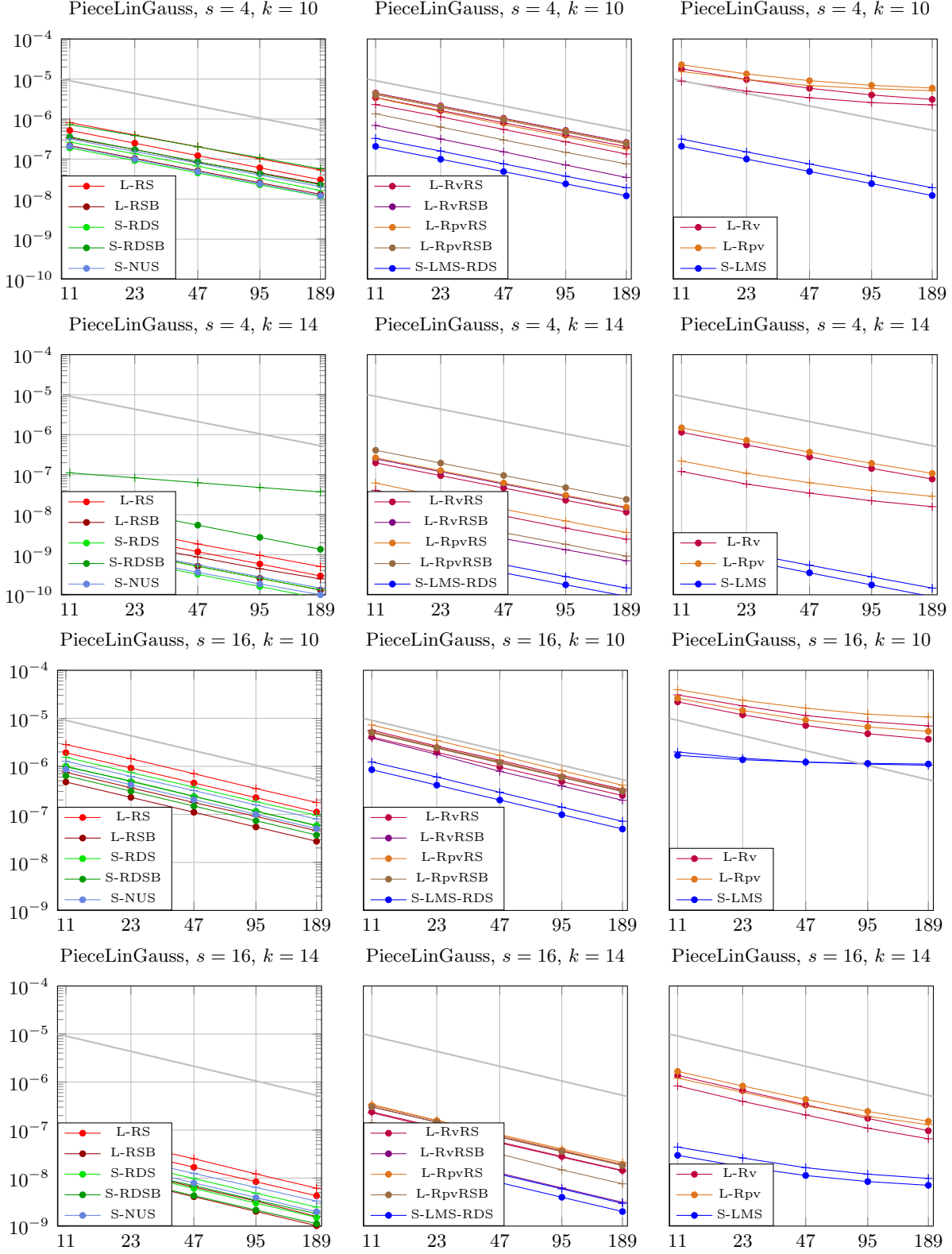


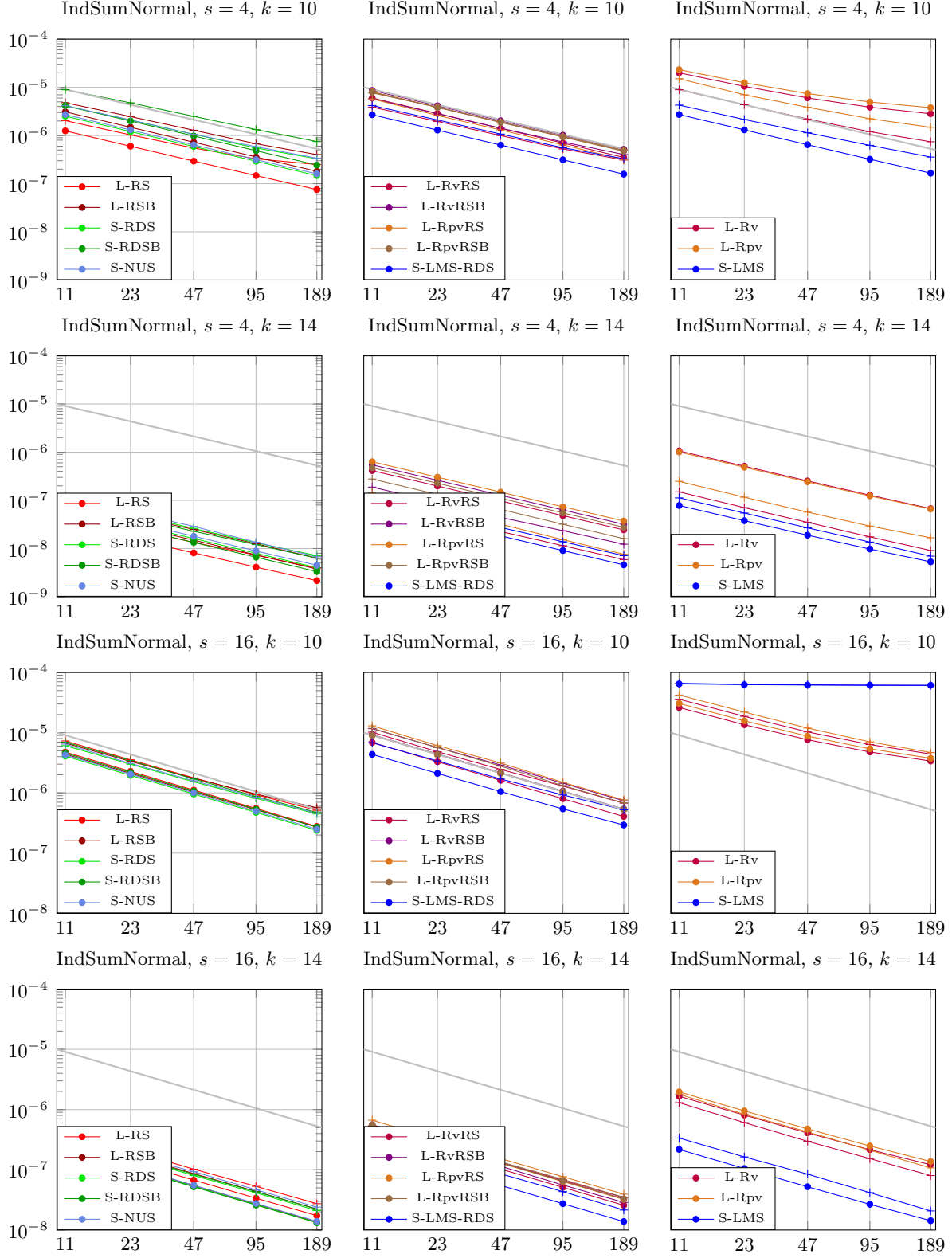












3.5 Convergence of the quantiles of A_r and M_r as functions of k

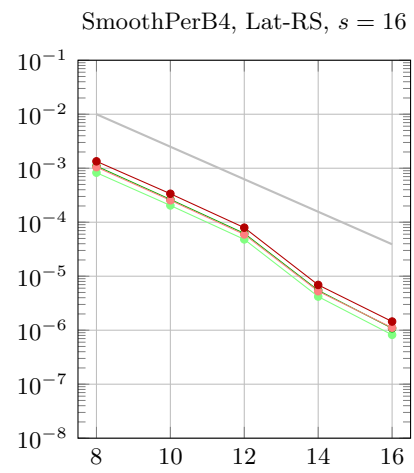
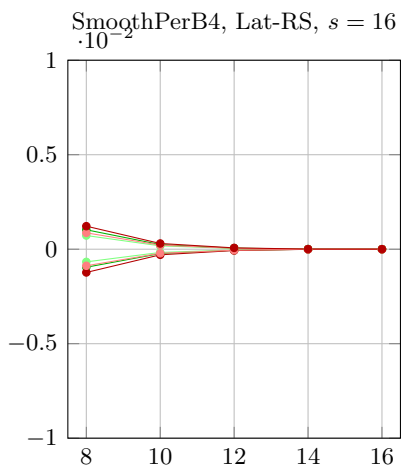
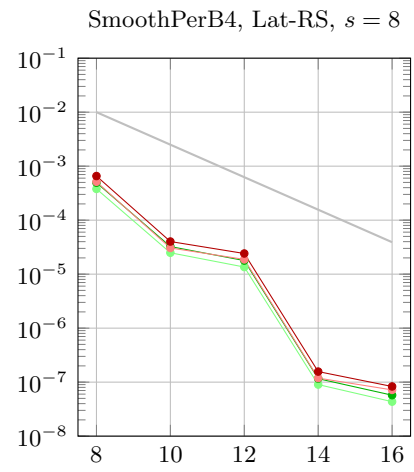
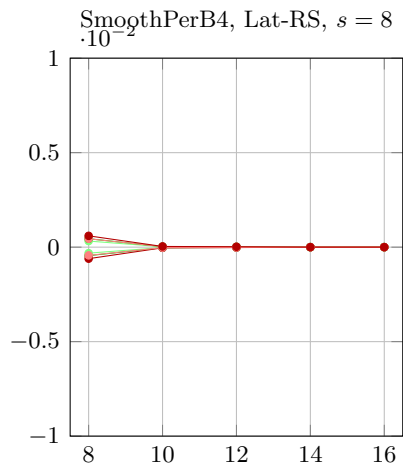
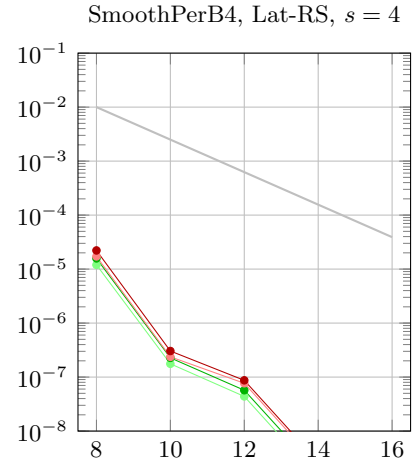
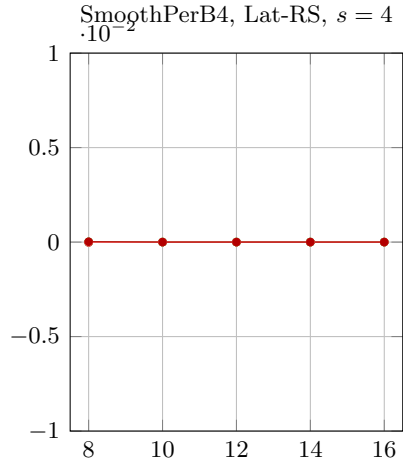
Our next series of plots shows how the quantiles of the empirical distributions of A_r and M_r and of the corresponding absolute errors, converge as functions of k . We show this for $r = 11$, in $s = 4, 8, 16$ dimensions, for the QMC/RQMC methods Lat-RS, Lat-RvRS, Lat-Rv, Lat-Rpv, Sob-LMS, and Sob-LMS-RDS. These six methods give a representative illustration of the different types of behavior that can be observed for the error quantiles. Each page shows the plots for one model and one method, with one row per dimension s .

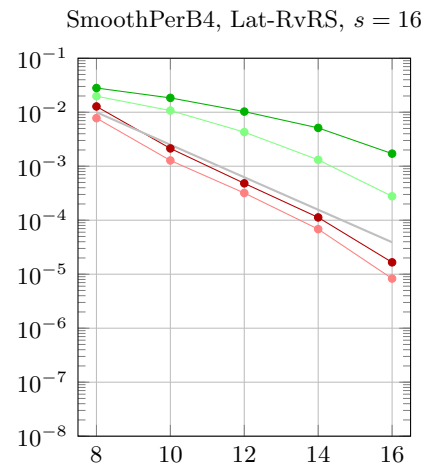
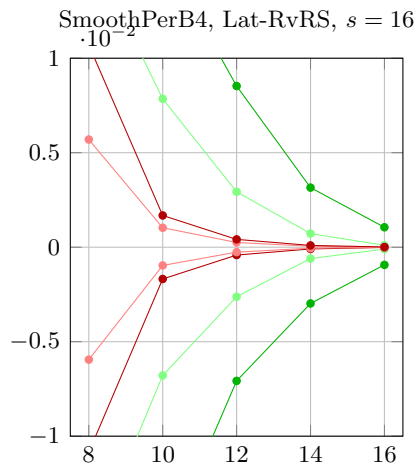
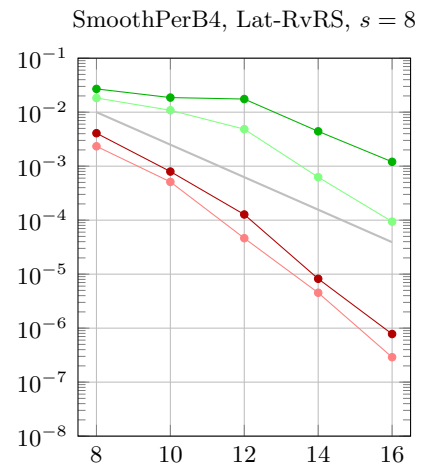
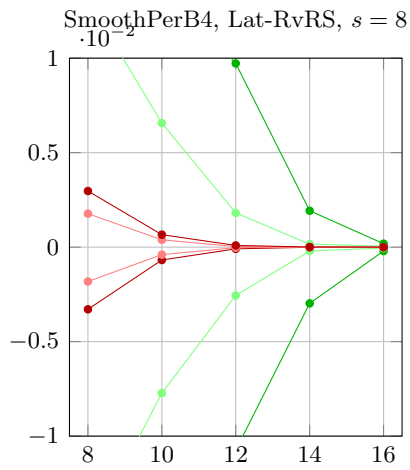
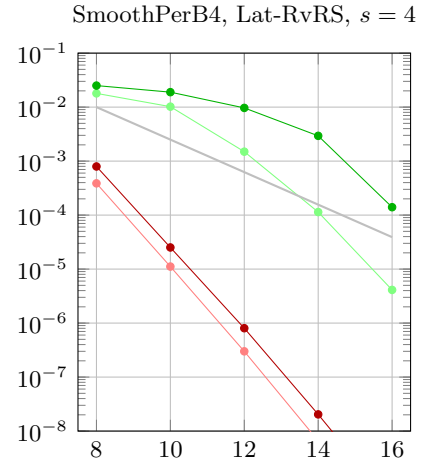
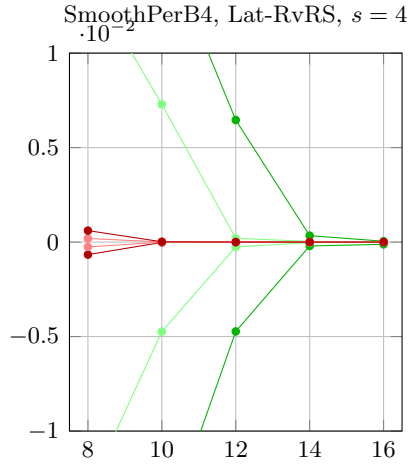
In each row, on the left picture, we see the 1%, 5%, 95%, 99% quantiles of the empirical distributions of A_r in green and M_r in red, as functions of k . The 1% and 99% quantiles are in dark while the 5% and 95% ones are in lighter color. Of course, the two lower lines are for 1% and 5%, and the two upper lines are for 95% and 99%. For each model, the vertical axis uses the same scale for all cases for those plots.

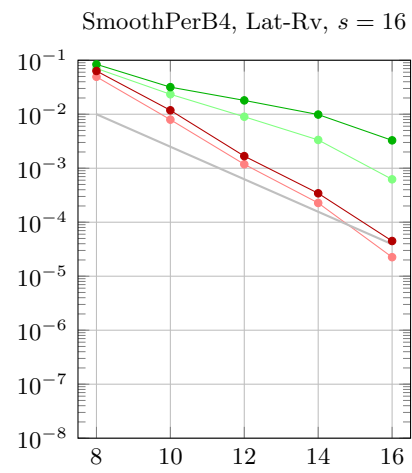
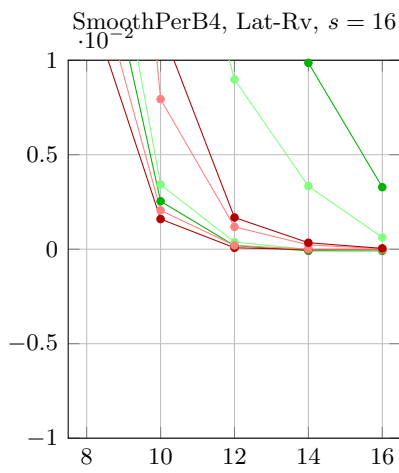
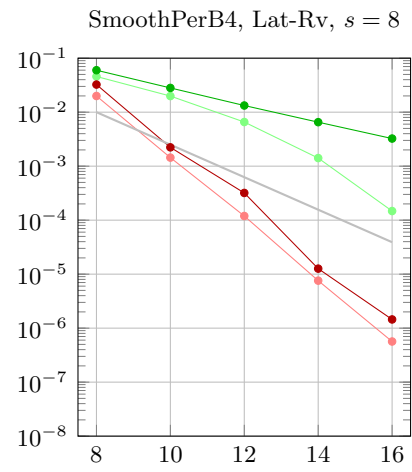
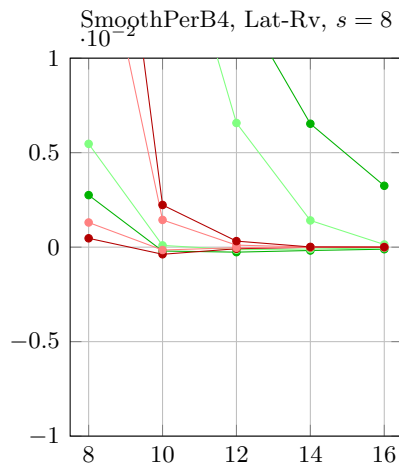
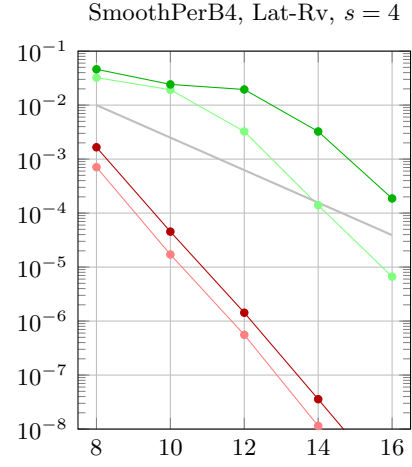
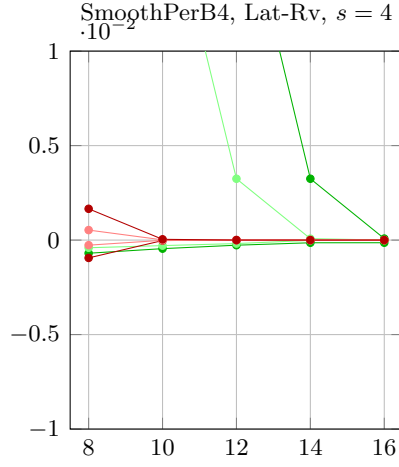
Looking at the quantiles in log scale would give a better visual idea of the convergence rate, but we cannot use a log scale for the 1% and 5% quantiles because these quantiles are usually negative. For this reason, it is more appropriate to show use the quantiles of the absolute error instead. We do that in the right picture of each row, which shows the 95% and 99% quantiles of the absolute error with the vertical axis in log scale, and using the same colors. For these plots, for each model, the vertical axis uses the same scale for all lattice cases and also the same scale for all Sobol' cases. These two scales differ only for some of the models.

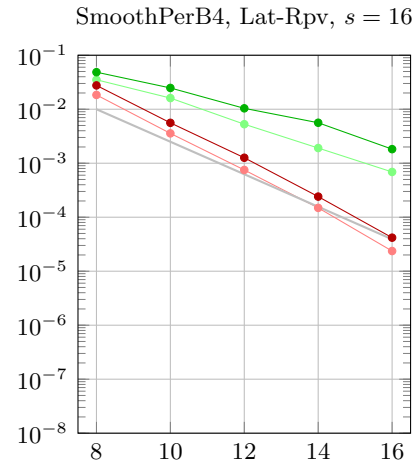
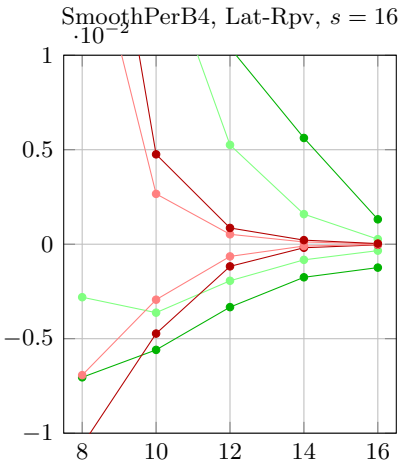
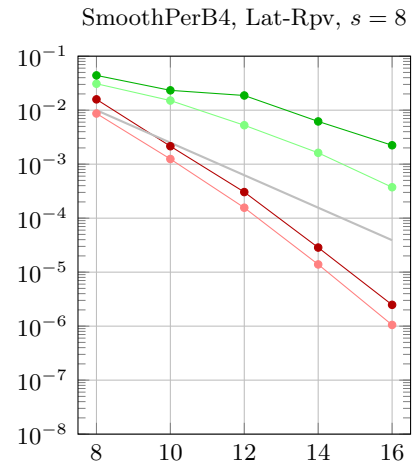
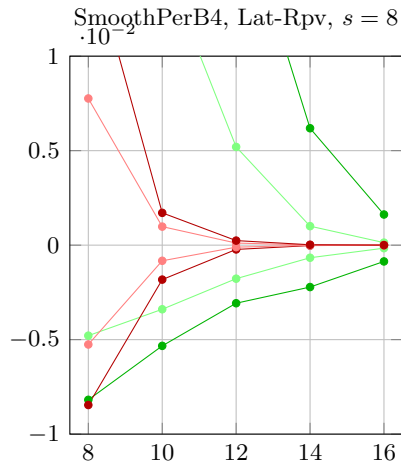
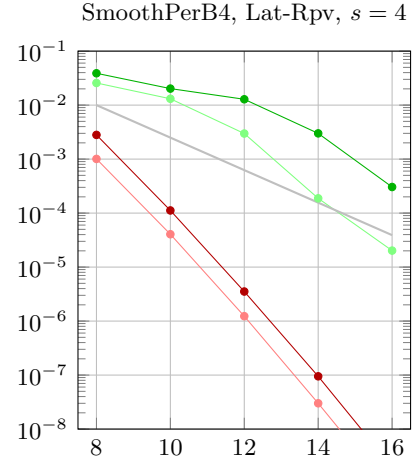
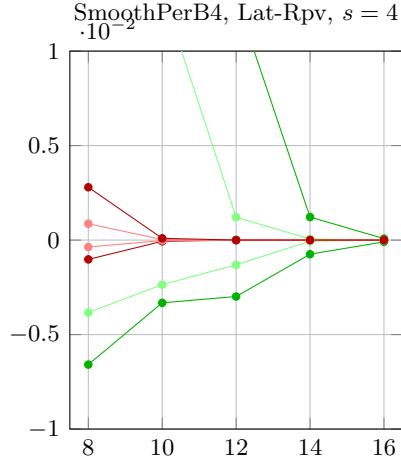
We see in these plots that for the random-parameter methods, the logs of the upper quantiles are usually smaller for M_r than for A_r , and their convergence rate is also often better and tends to improve with k .

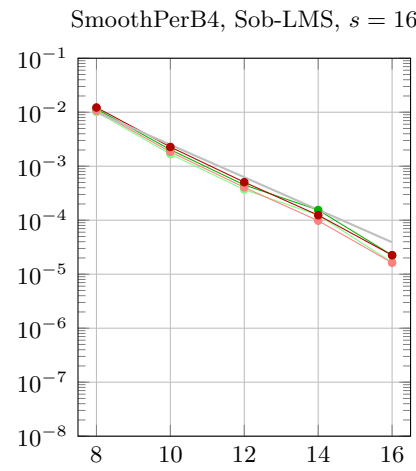
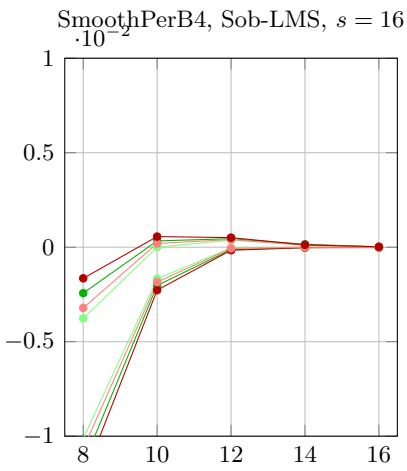
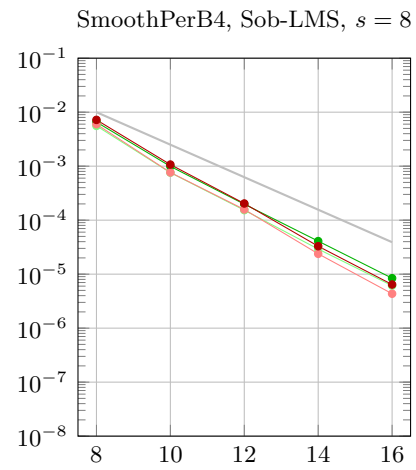
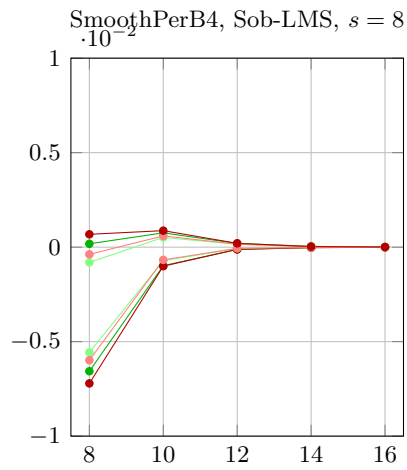
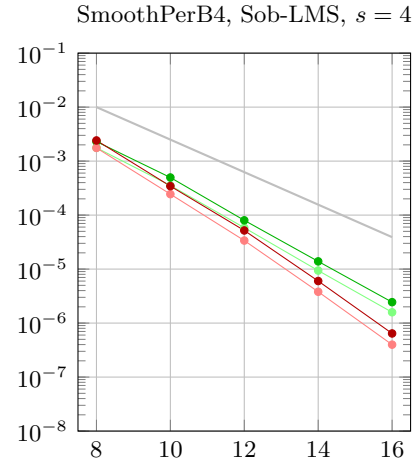
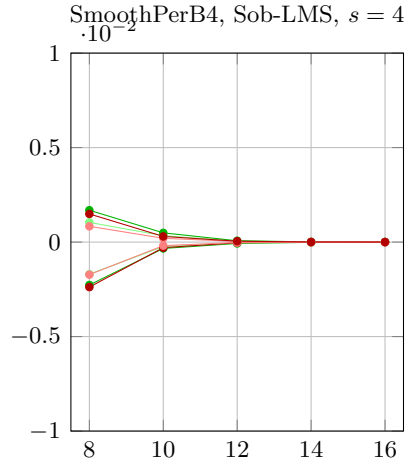
In some of the plots, for example for `SumUeU` with Lat-Rv, we see only a red line. The reason is that in this case, the error is a constant, therefore A_r and M_r and exactly the same and the green line hides under the red one.

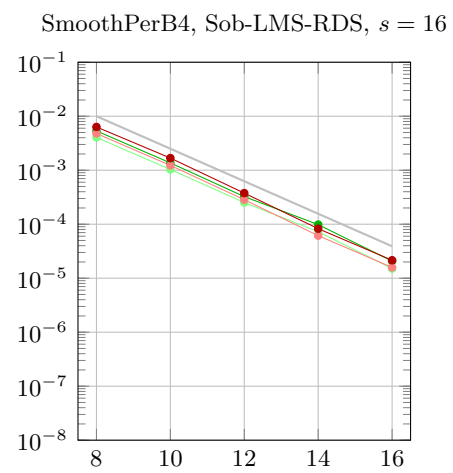
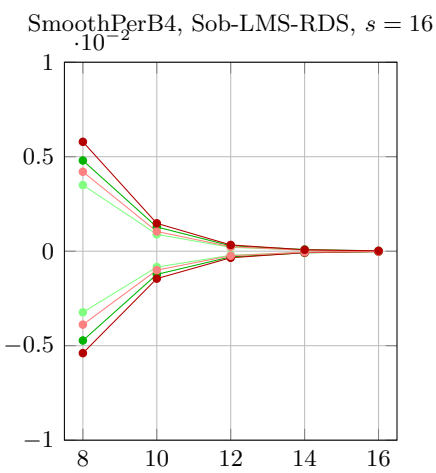
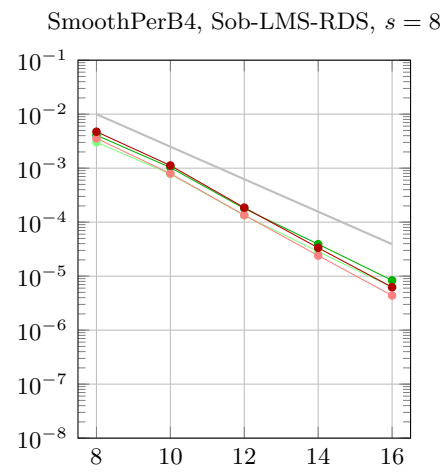
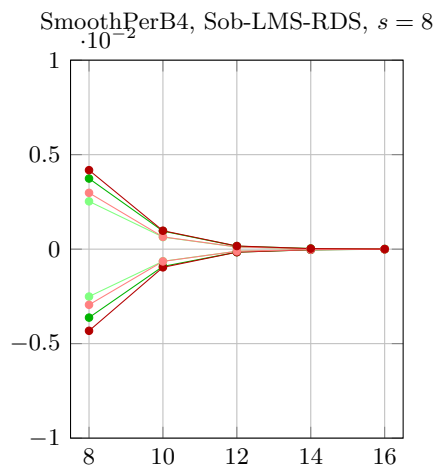
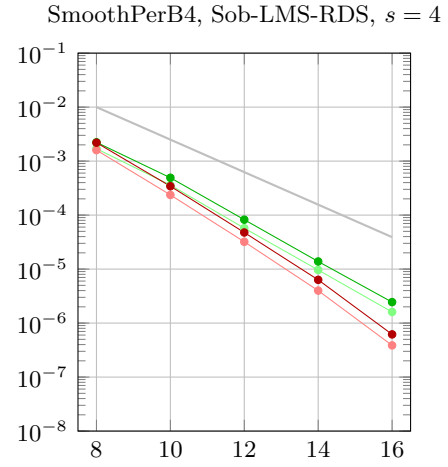
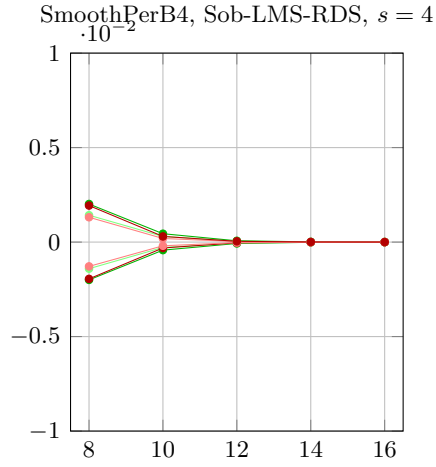


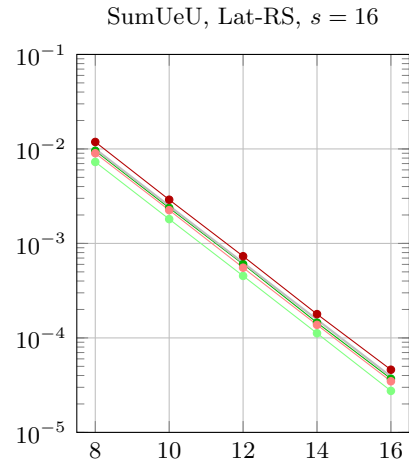
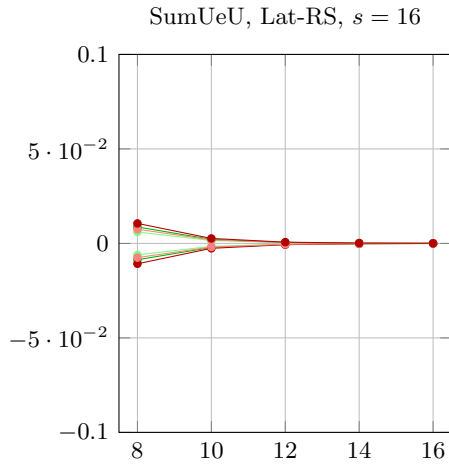
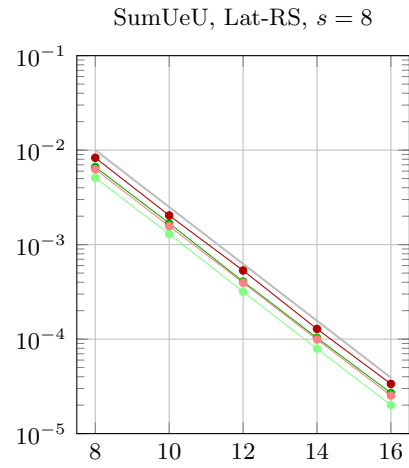
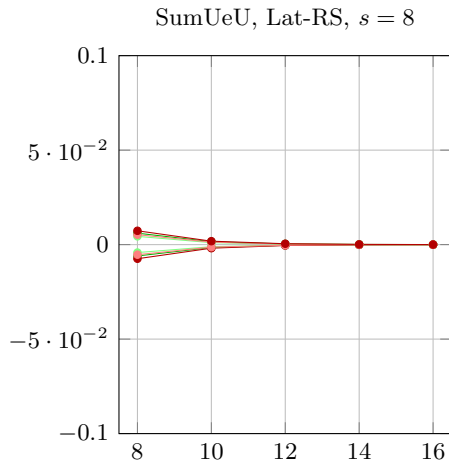
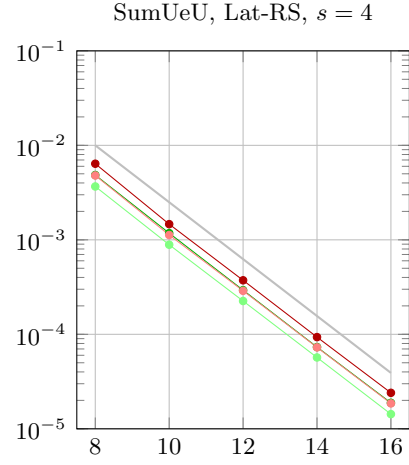
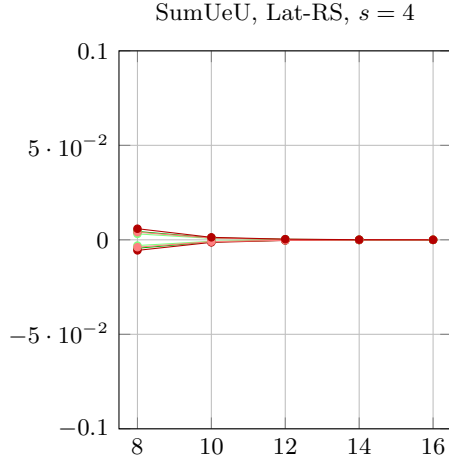


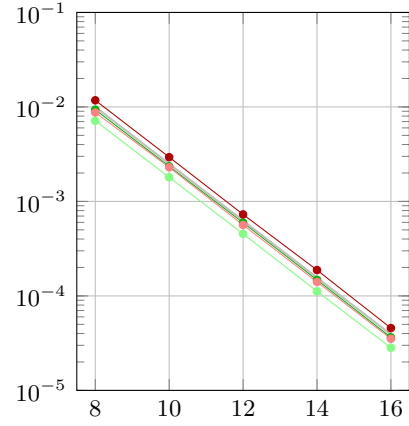
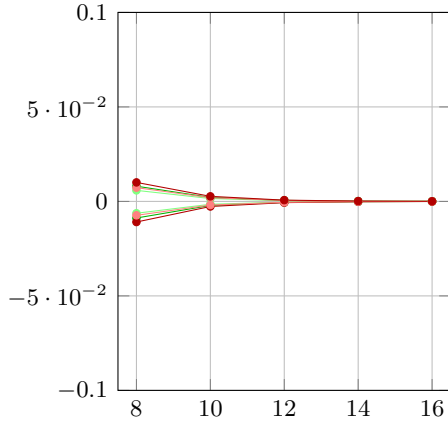
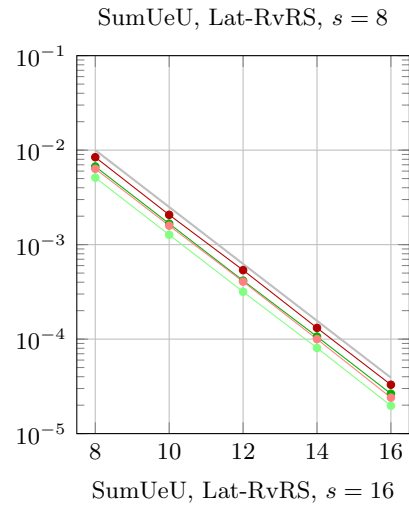
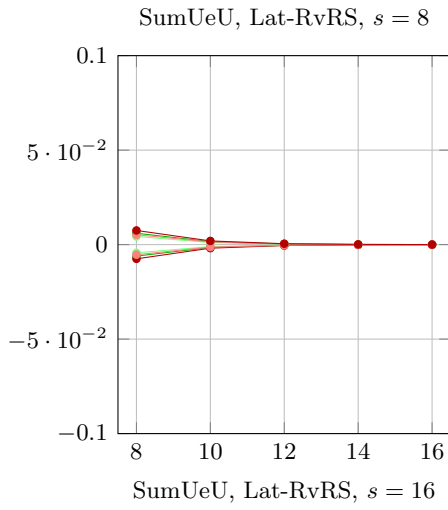
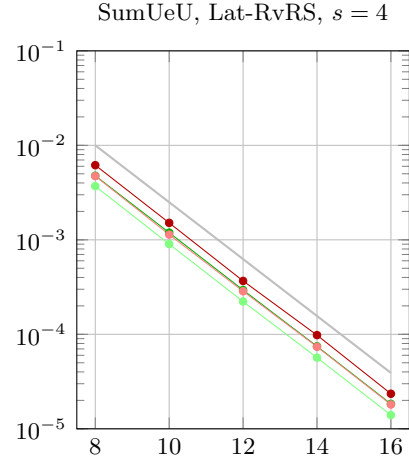
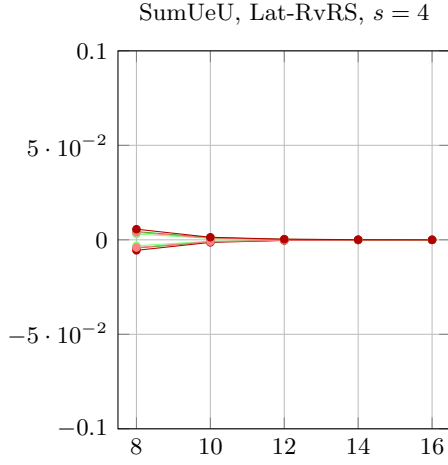


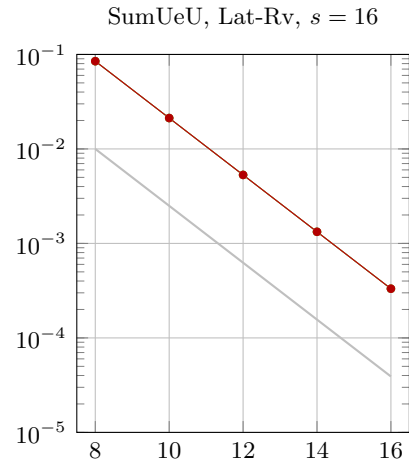
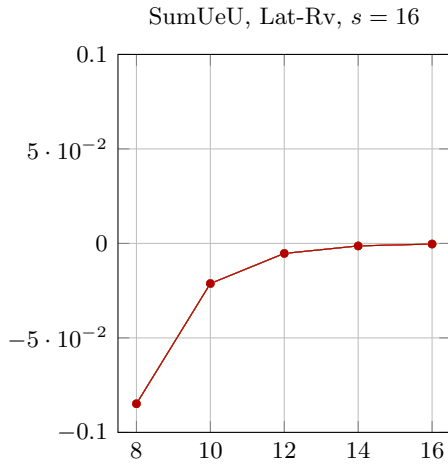
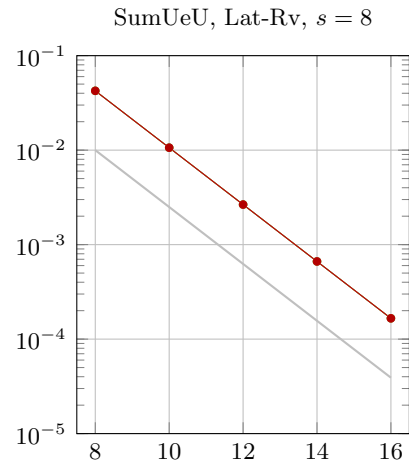
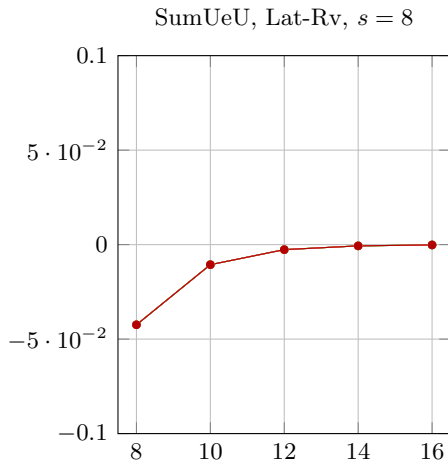
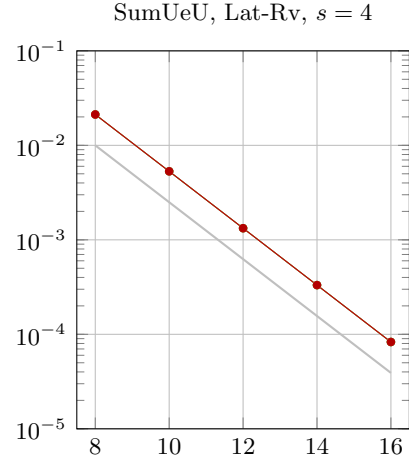
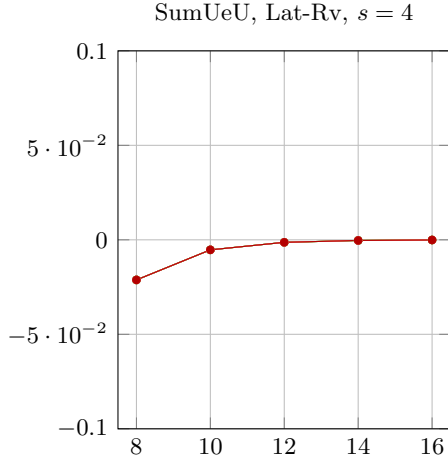


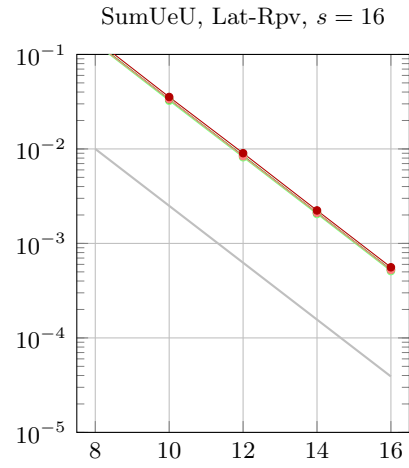
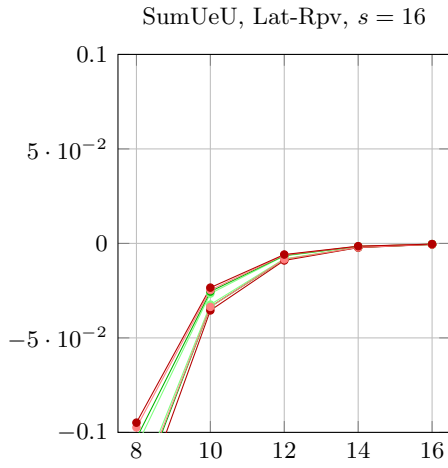
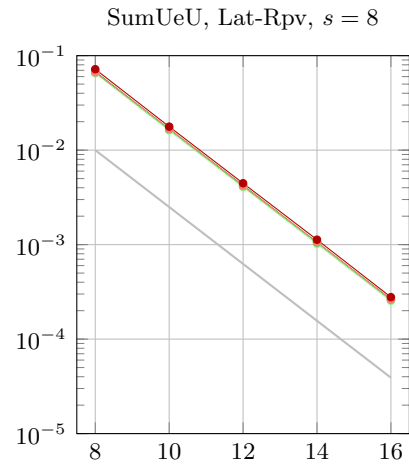
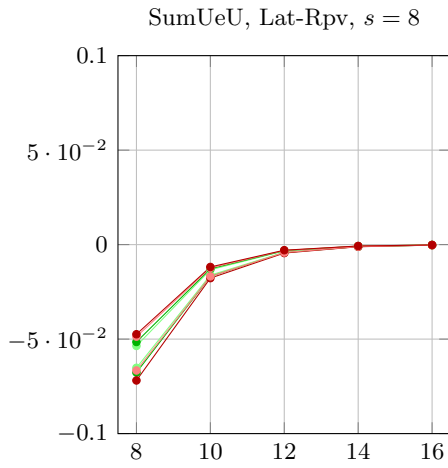
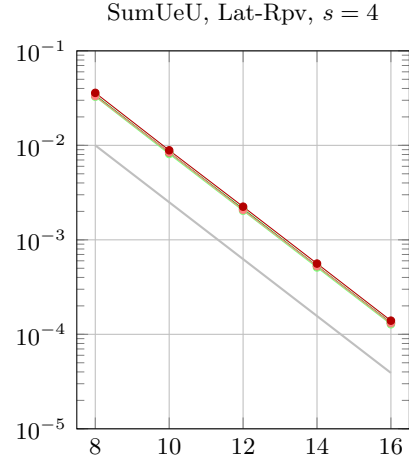
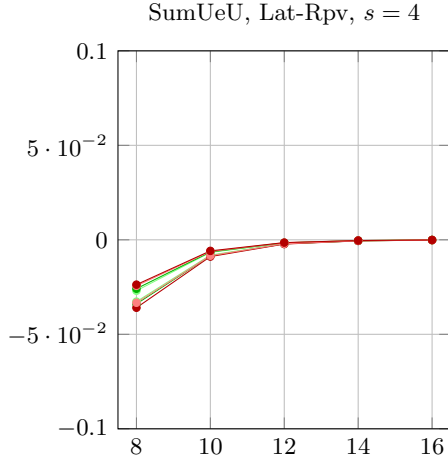


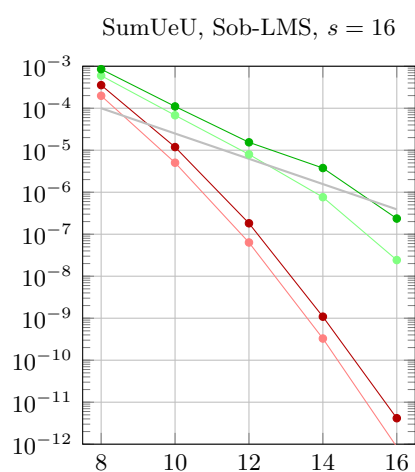
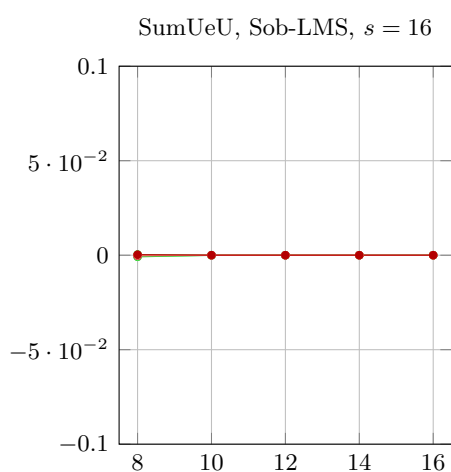
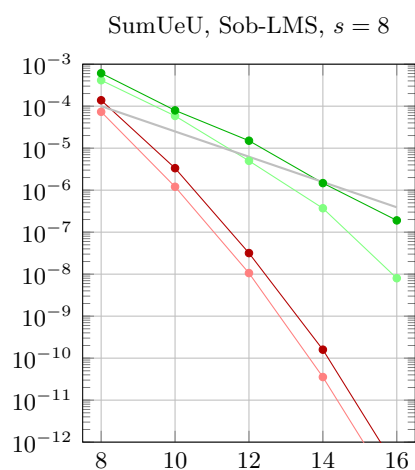
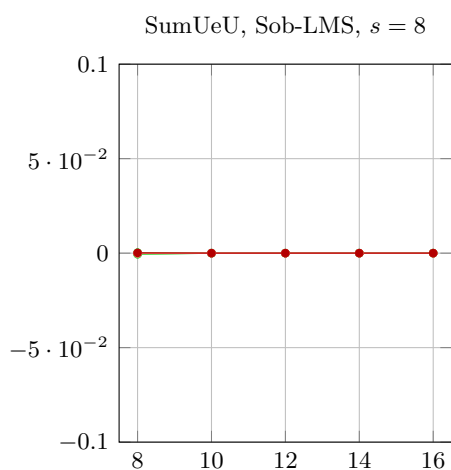
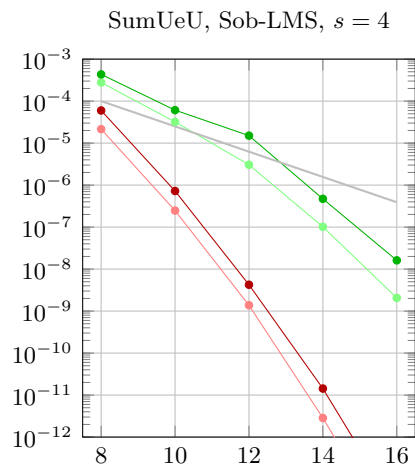
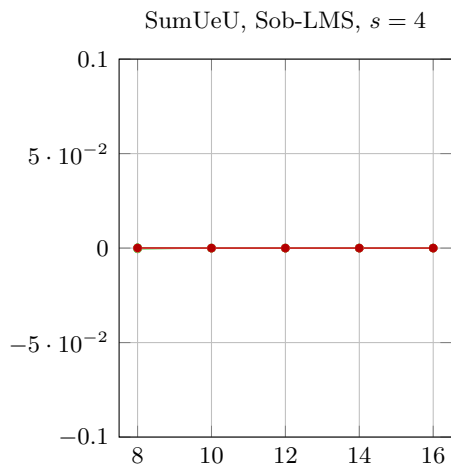


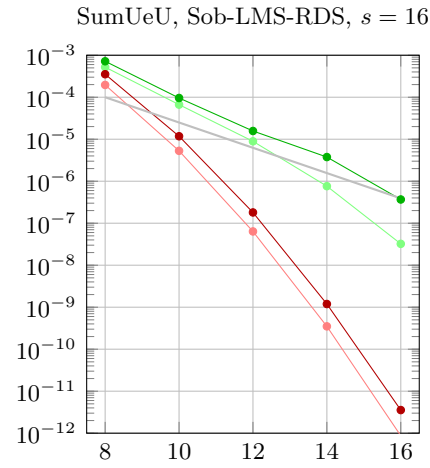
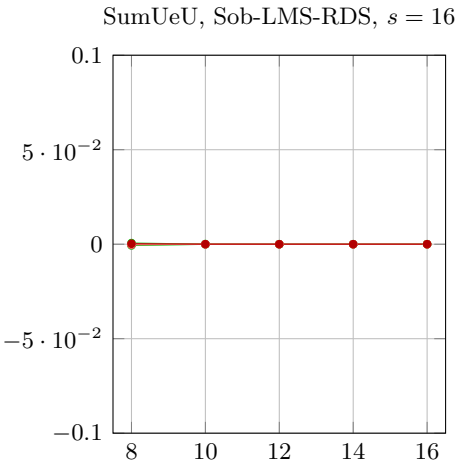
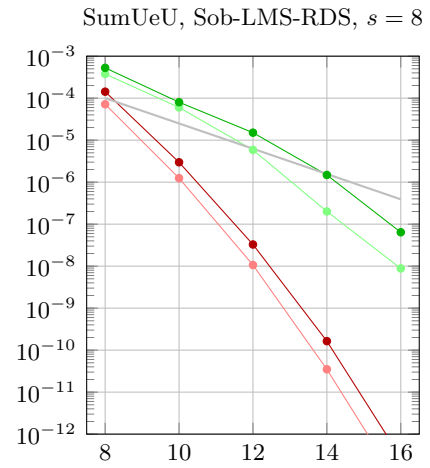
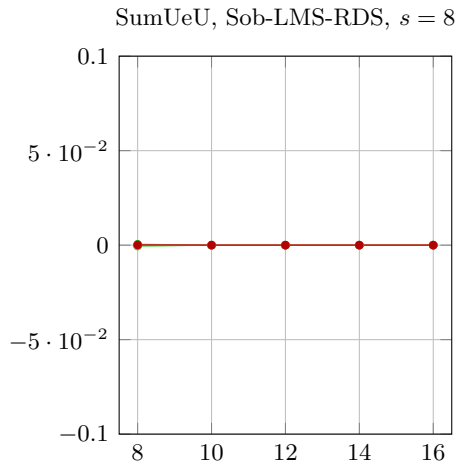
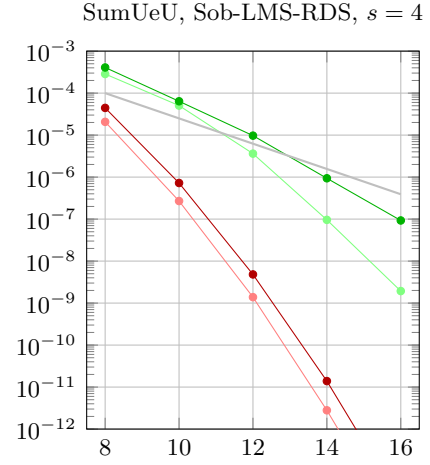
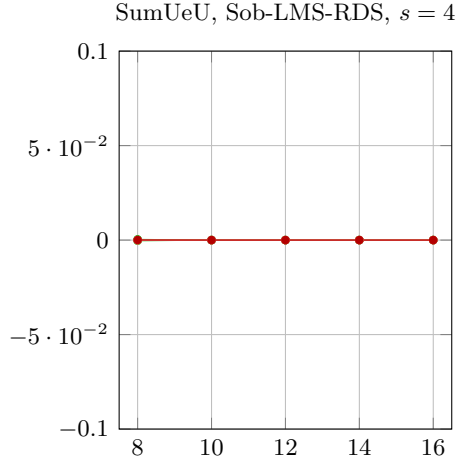


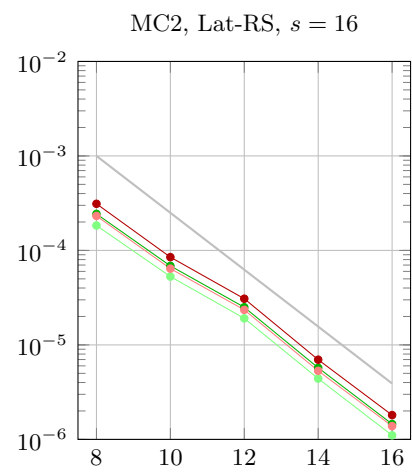
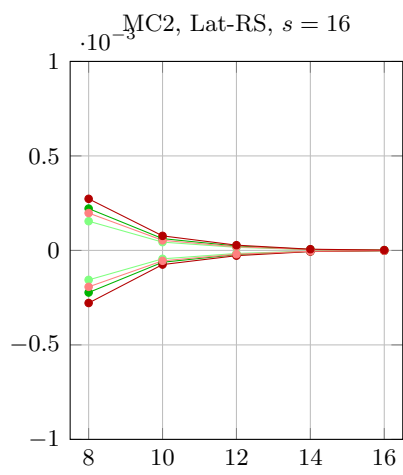
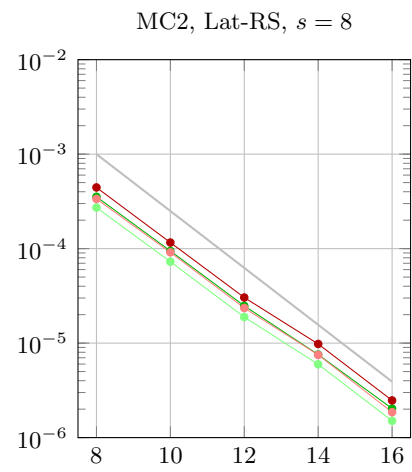
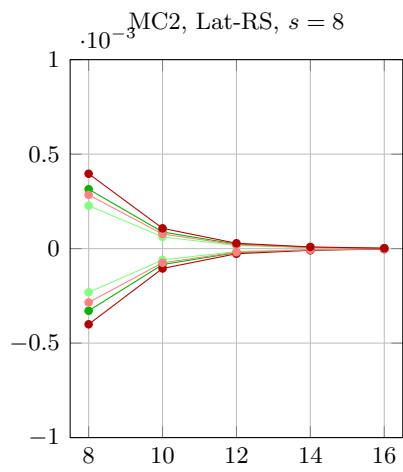
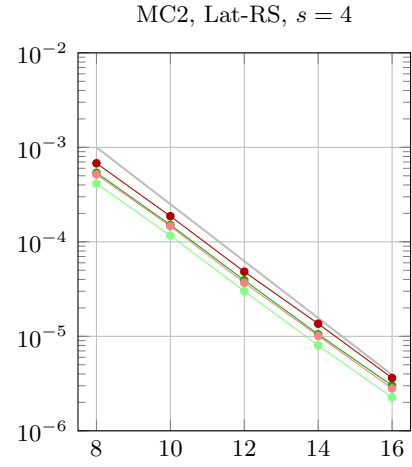
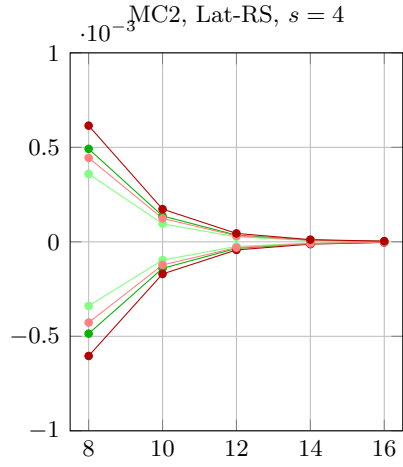


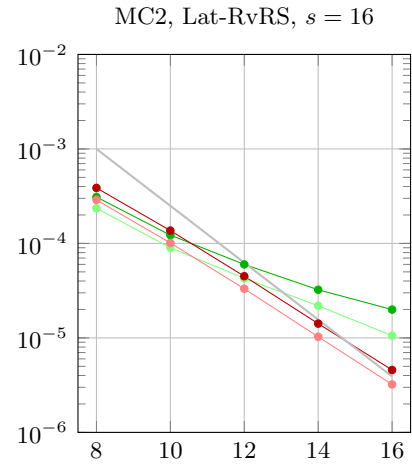
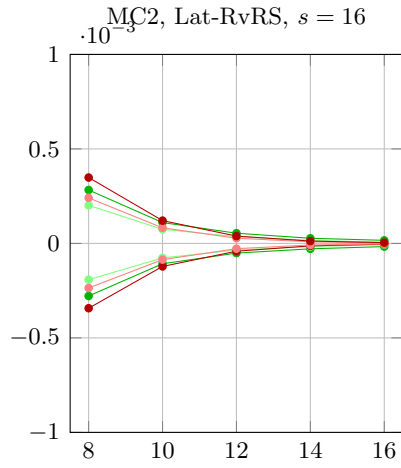
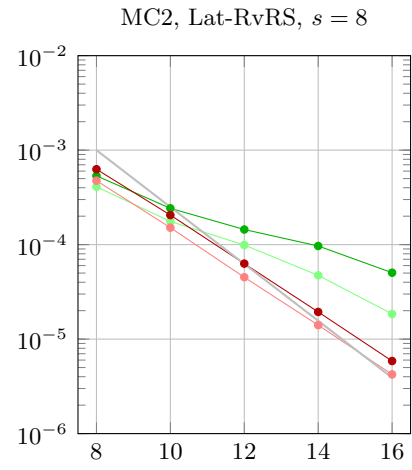
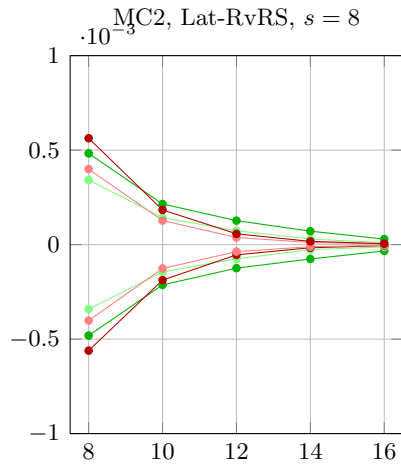
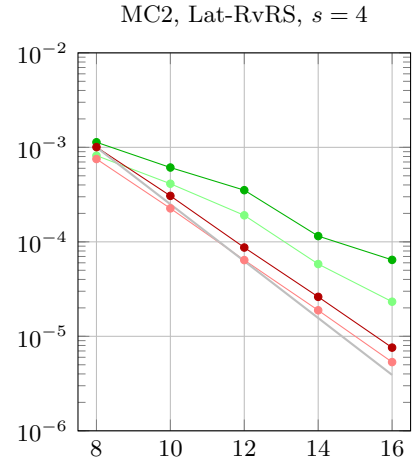
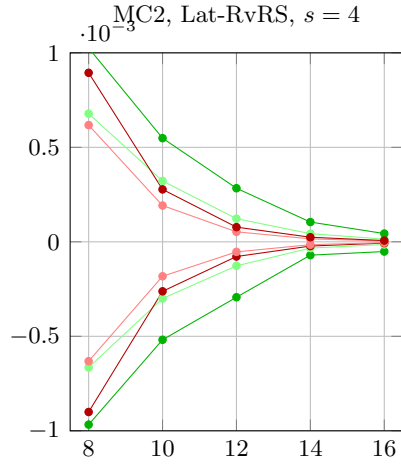


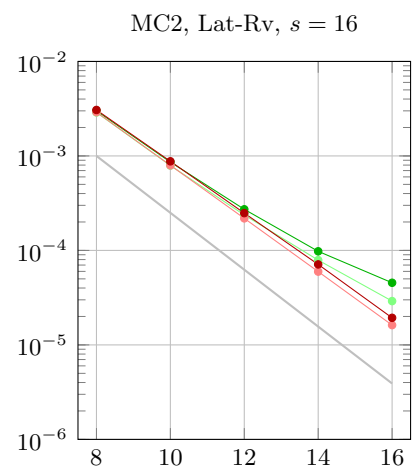
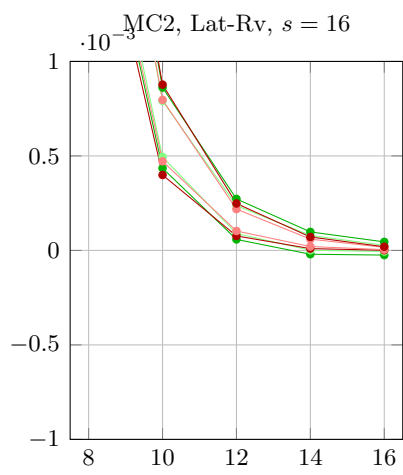
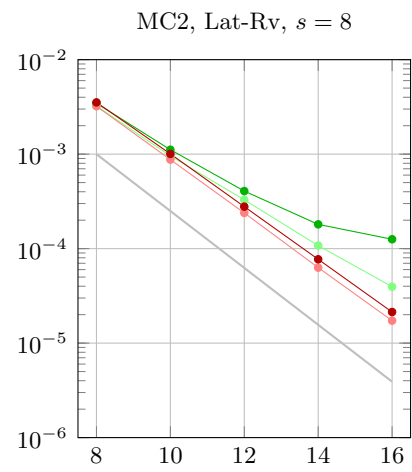
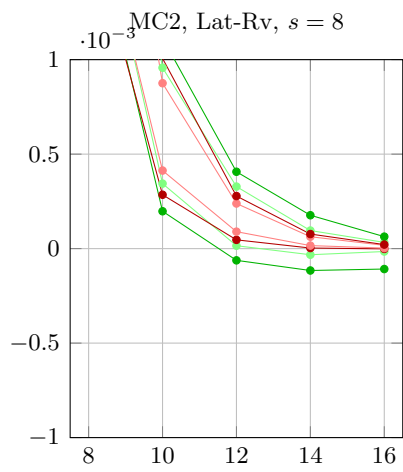
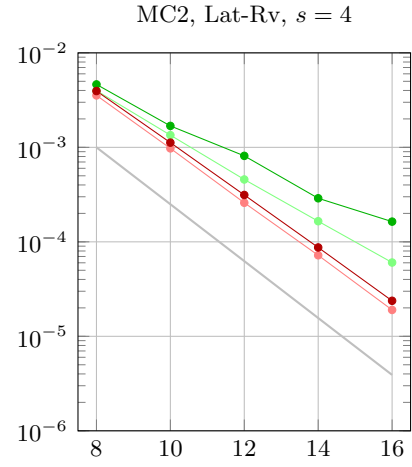
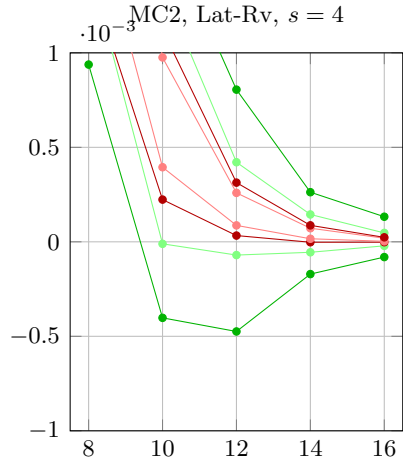


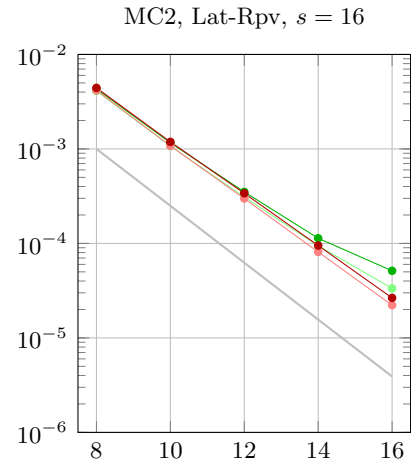
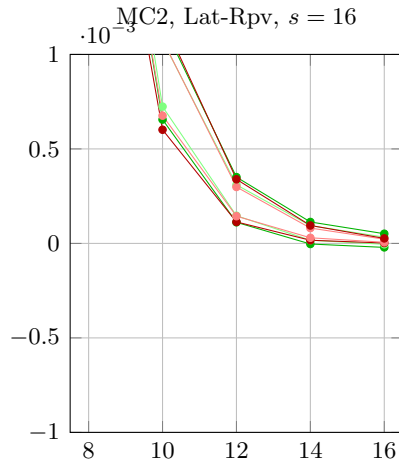
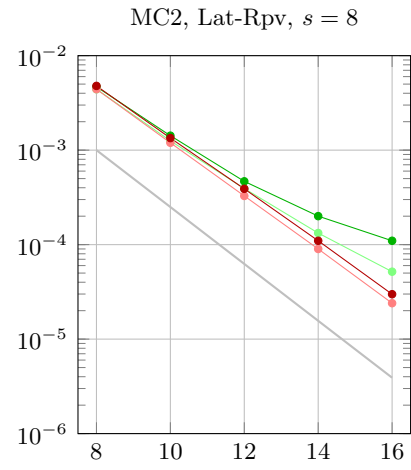
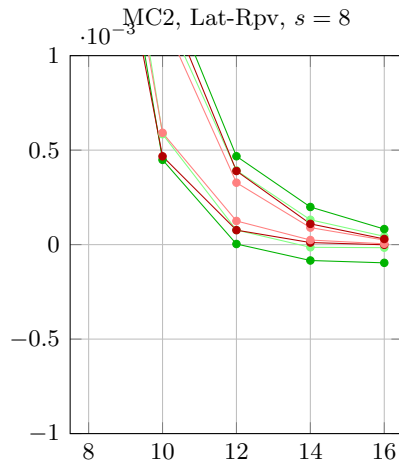
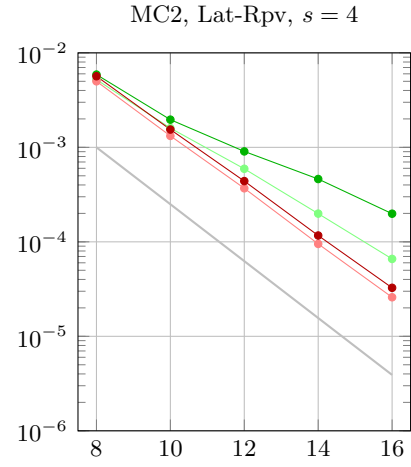
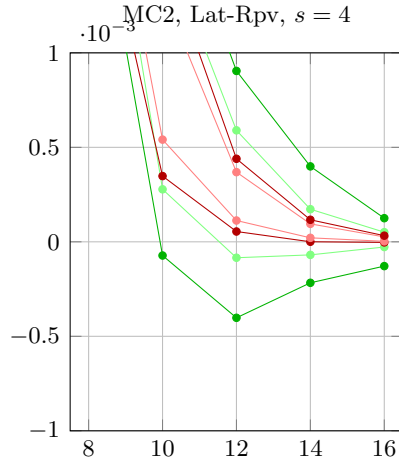


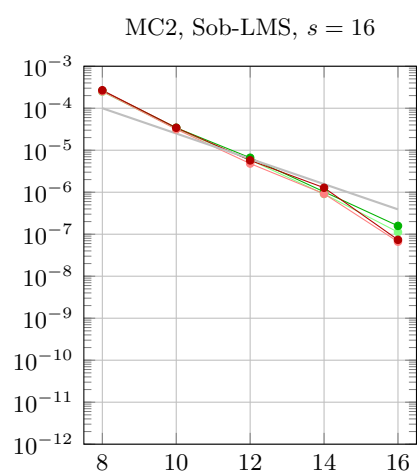
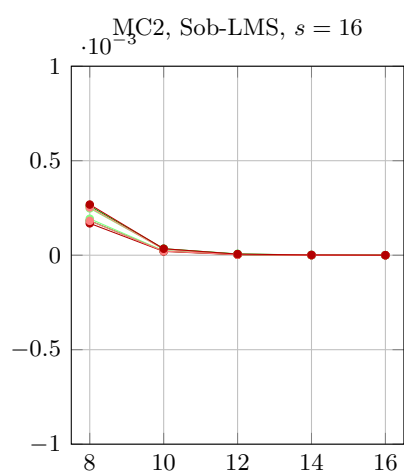
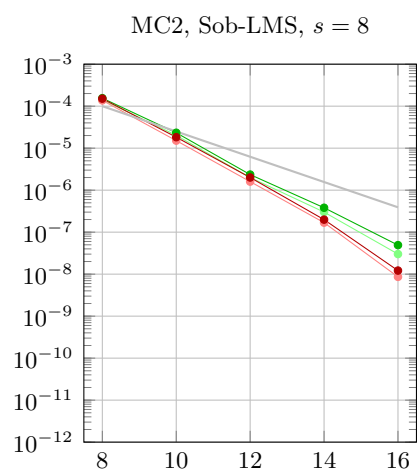
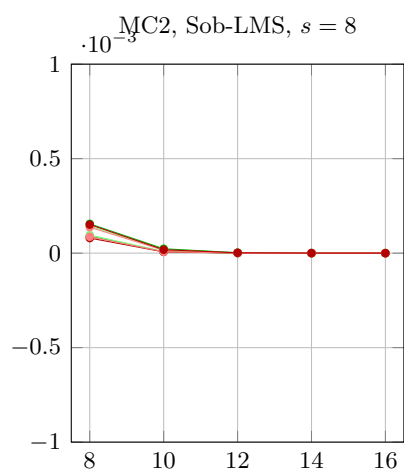
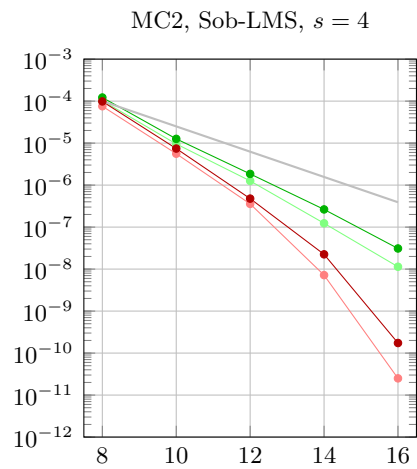
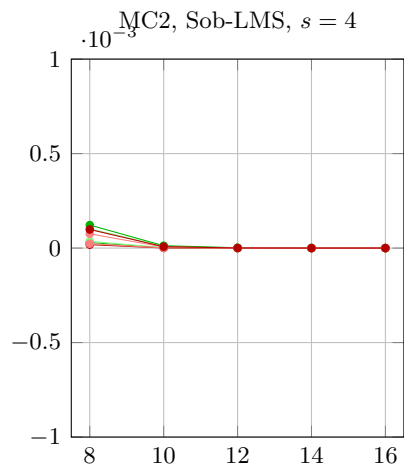


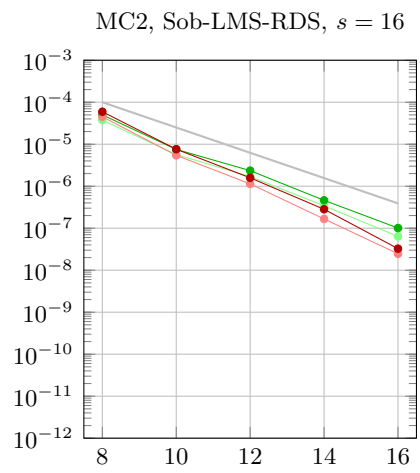
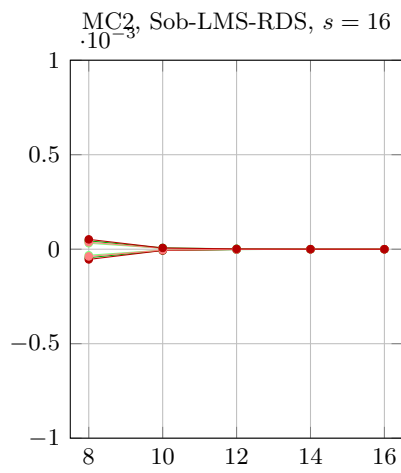
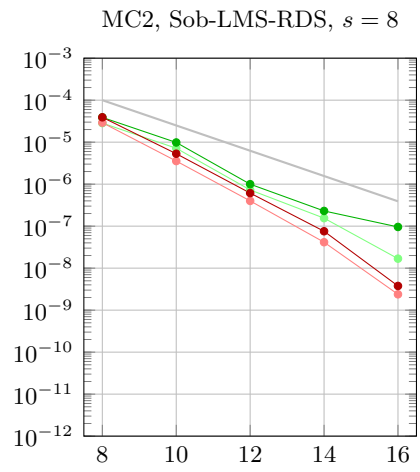
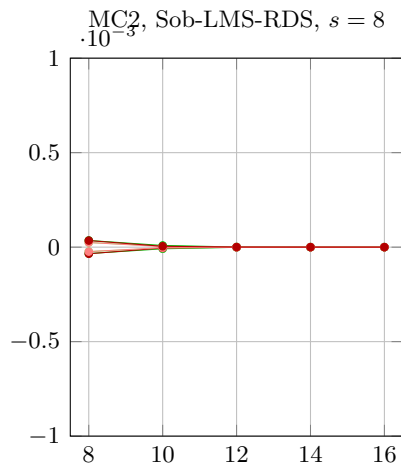
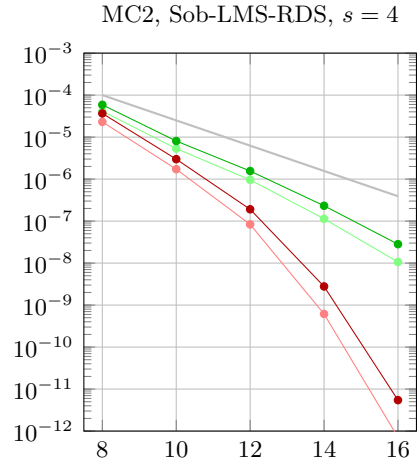
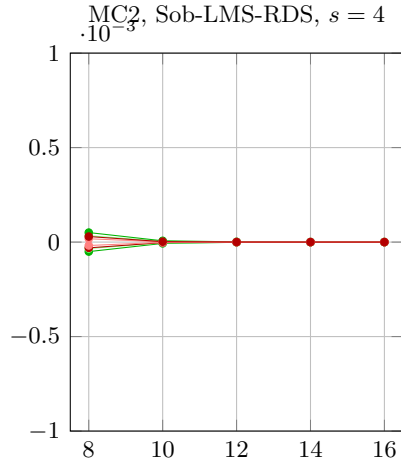


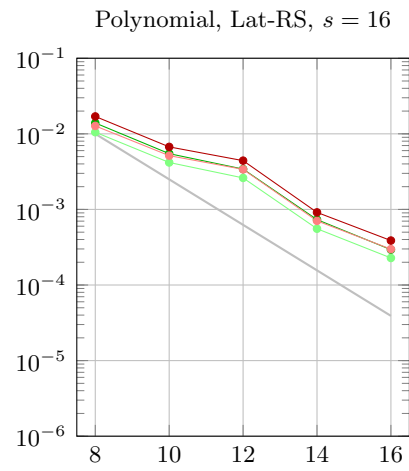
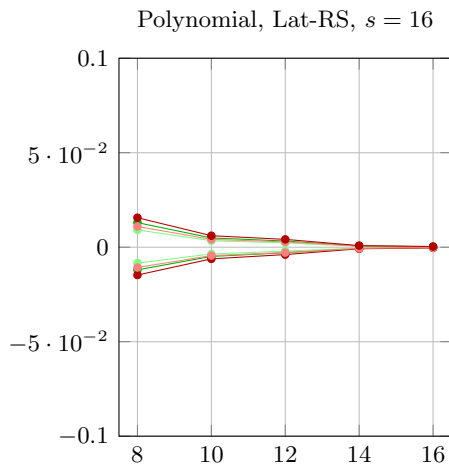
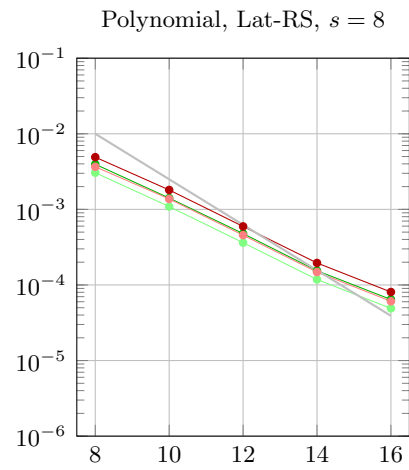
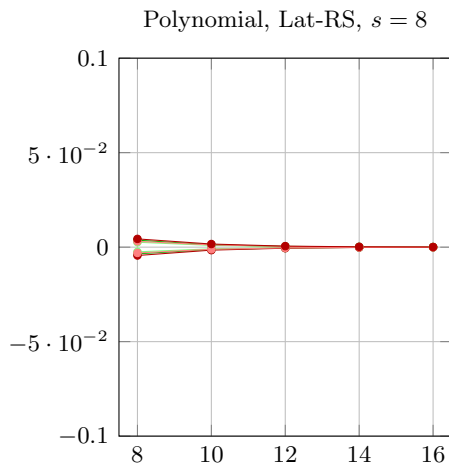
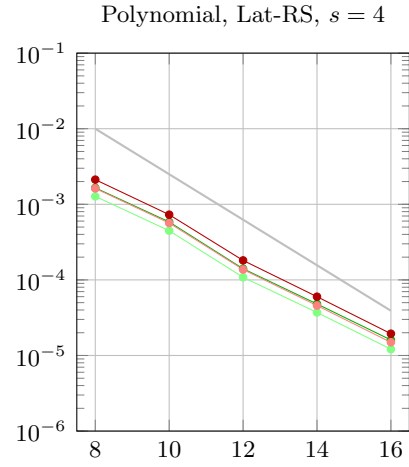
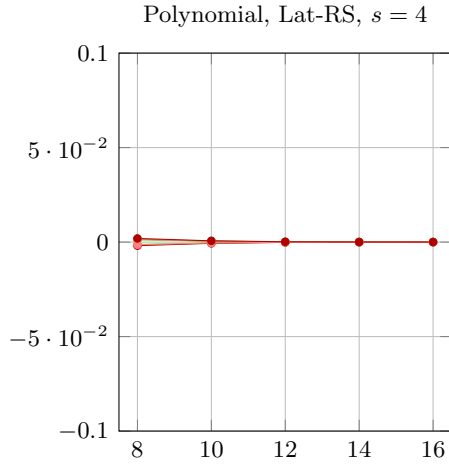


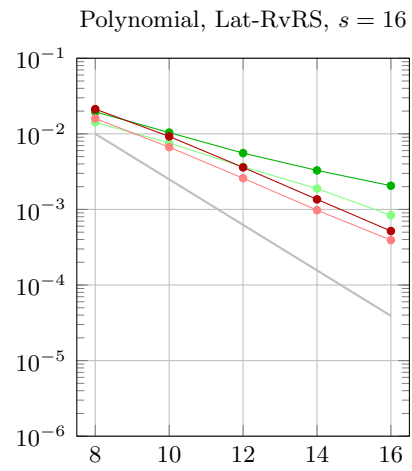
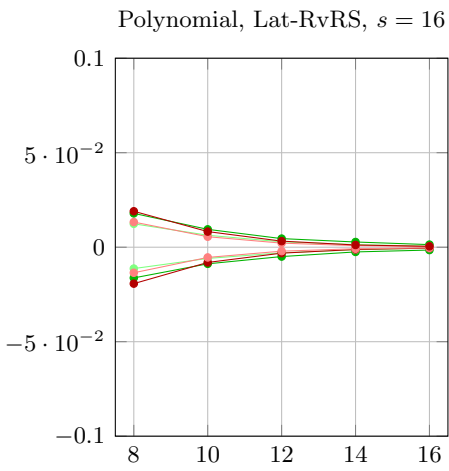
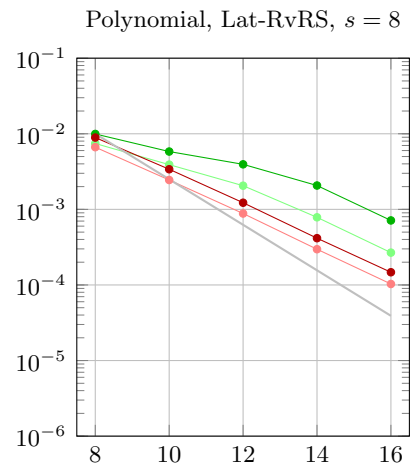
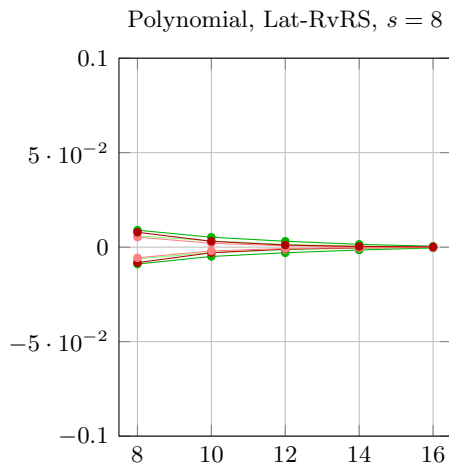
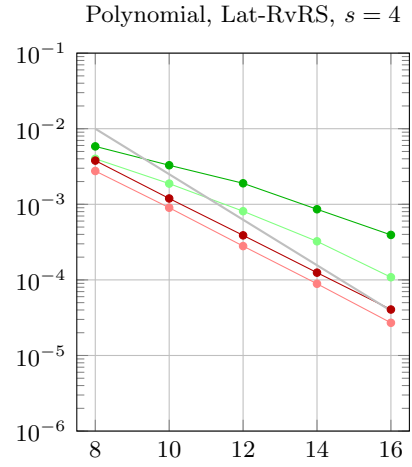
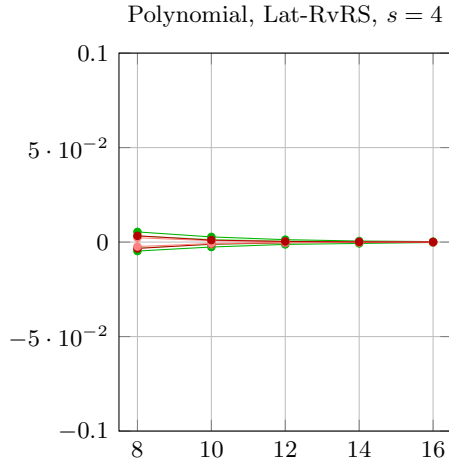


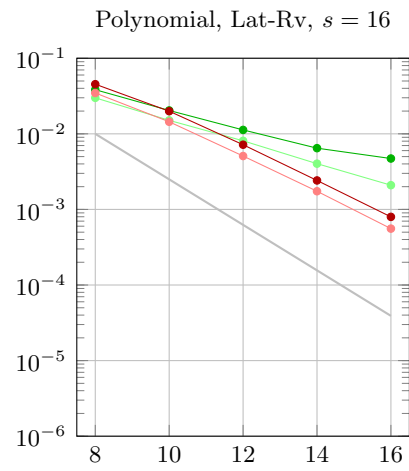
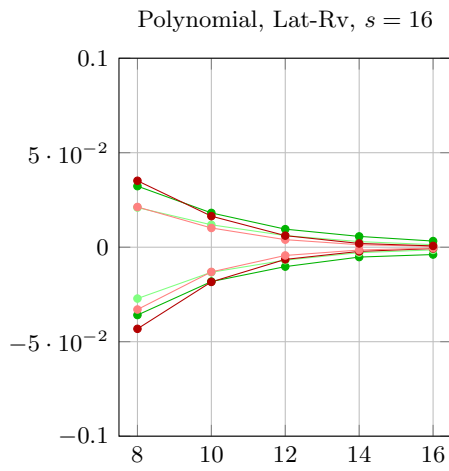
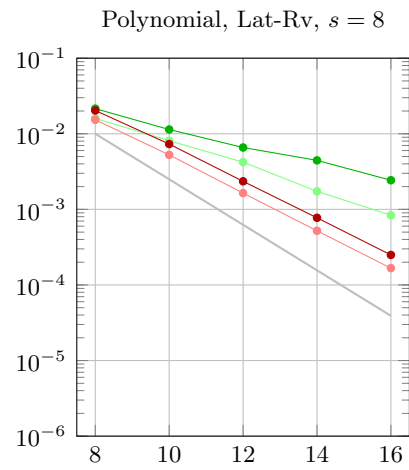
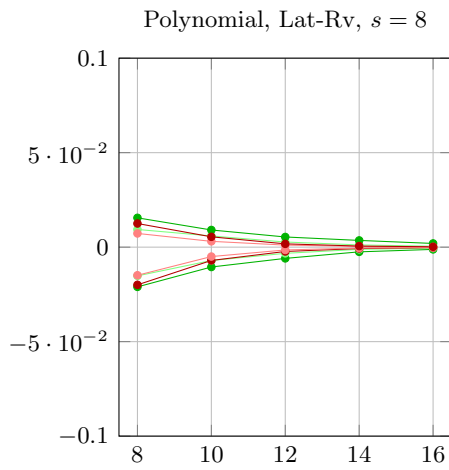
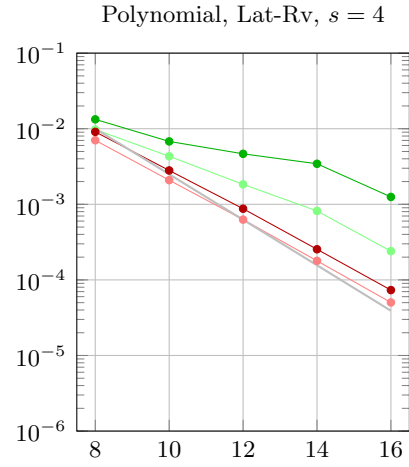
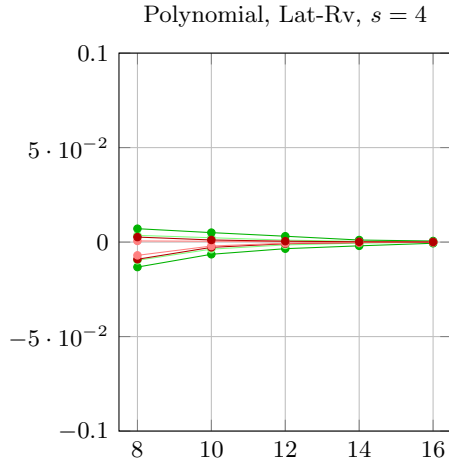


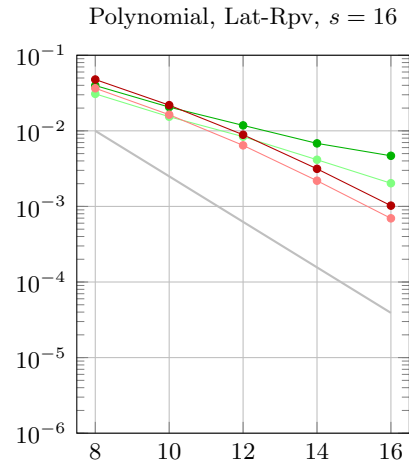
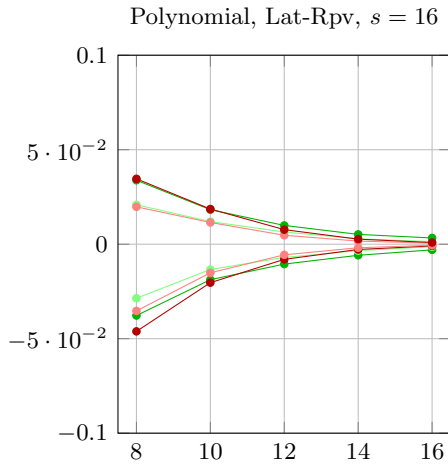
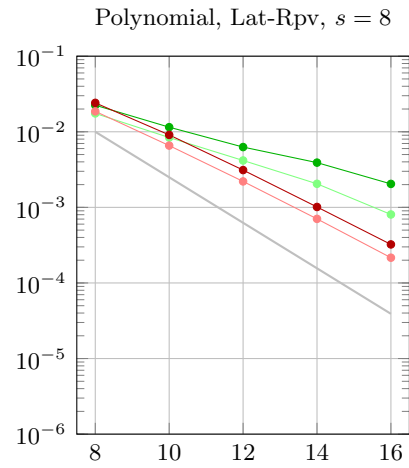
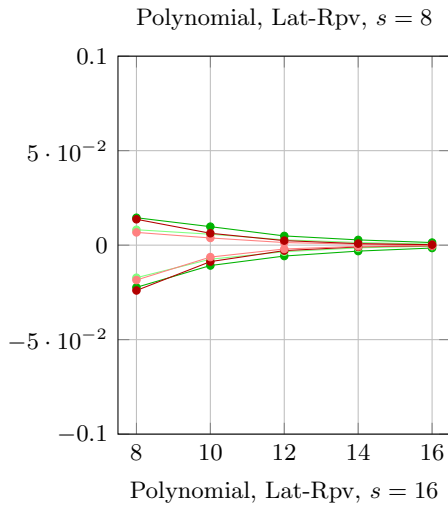
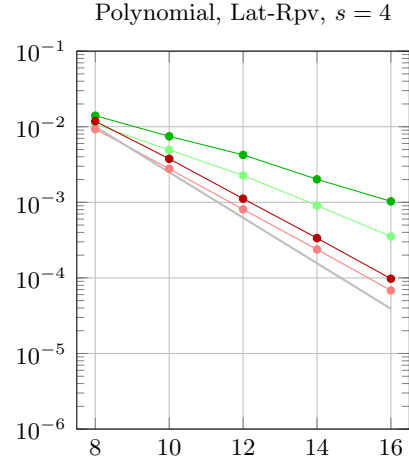
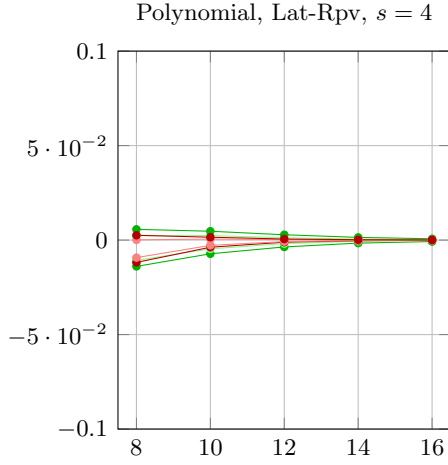


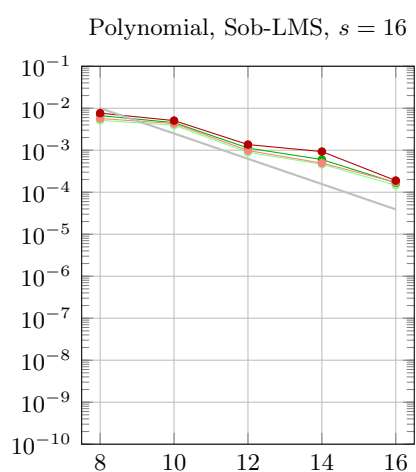
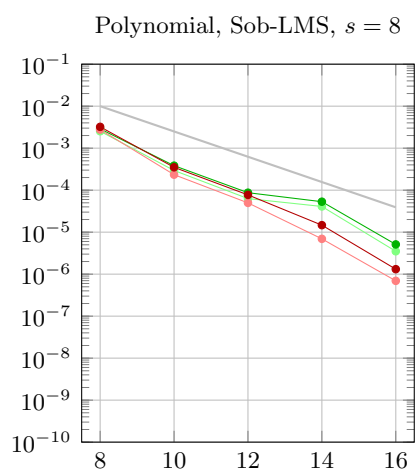
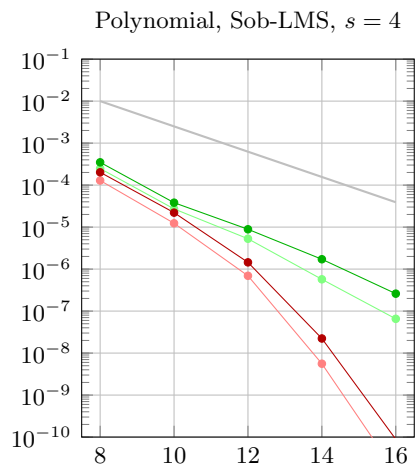
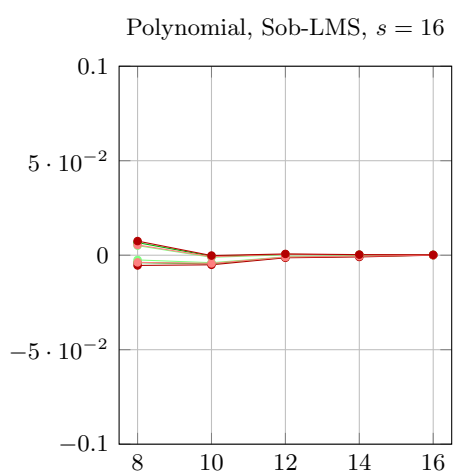
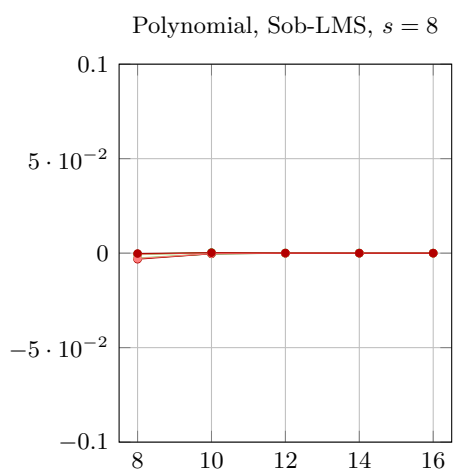
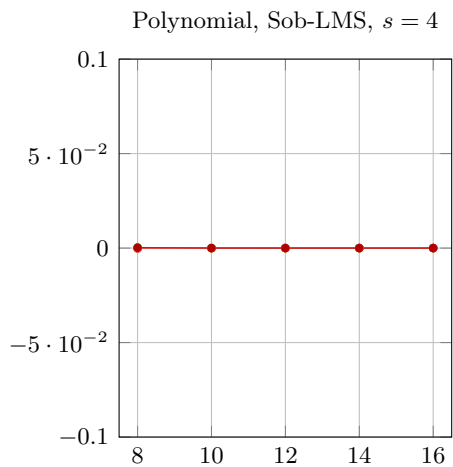




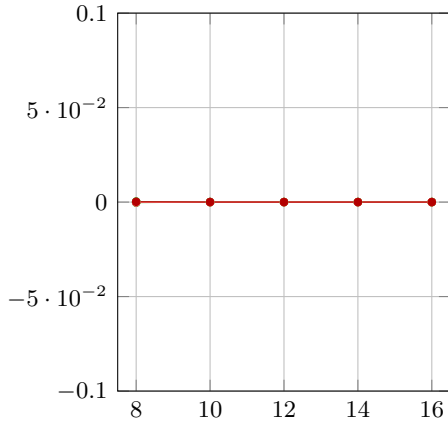




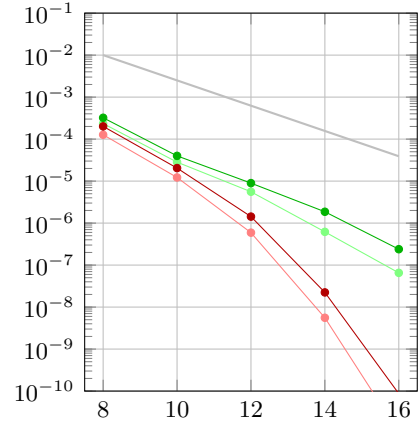




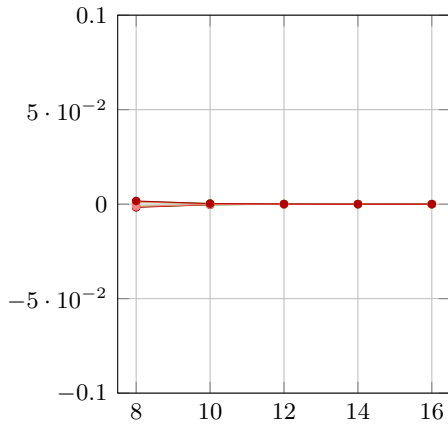
Polynomial, Sob-LMS-RDS, $s = 4$



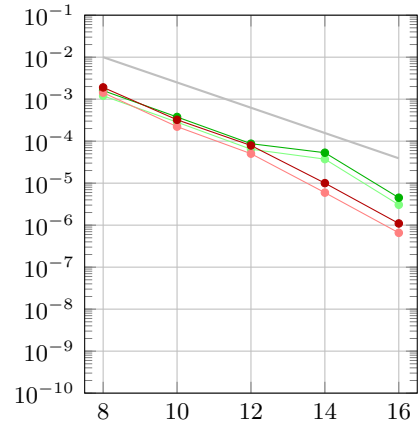
Polynomial, Sob-LMS-RDS, $s = 4$



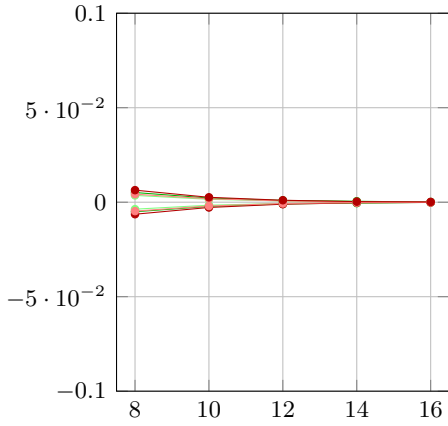
Polynomial, Sob-LMS-RDS, $s = 8$



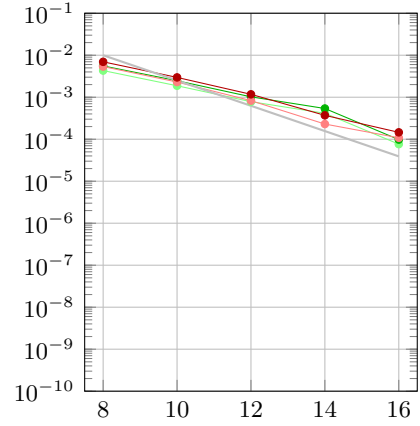
Polynomial, Sob-LMS-RDS, $s = 8$

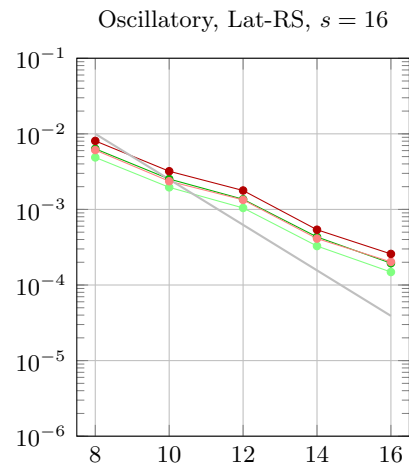
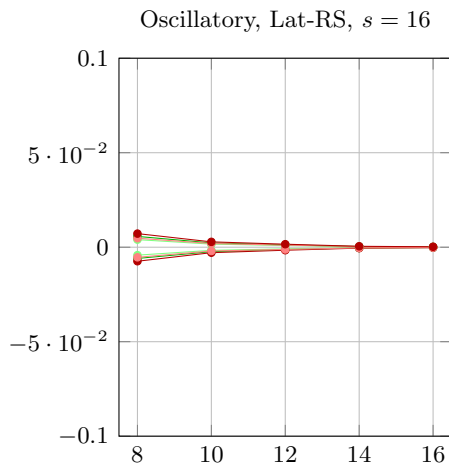
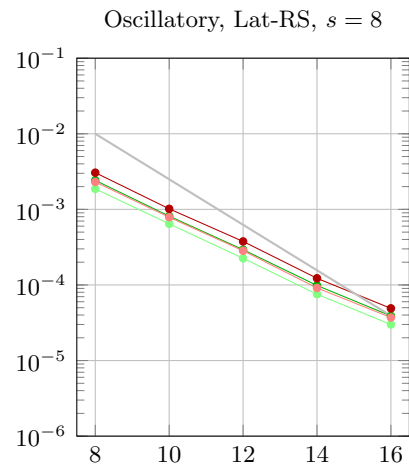
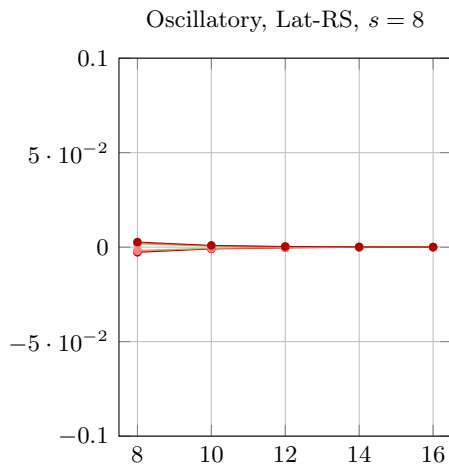
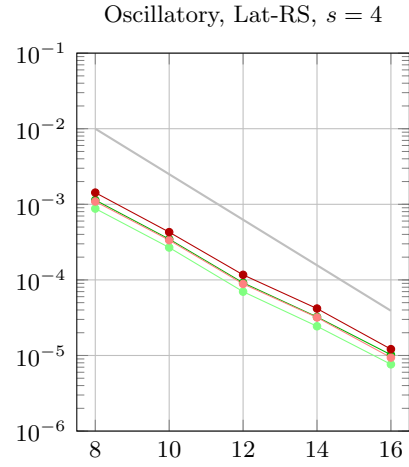
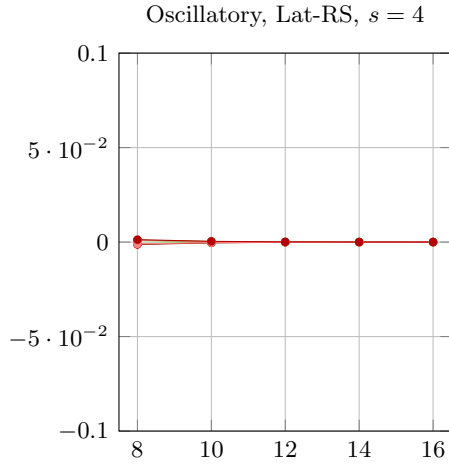


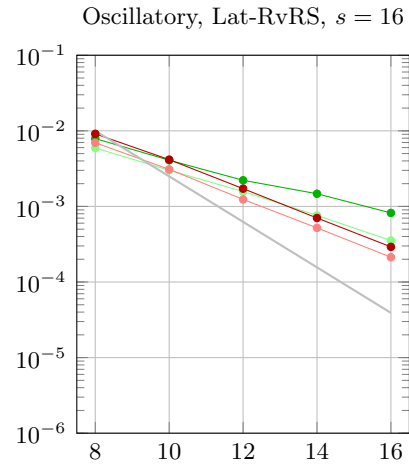
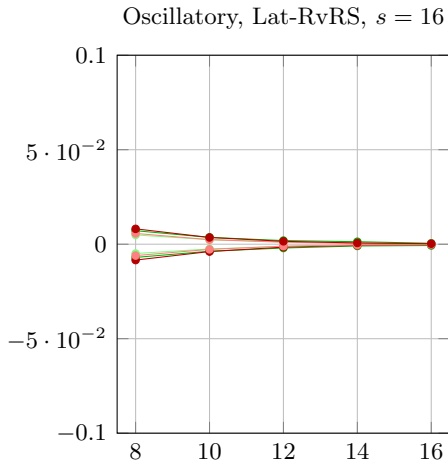
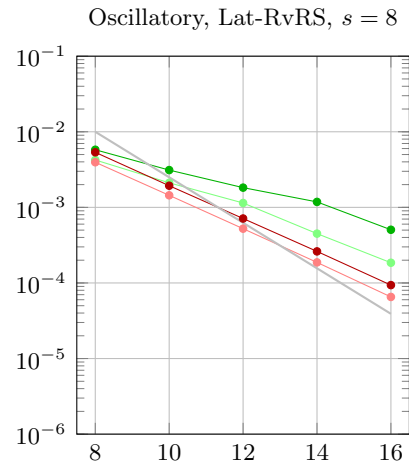
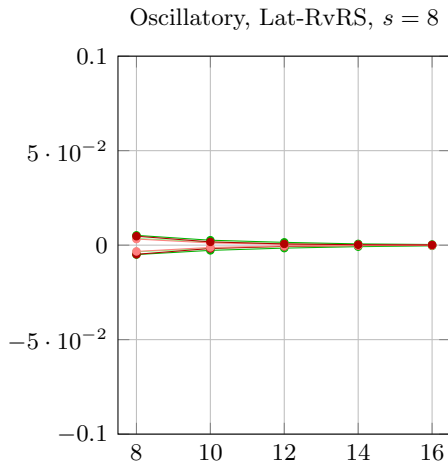
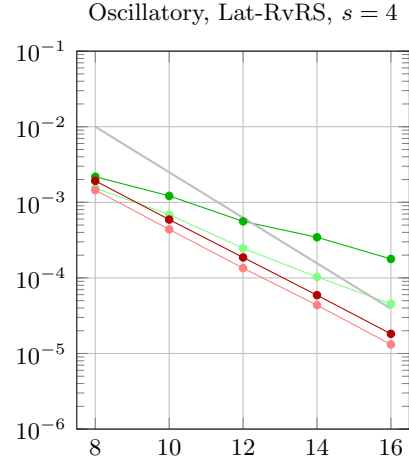
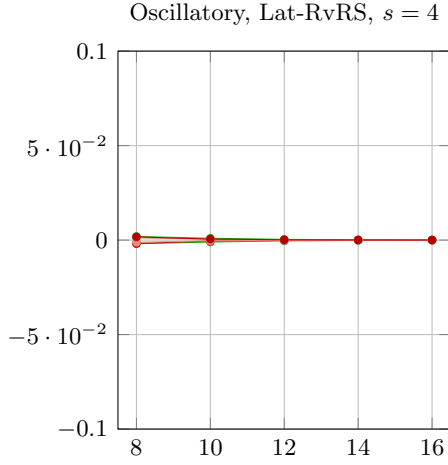
Polynomial, Sob-LMS-RDS, $s = 16$

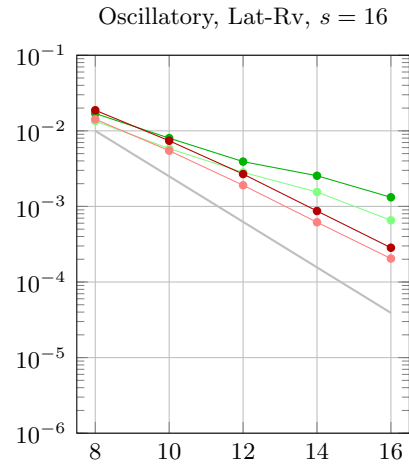
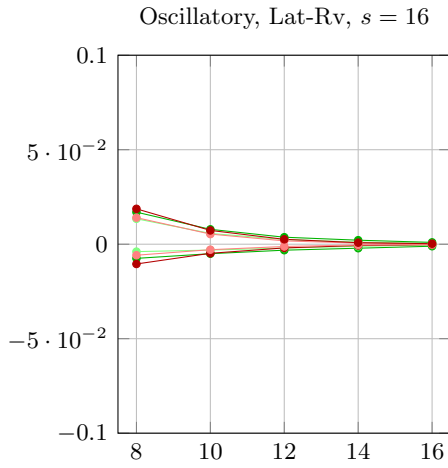
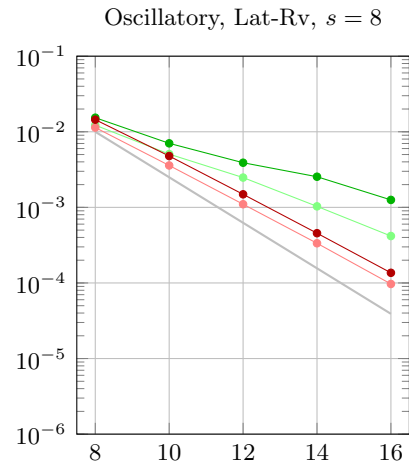
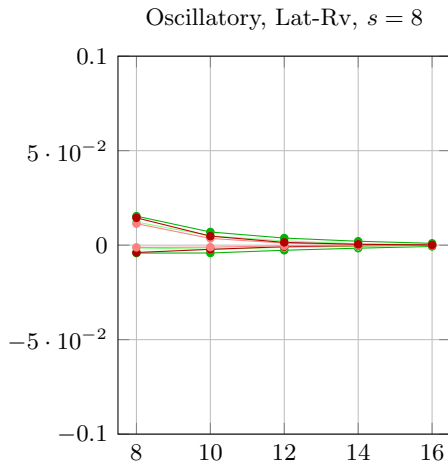
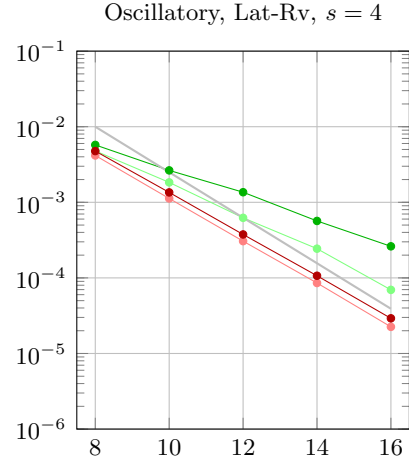
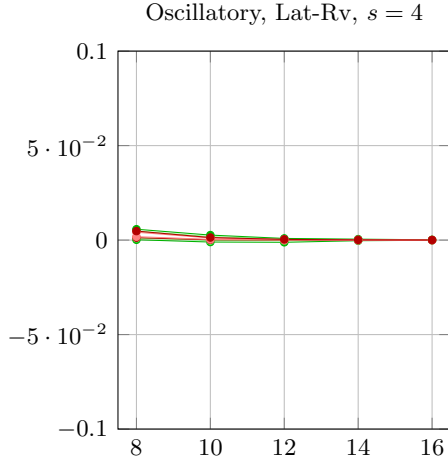


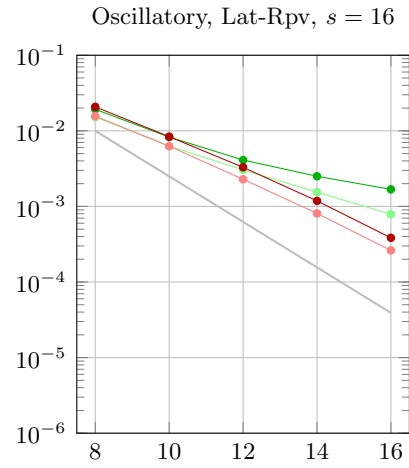
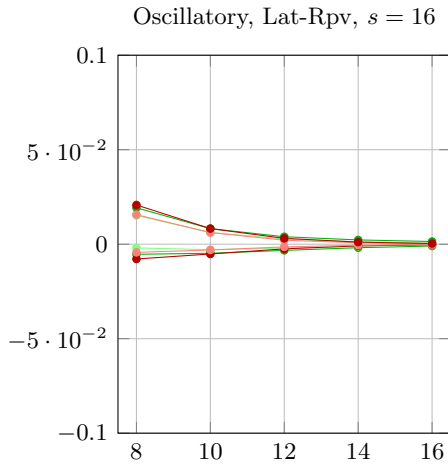
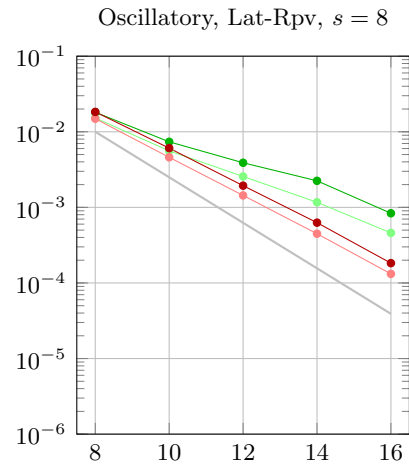
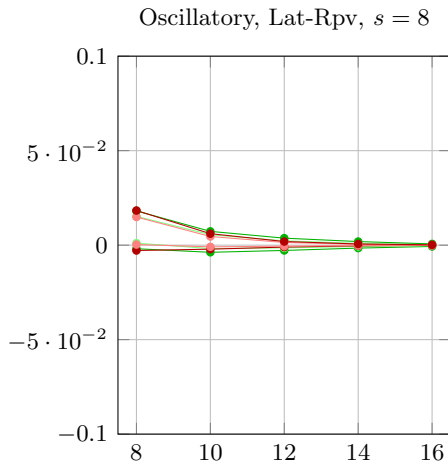
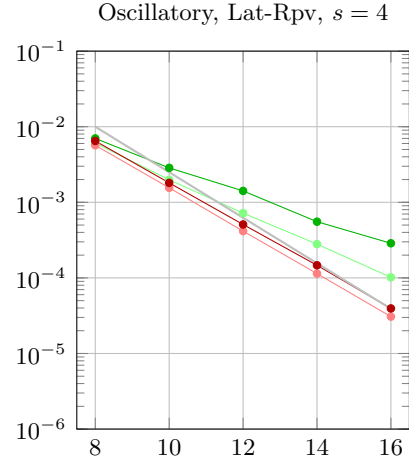
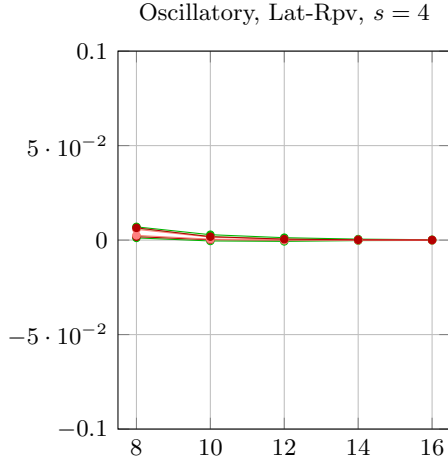
Polynomial, Sob-LMS-RDS, $s = 16$

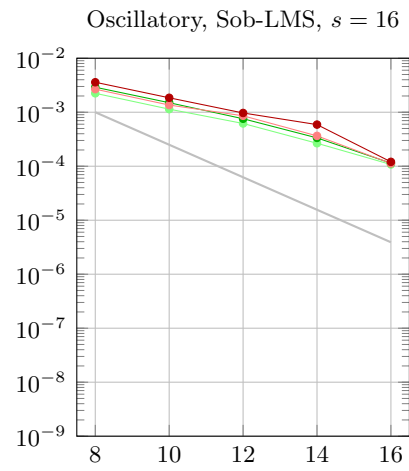
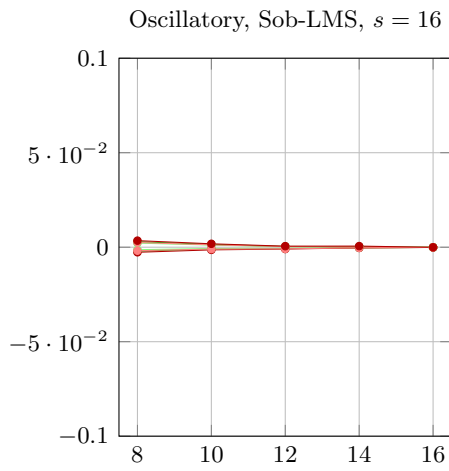
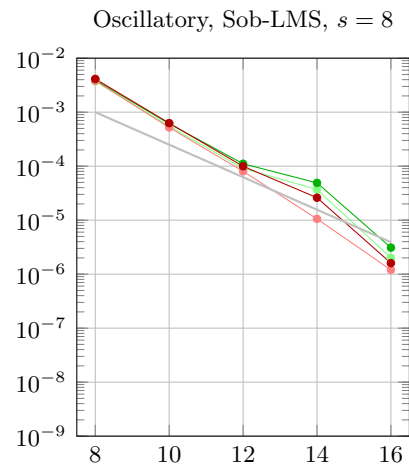
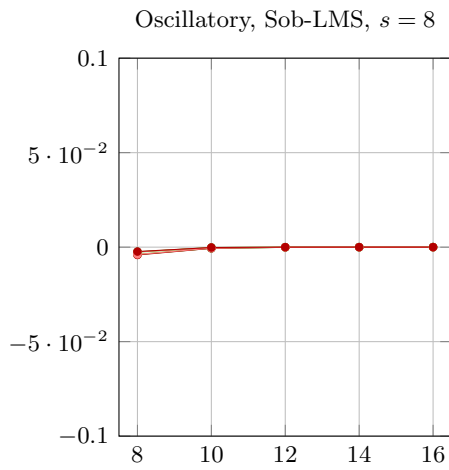
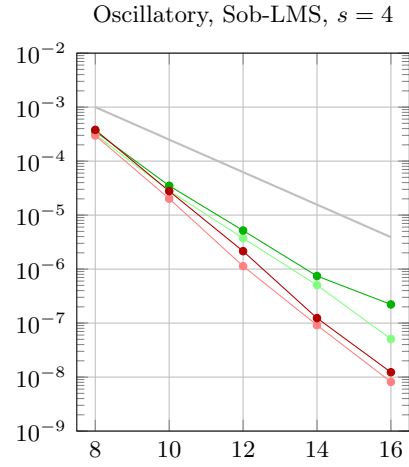
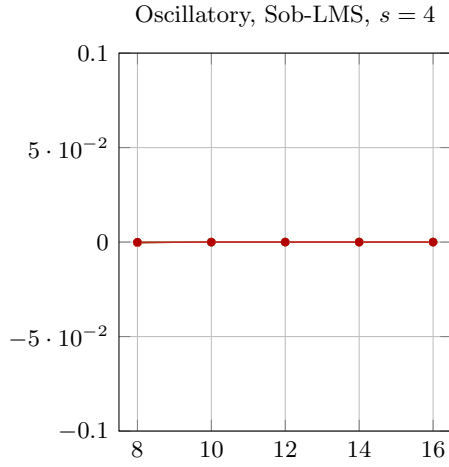




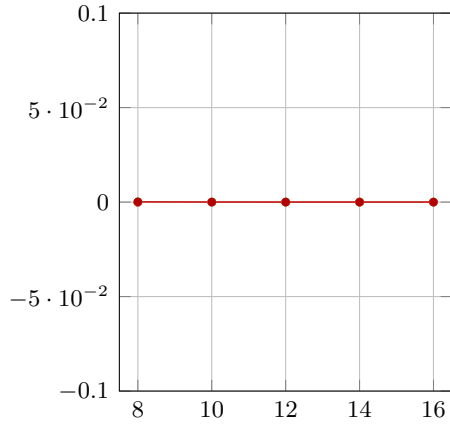




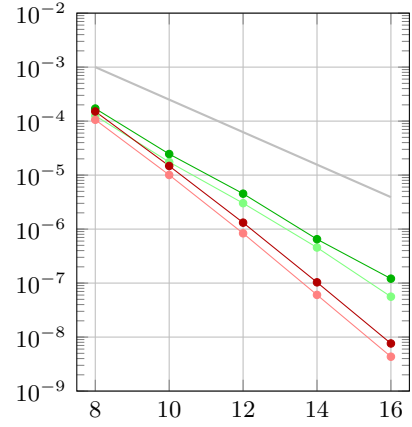




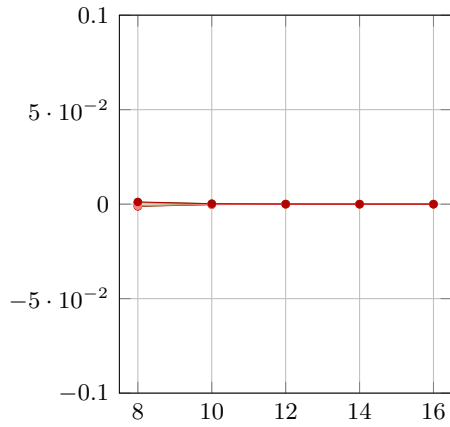
Oscillatory, Sob-LMS-RDS, $s = 4$



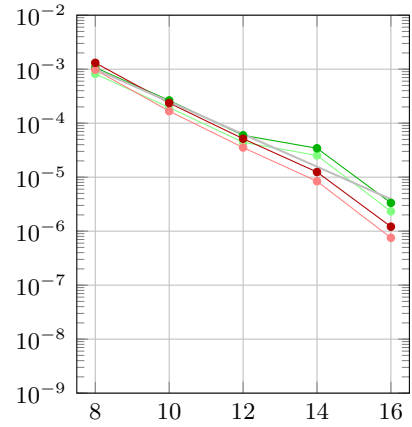
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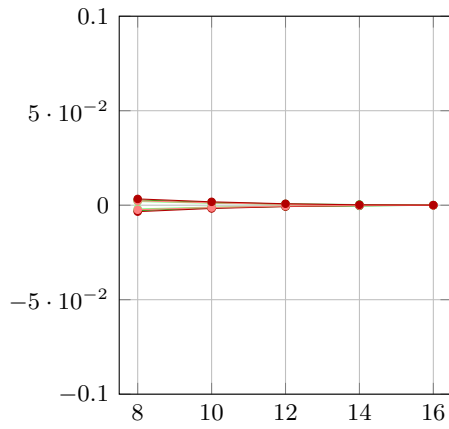
Oscillatory, Sob-LMS-RDS, $s = 8$



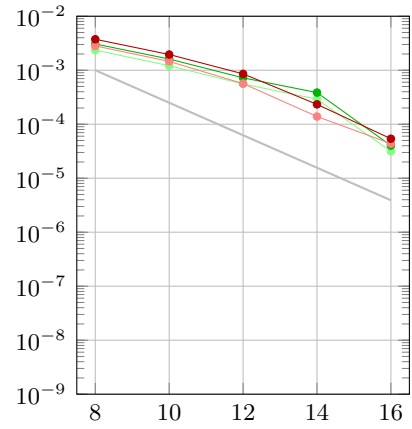
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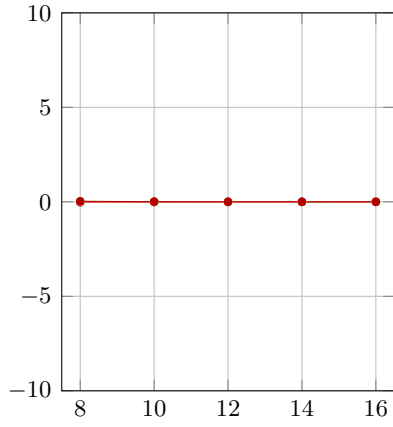
Oscillatory, Sob-LMS-RDS, $s = 16$



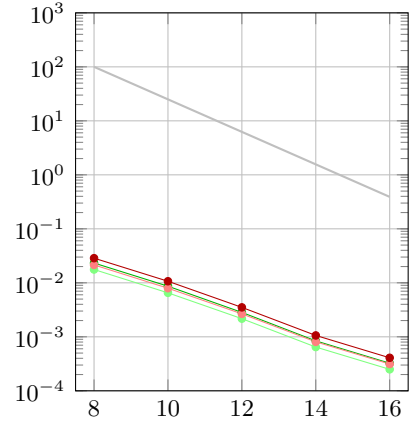
Oscillatory, Sob-LMS-RDS, $s = 16$



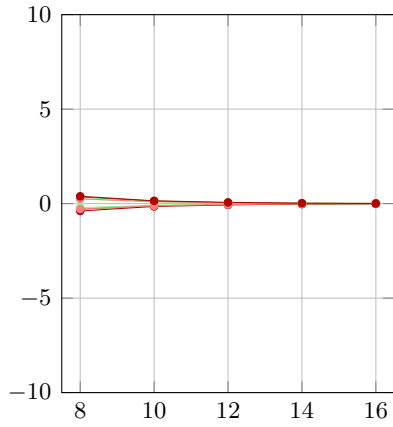
Gaussian, Lat-RS, $s = 4$



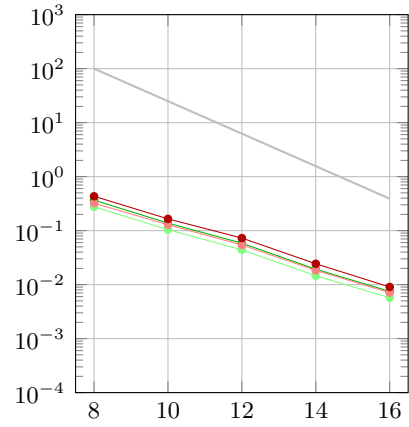
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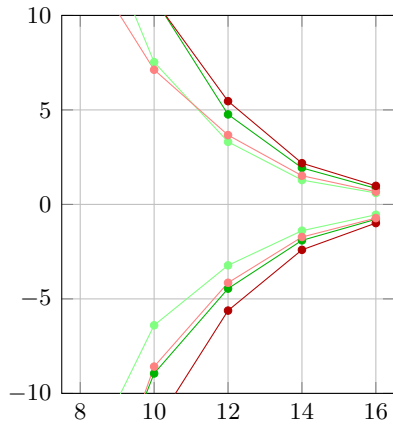
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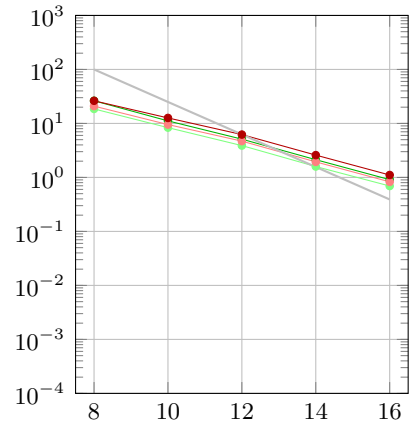
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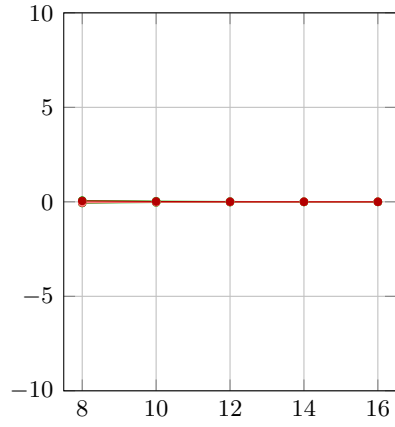
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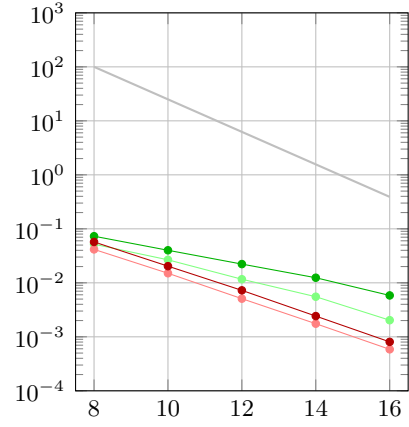
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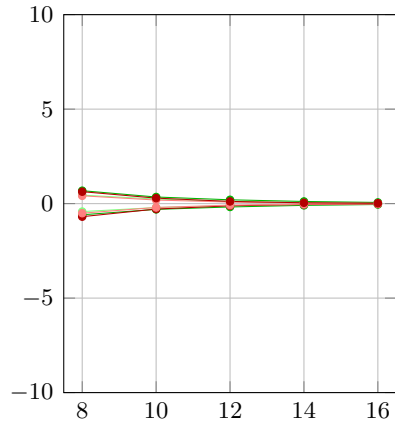
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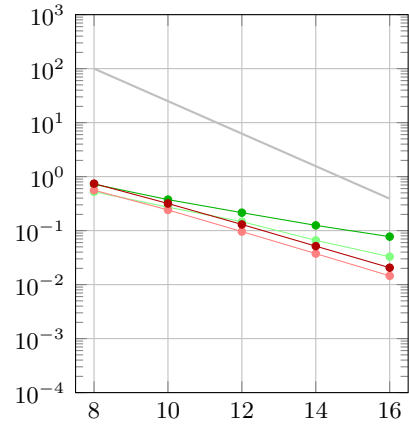
Gaussian, Lat-RvRS, $s = 4$



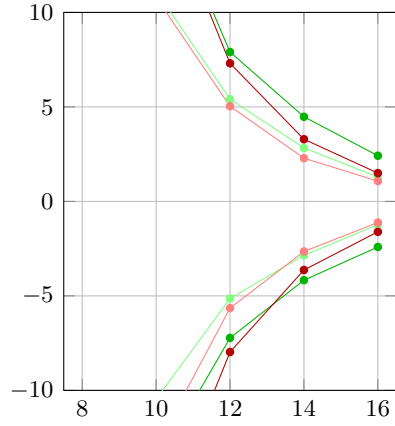
Gaussian, Lat-RvRS, $s = 8$



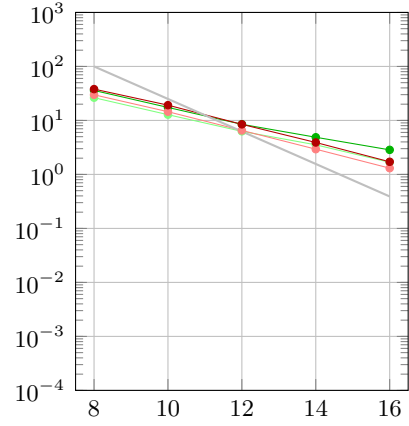
Gaussian, Lat-RvRS, $s = 8$

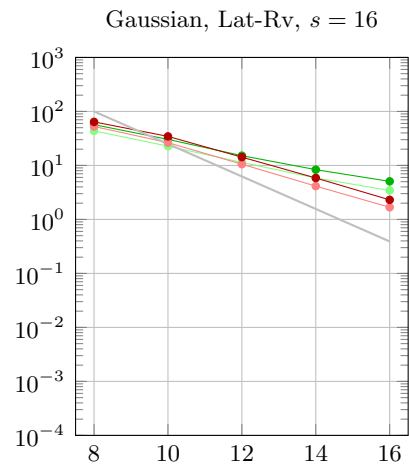
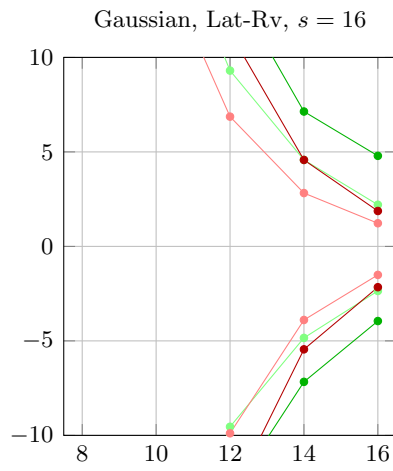
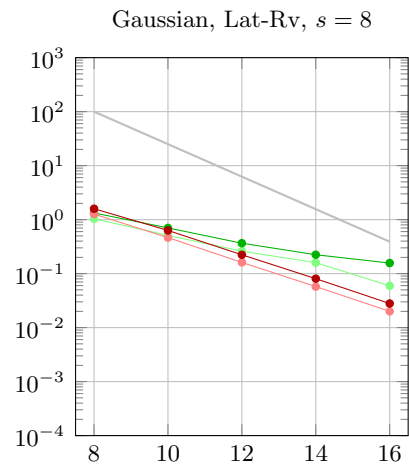
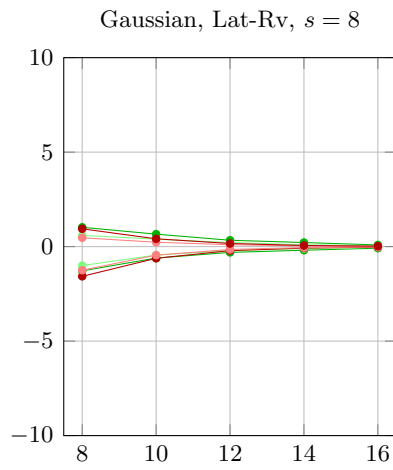
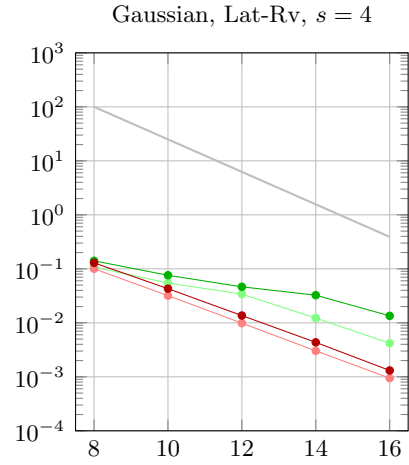
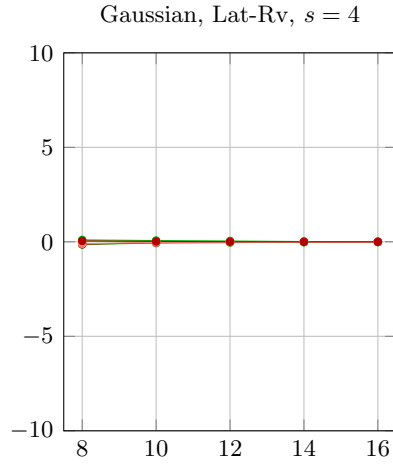


Gaussian, Lat-RvRS, $s = 16$

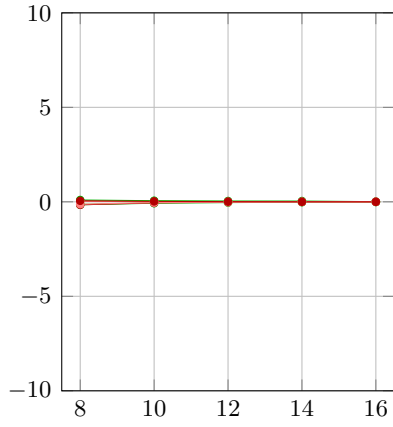


Gaussian, Lat-RvRS, $s = 16$

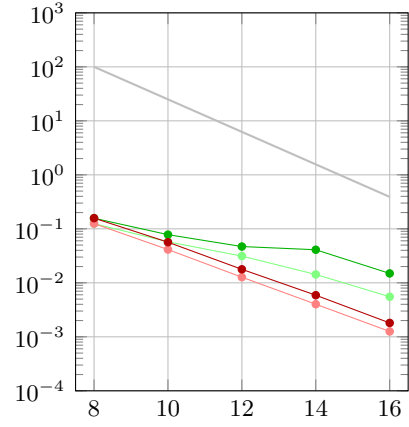




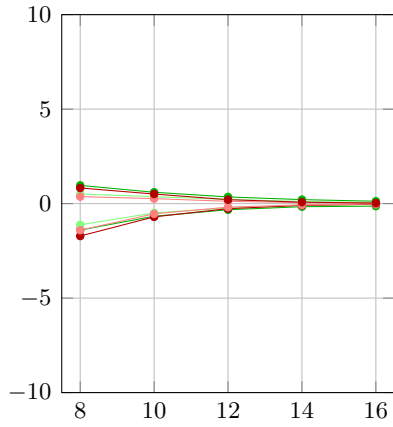
Gaussian, Lat-Rpv, $s = 4$



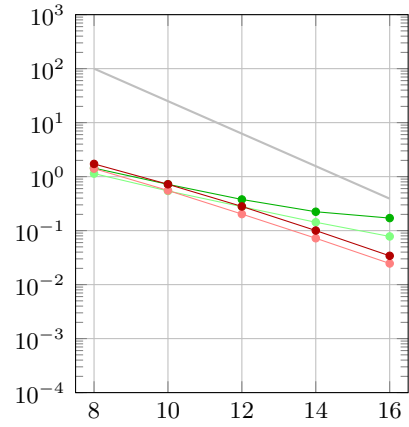
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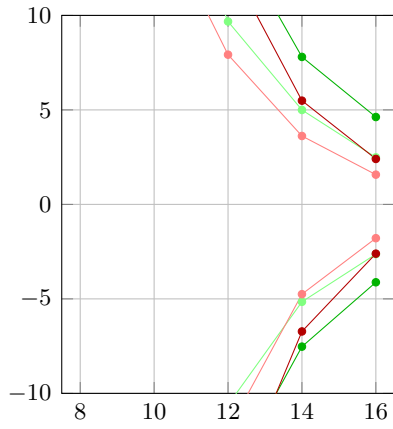
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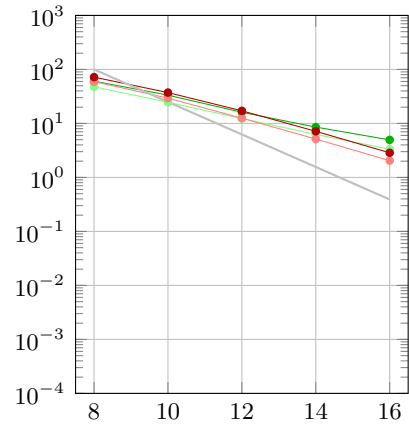
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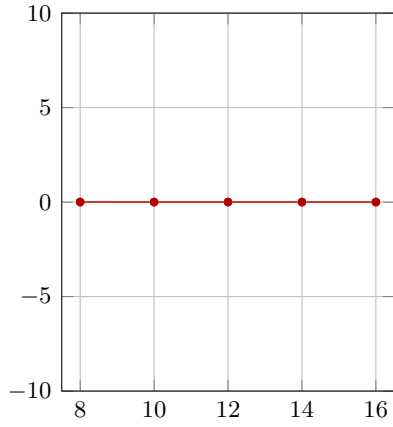
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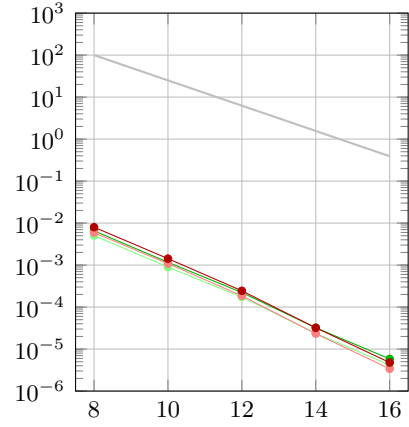
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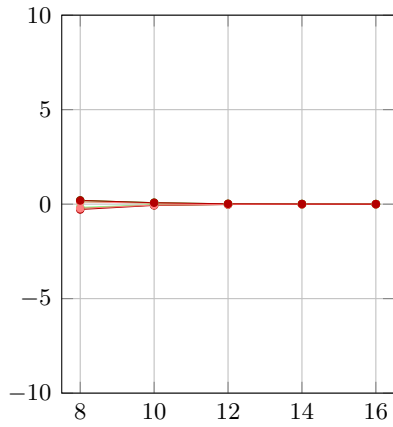
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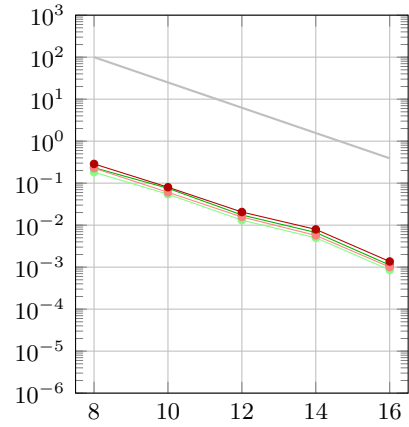
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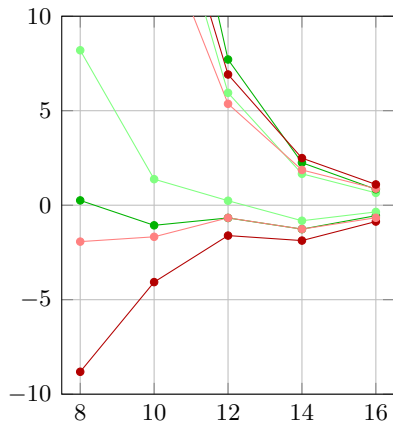
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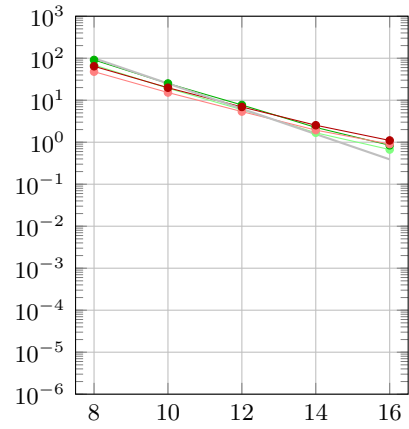
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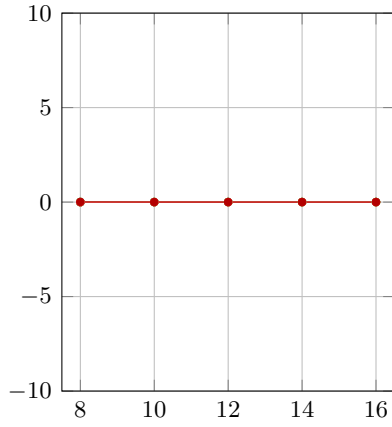
Gaussian, Sob-LMS, $s = 16$



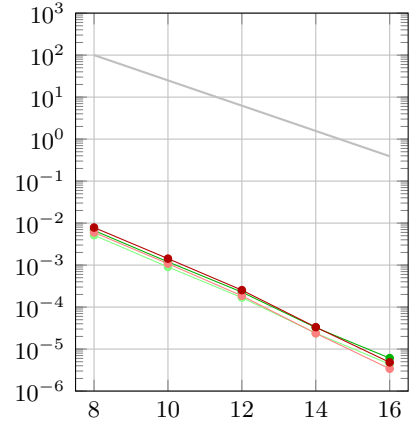
Gaussian, Sob-LMS, $s = 16$



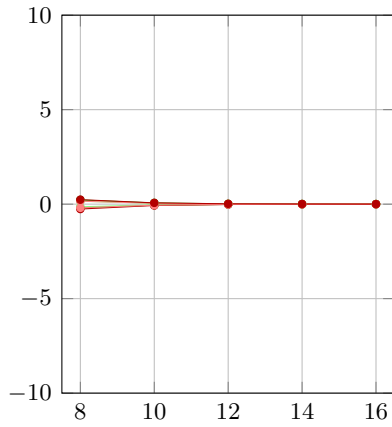
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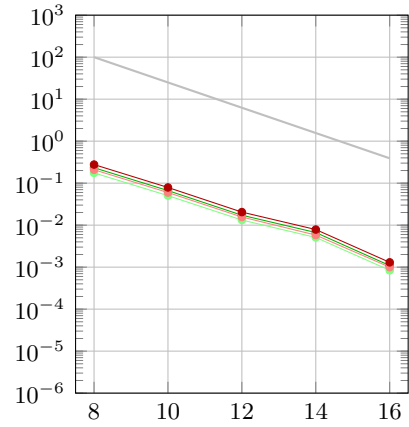
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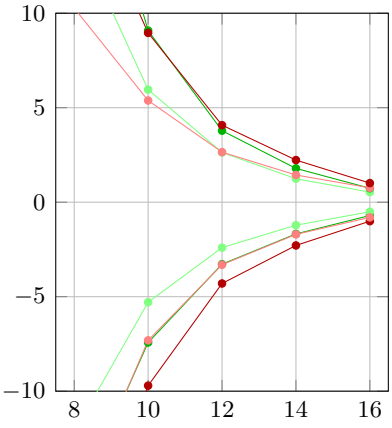
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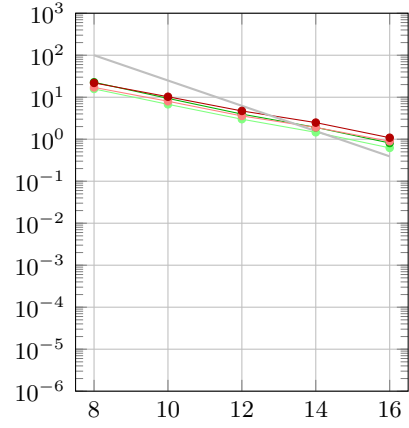
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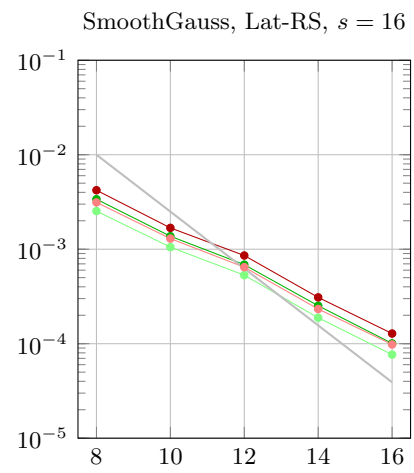
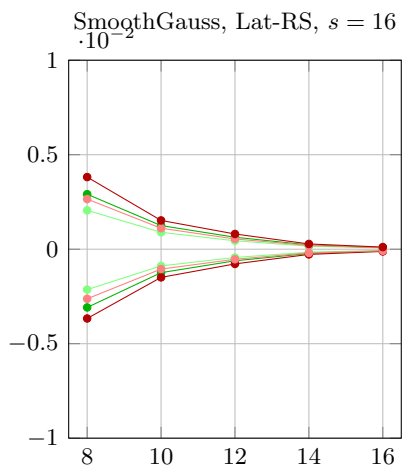
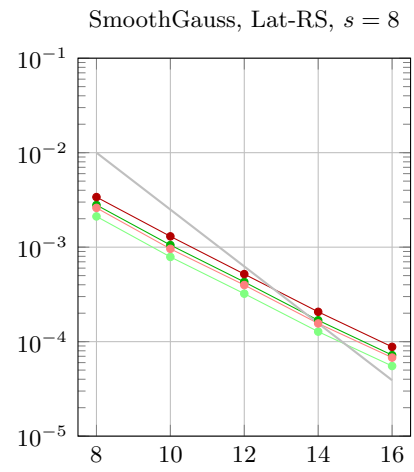
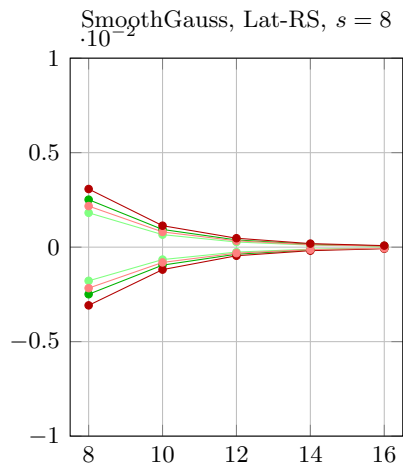
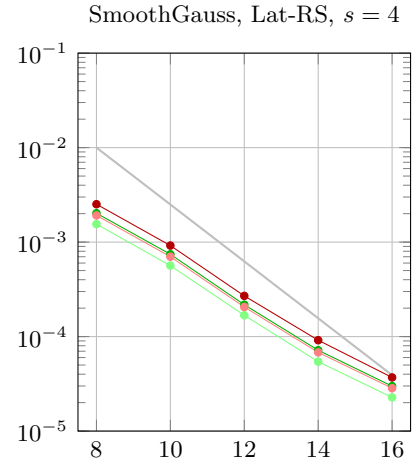
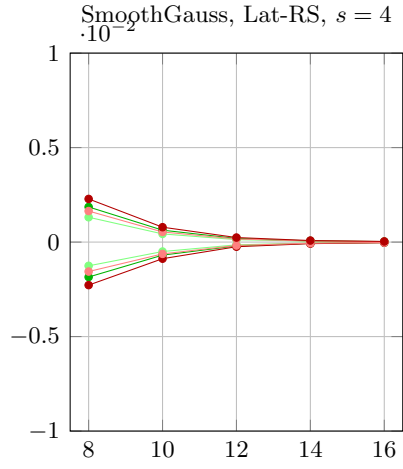


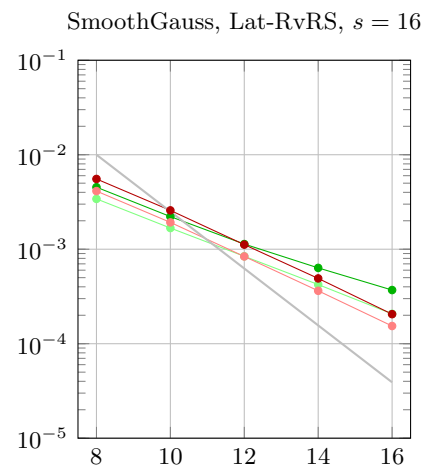
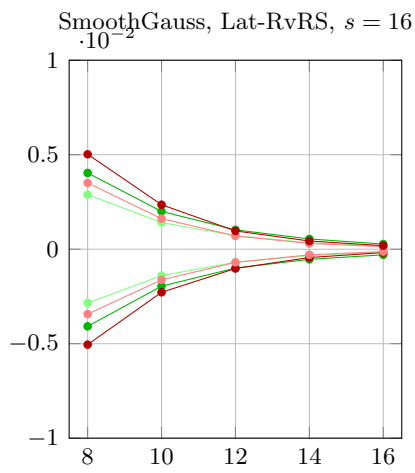
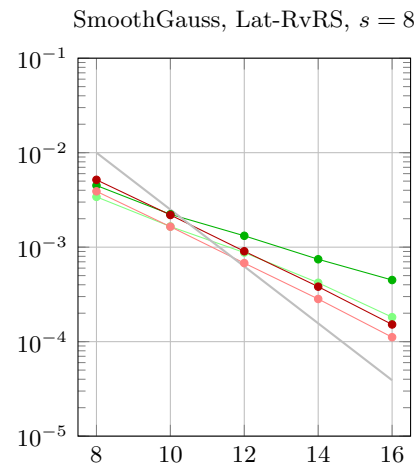
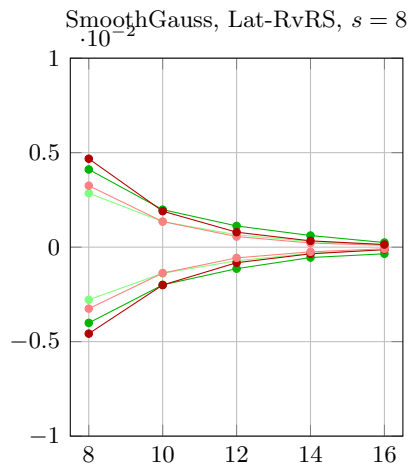
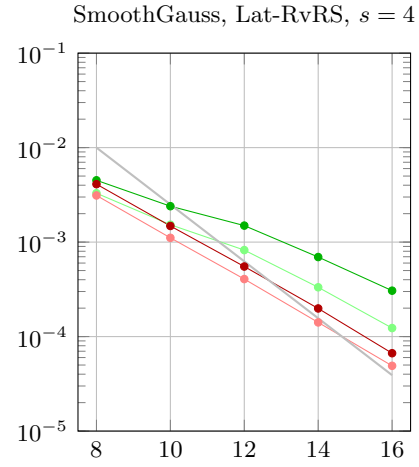
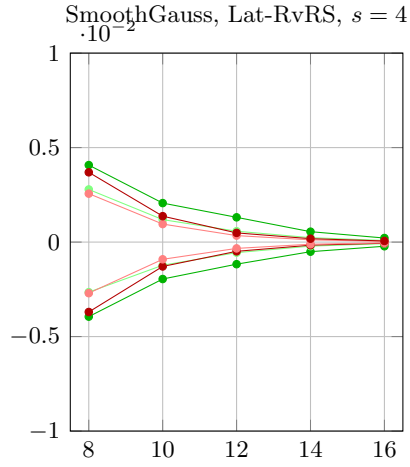
Gaussian, Sob-LMS-RDS, $s = 16$

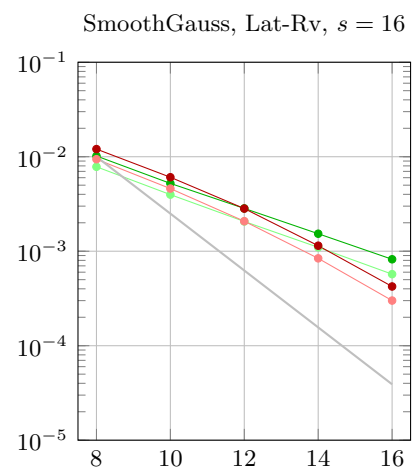
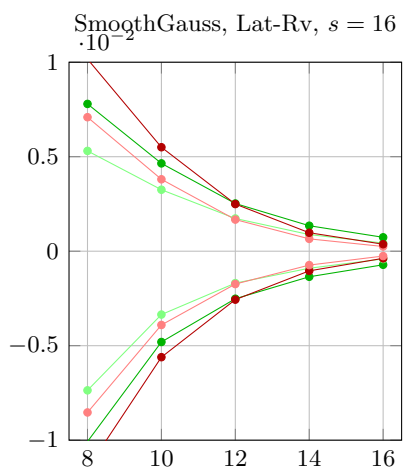
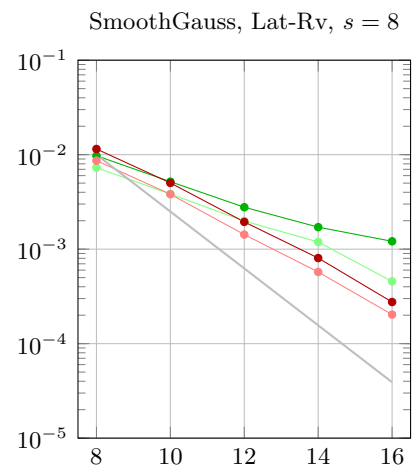
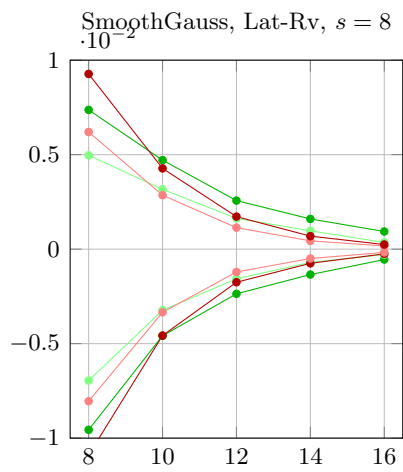
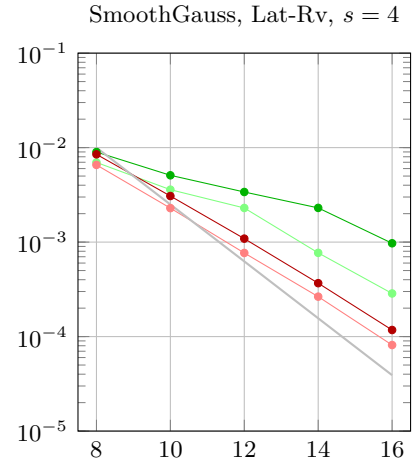
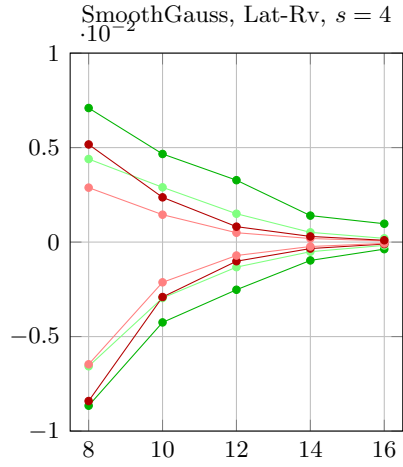


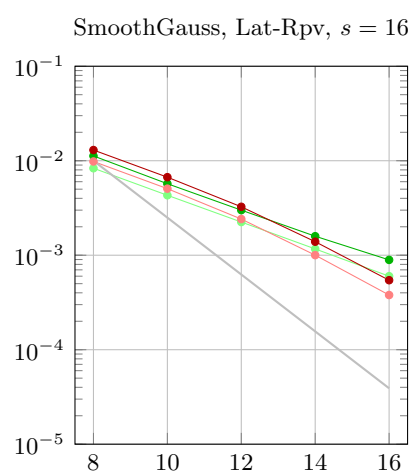
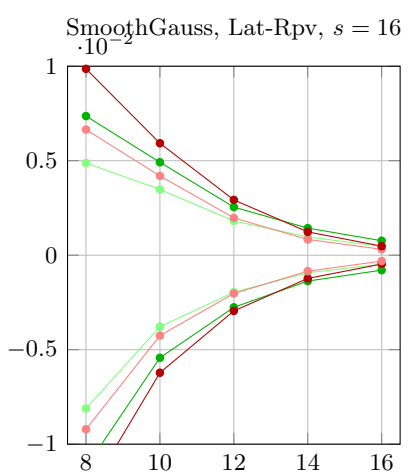
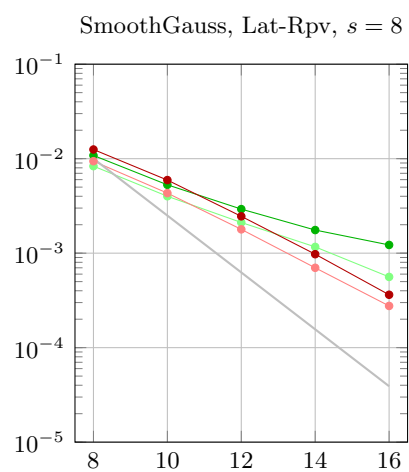
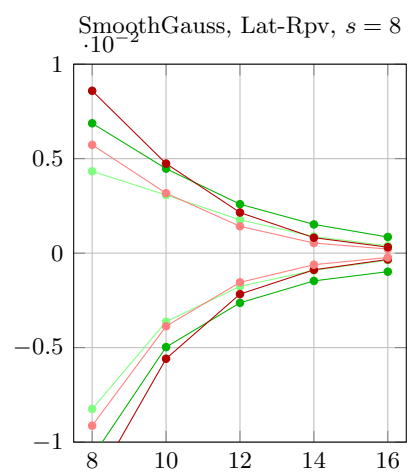
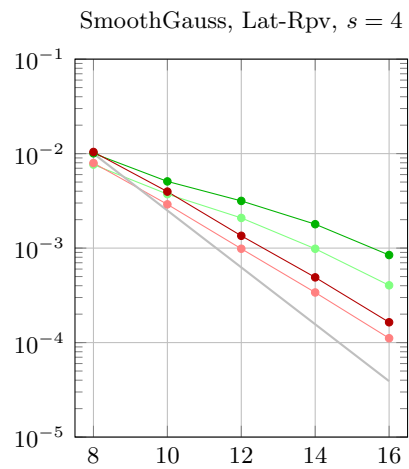
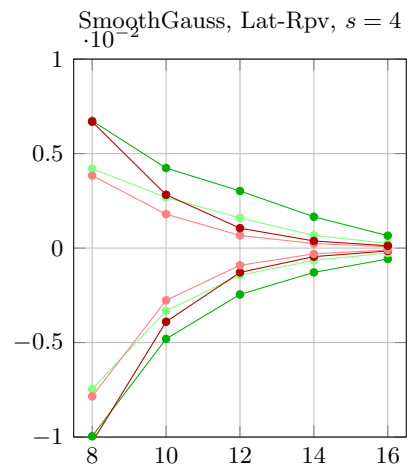
Gaussian, Sob-LMS-RDS, $s = 16$

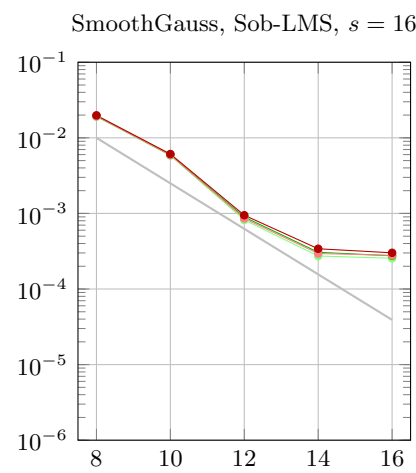
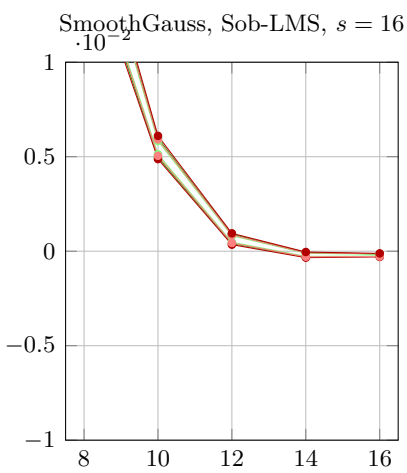
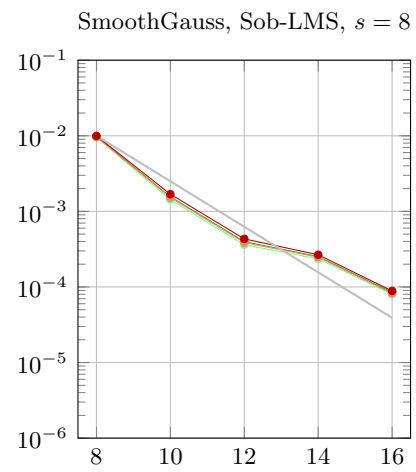
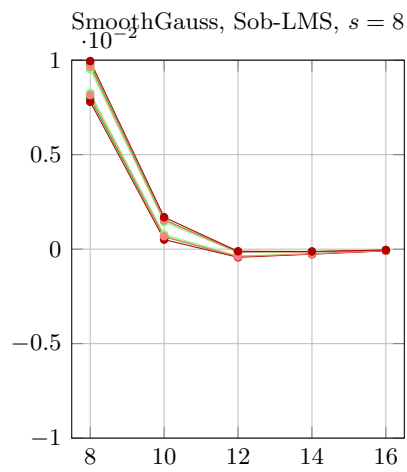
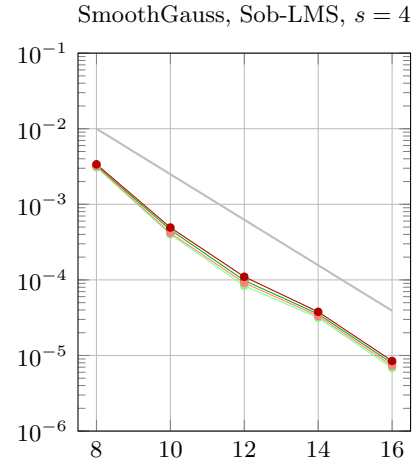
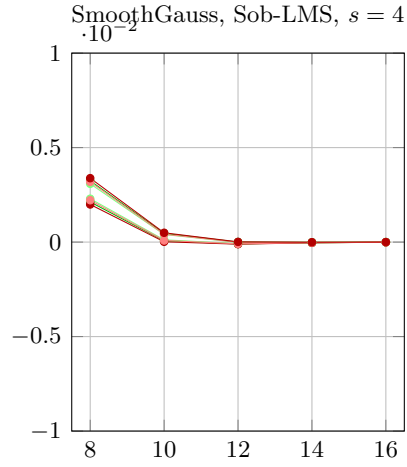




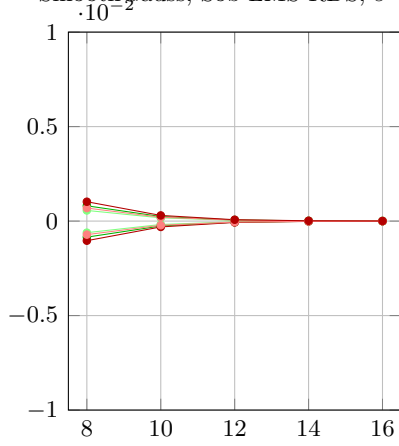




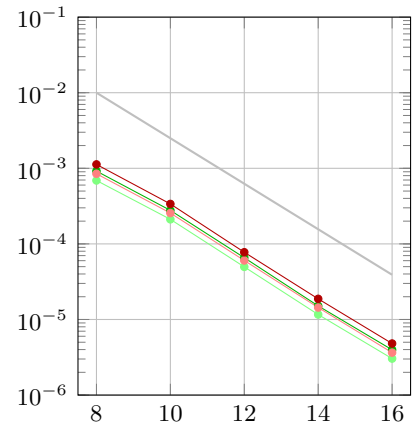




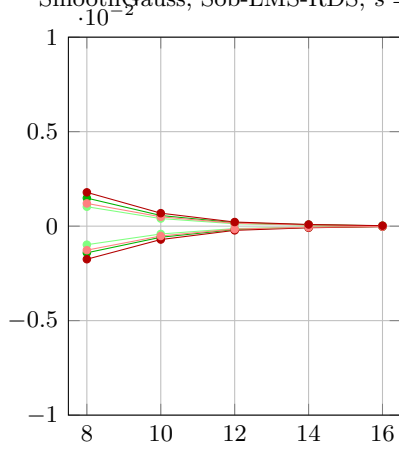
SmoothGauss, Sob-LMS-RDS, $s = 4$



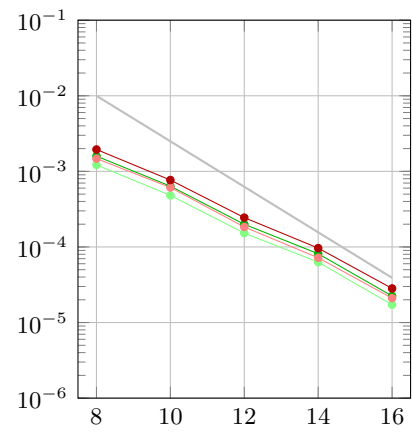
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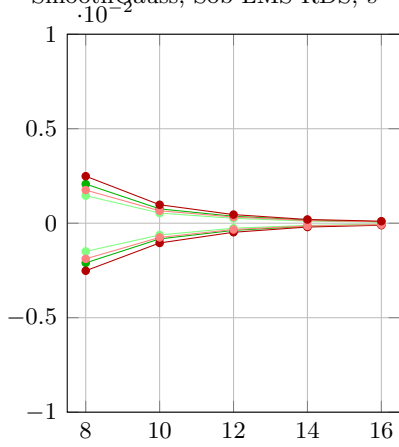
SmoothGauss, Sob-LMS-RDS, $s = 8$



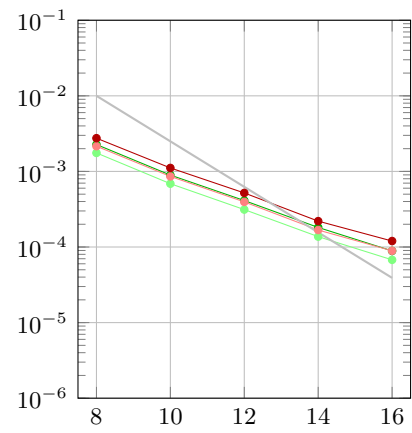
SmoothGauss, Sob-LMS-RDS, $s = 8$

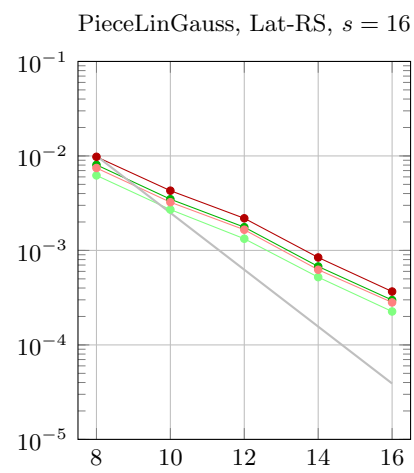
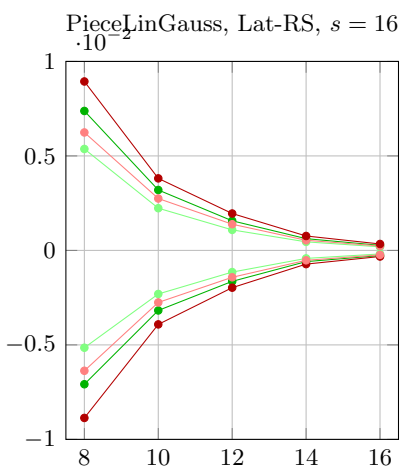
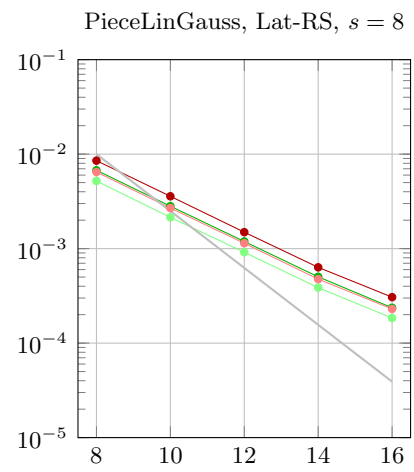
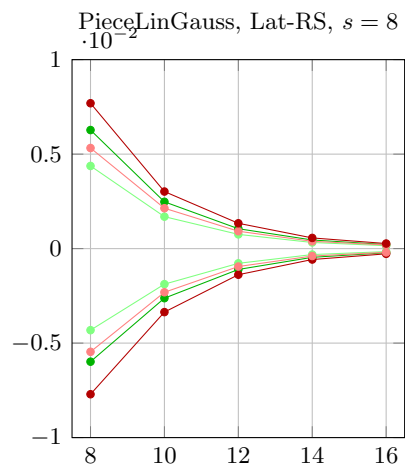
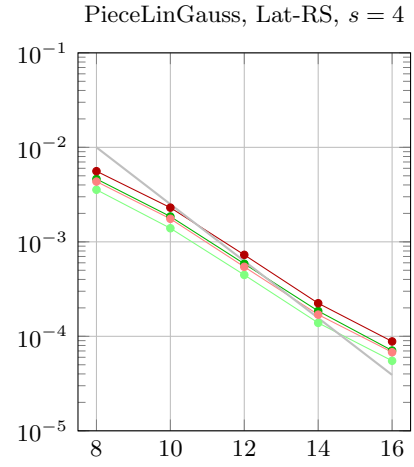
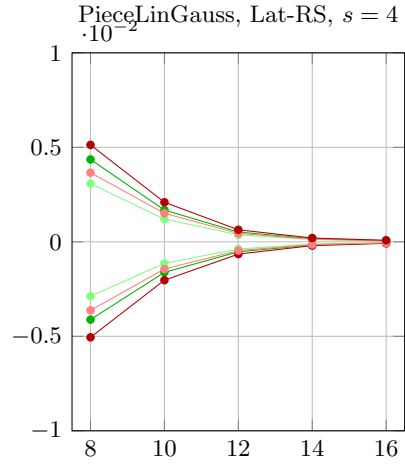


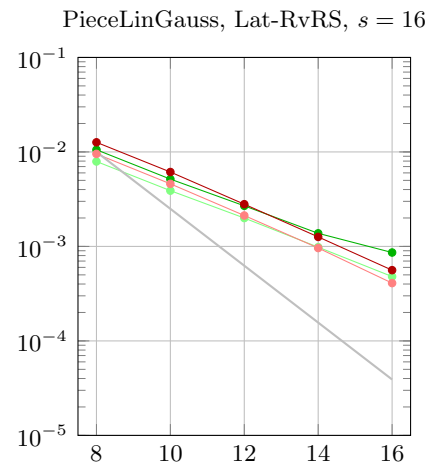
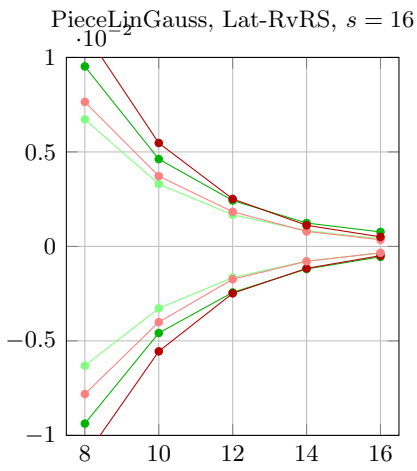
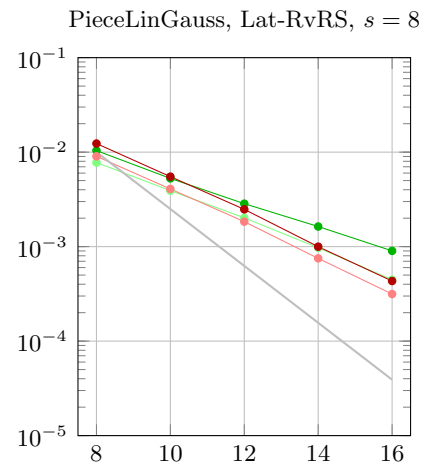
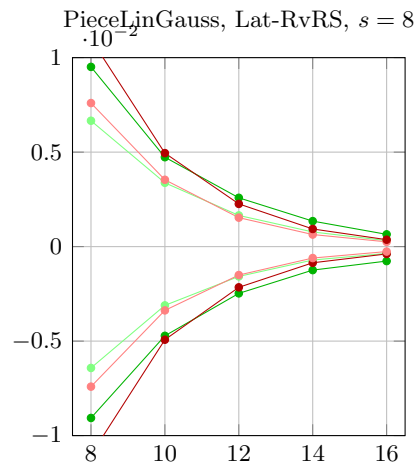
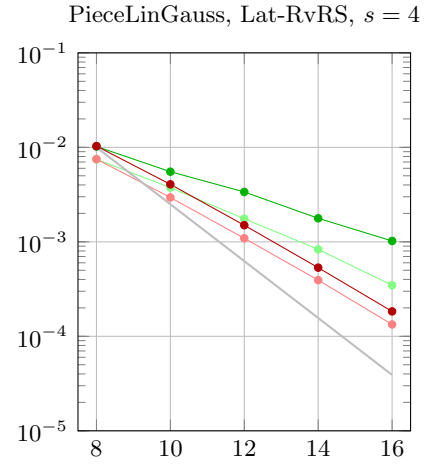
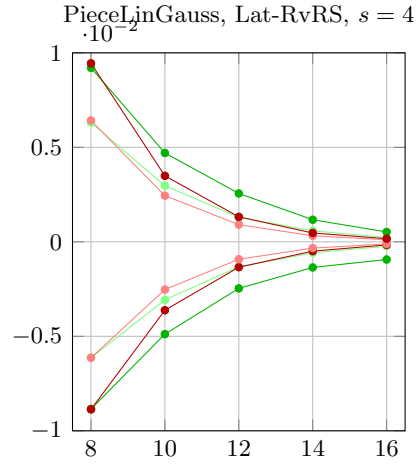
SmoothGauss, Sob-LMS-RDS, $s = 16$

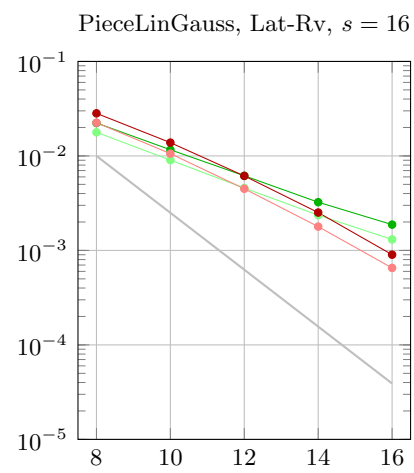
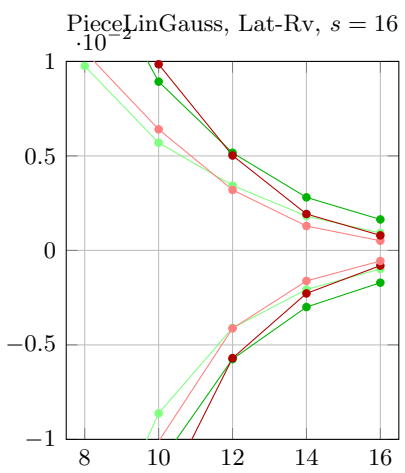
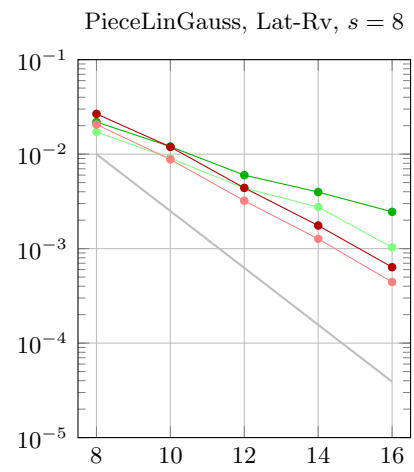
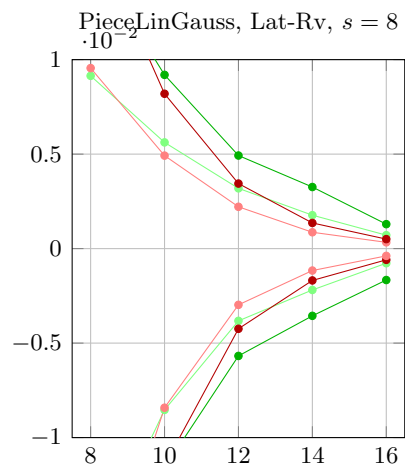
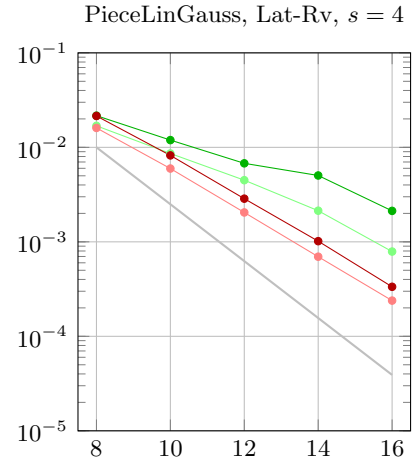
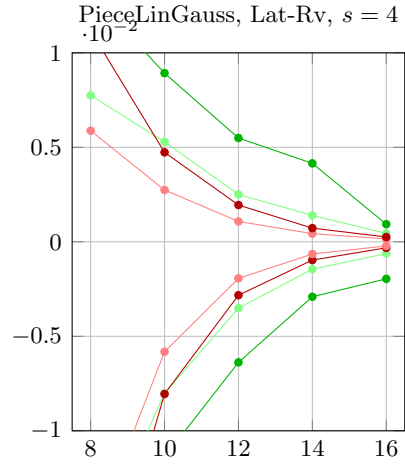


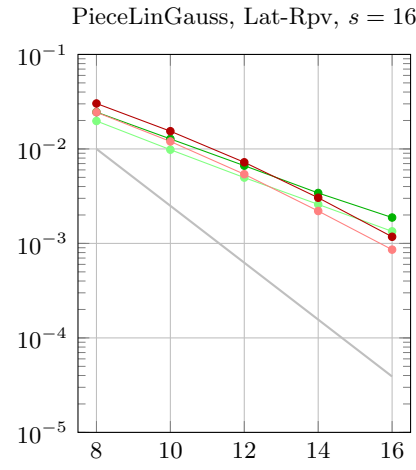
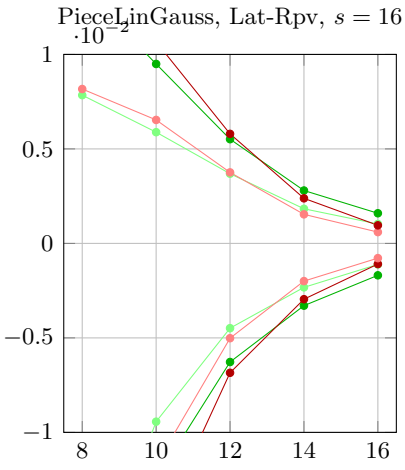
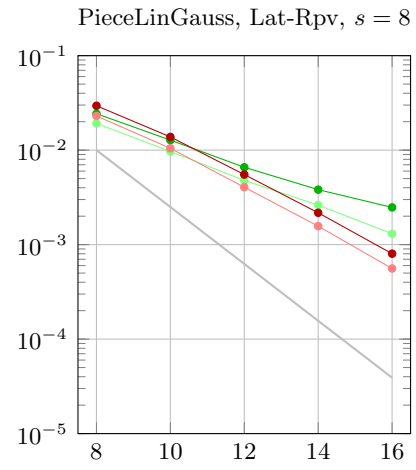
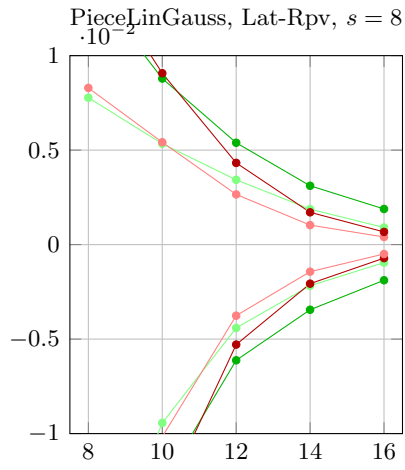
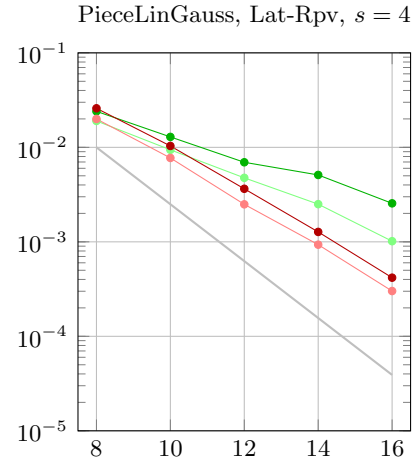
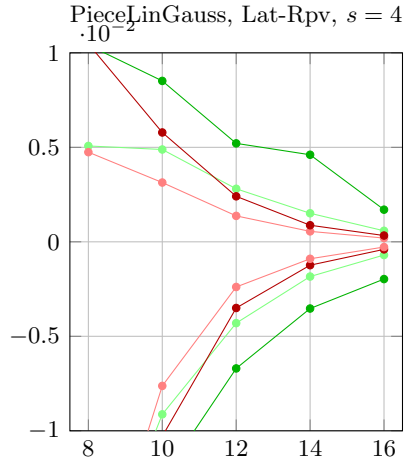
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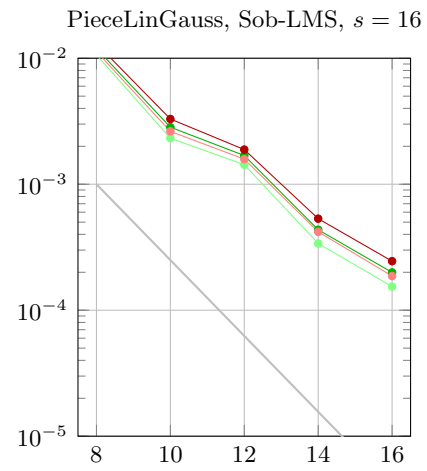
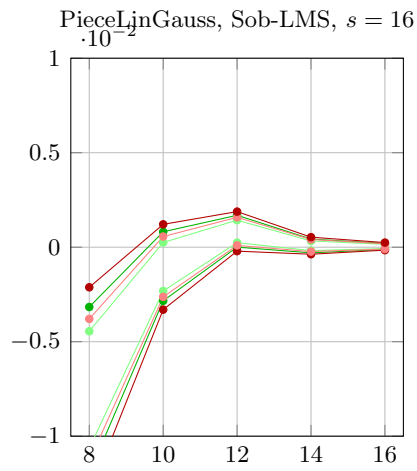
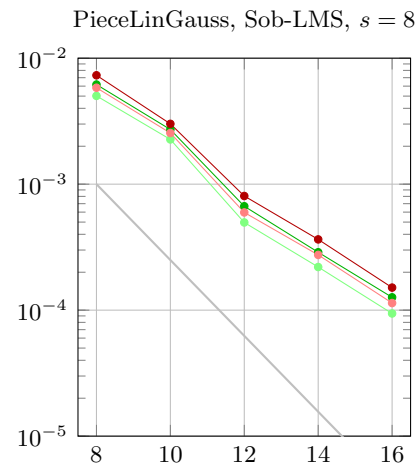
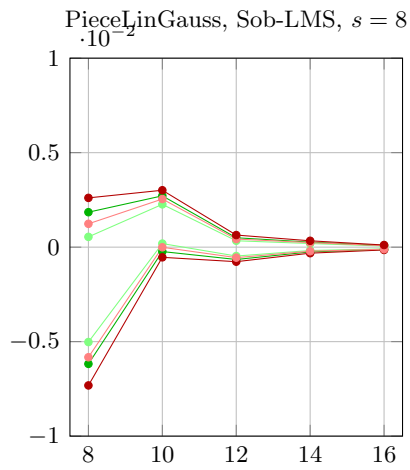
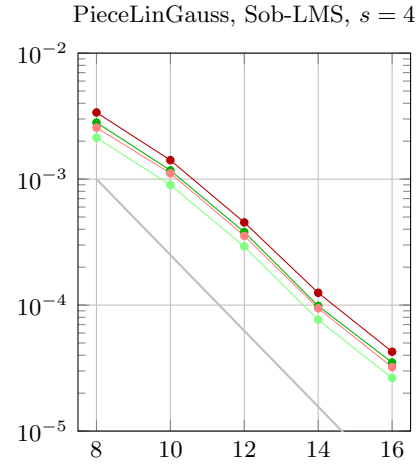
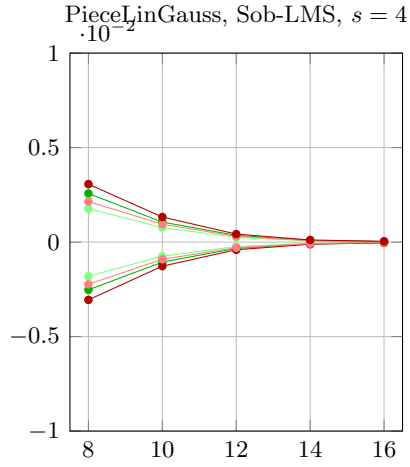


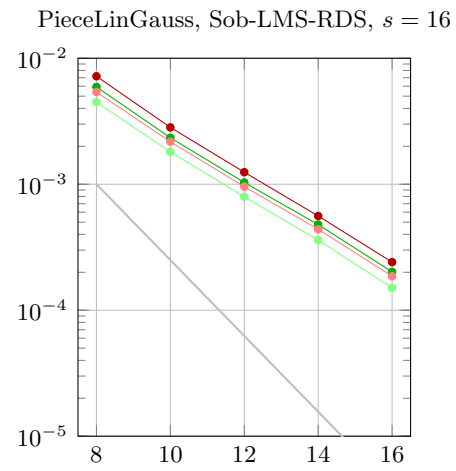
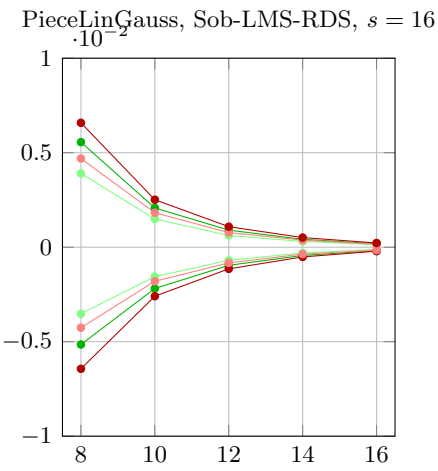
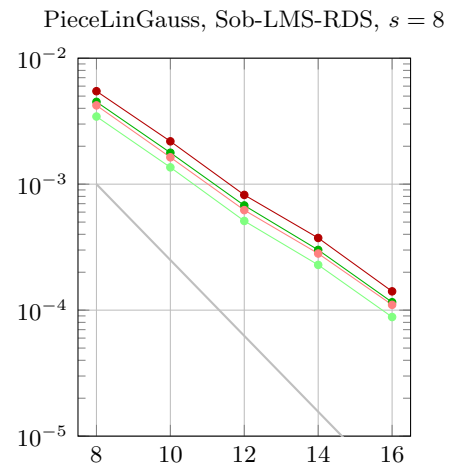
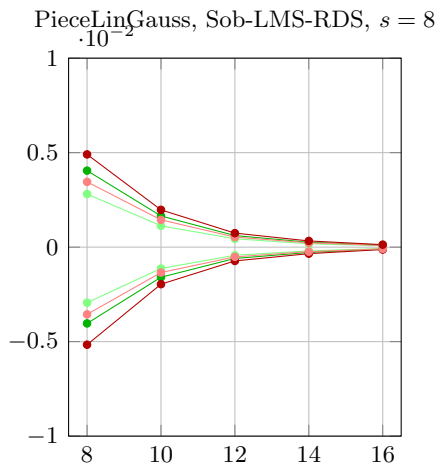
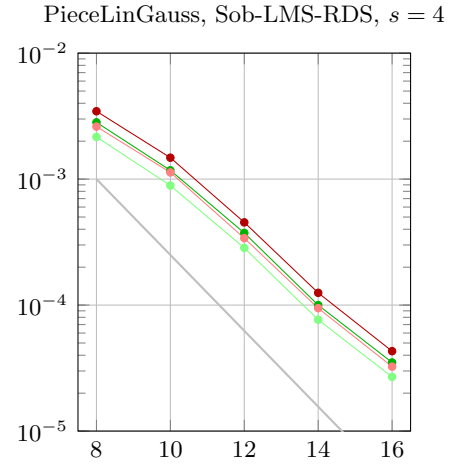
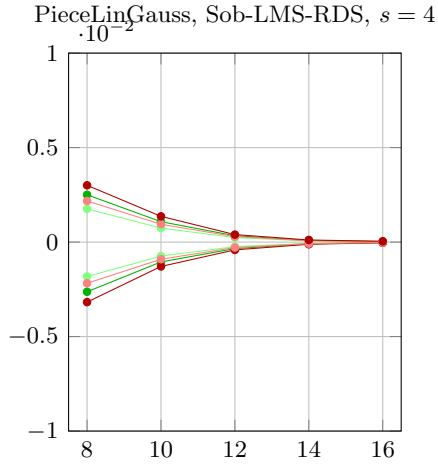


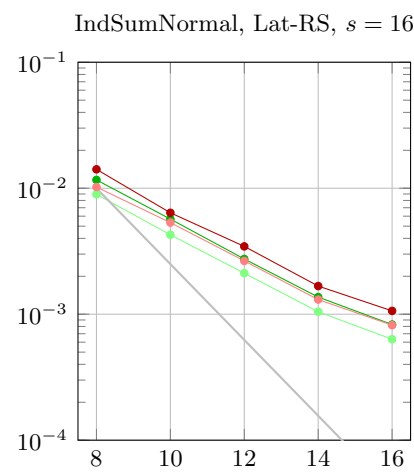
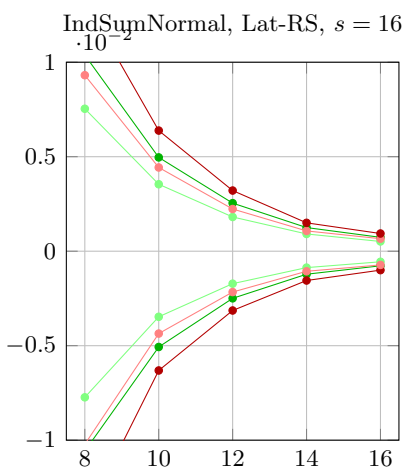
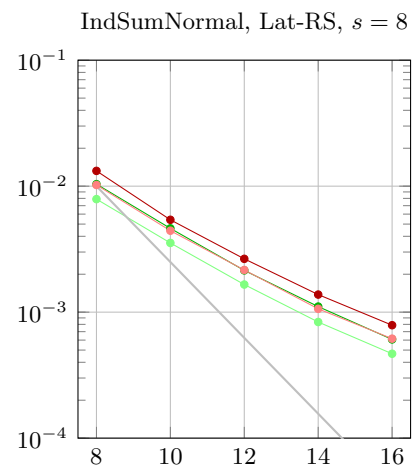
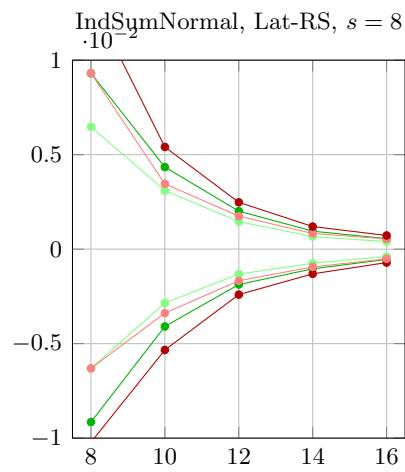
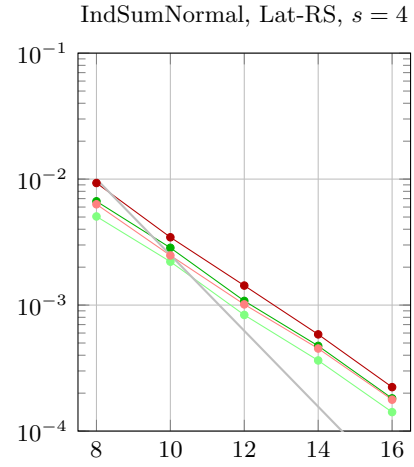
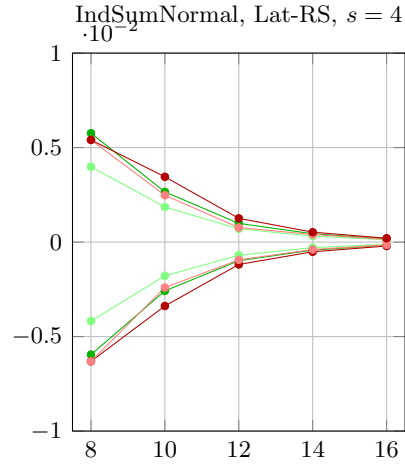


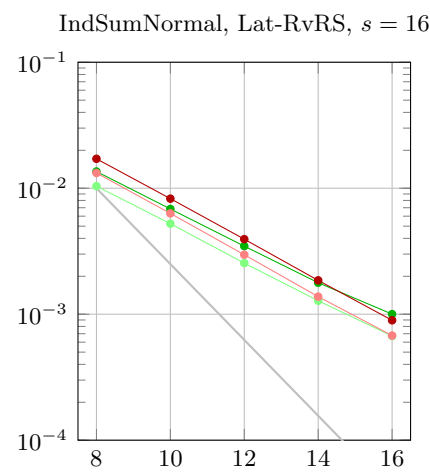
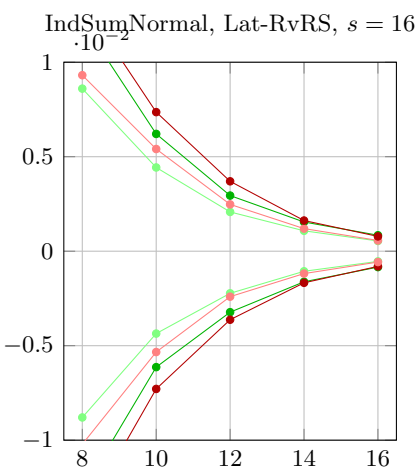
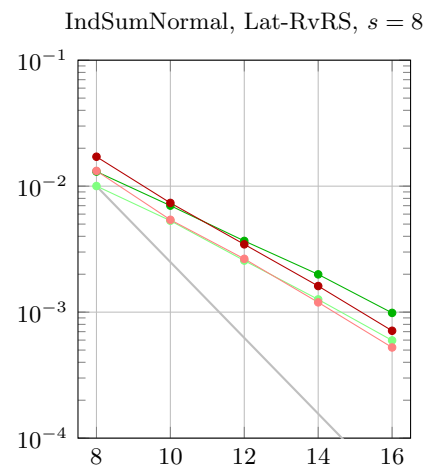
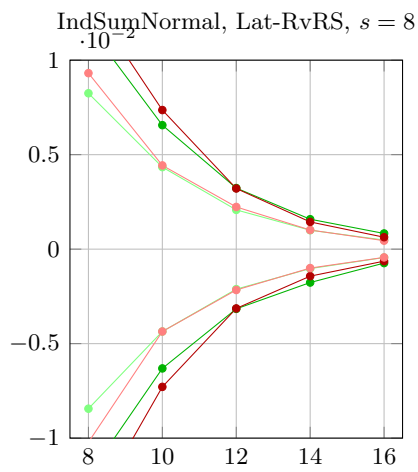
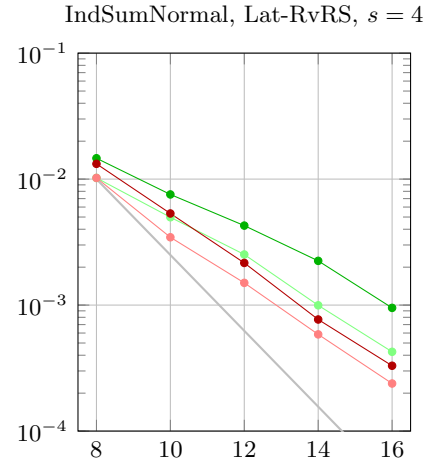
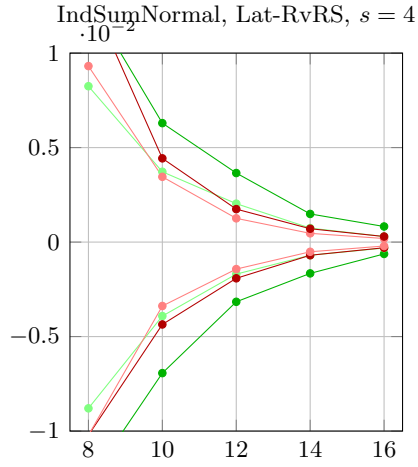


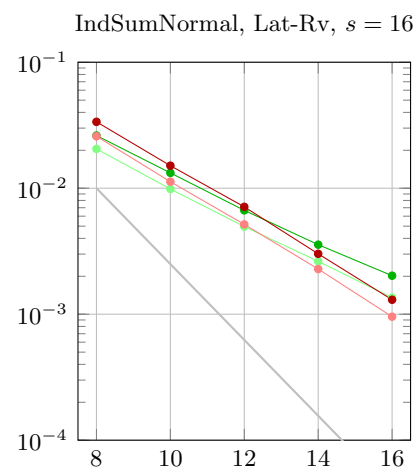
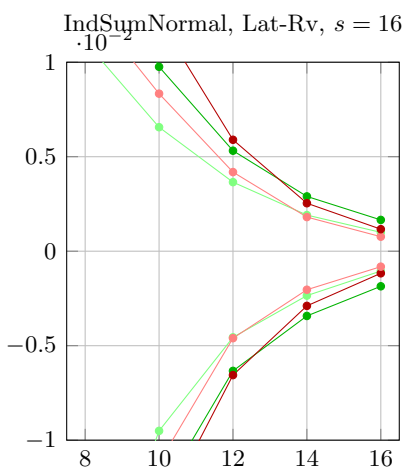
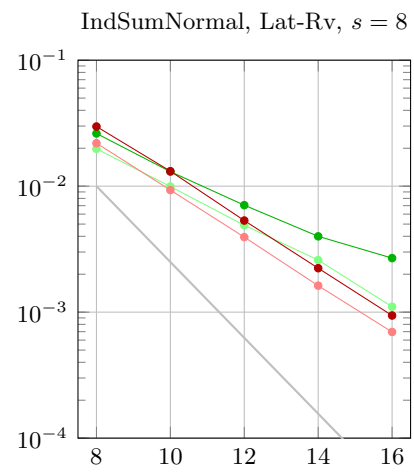
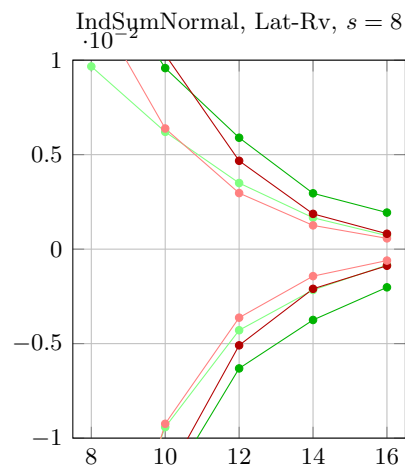
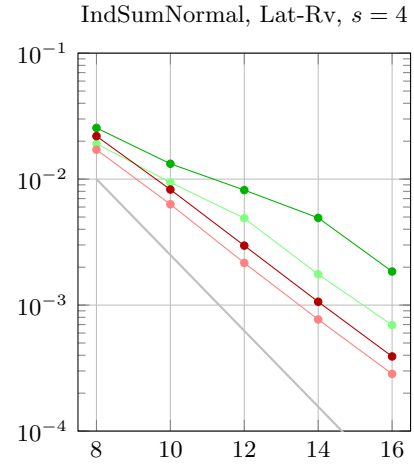
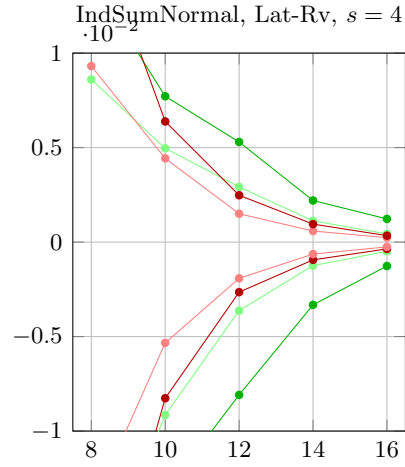


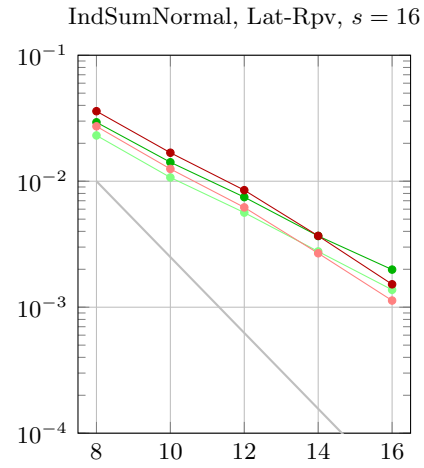
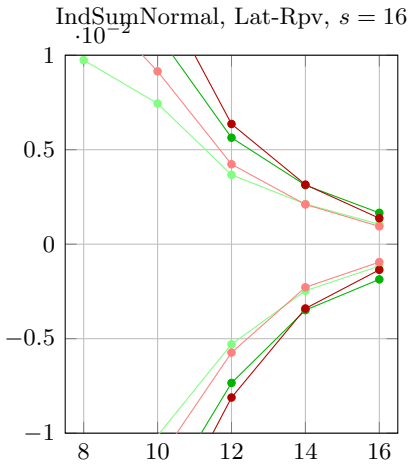
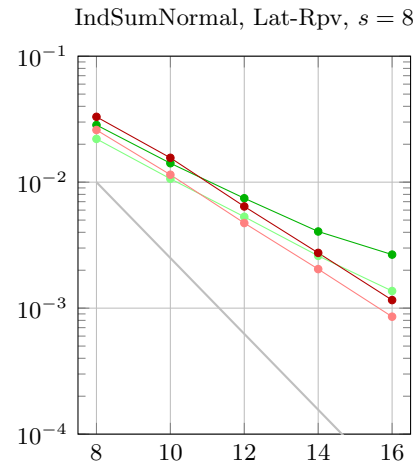
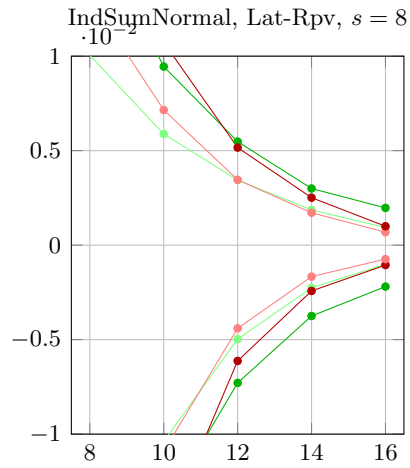
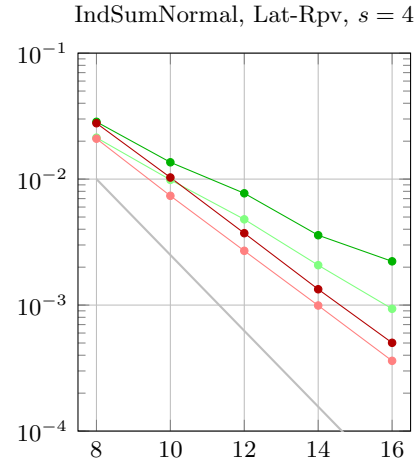
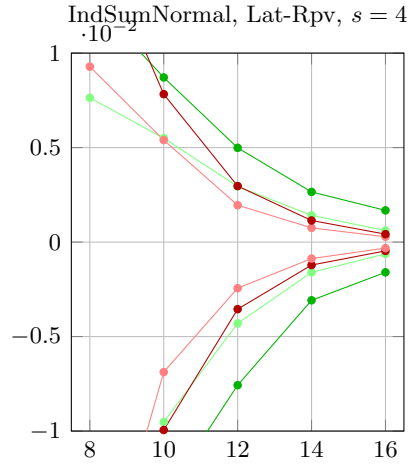


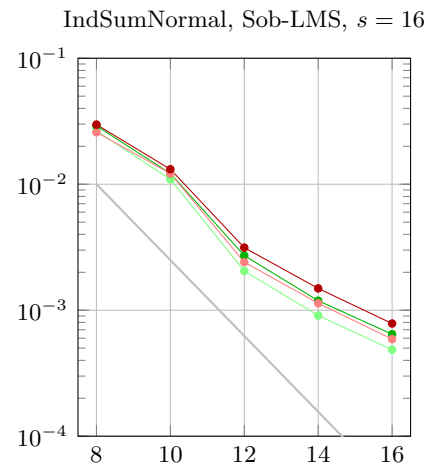
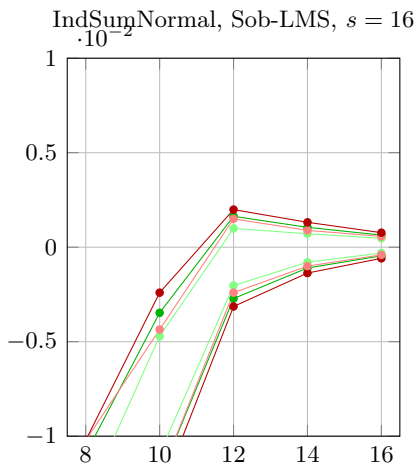
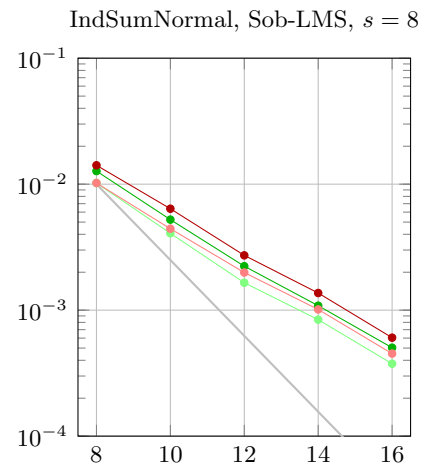
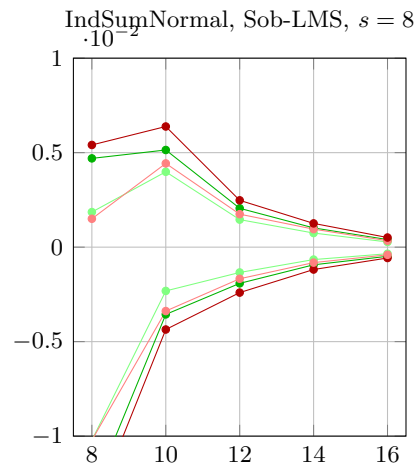
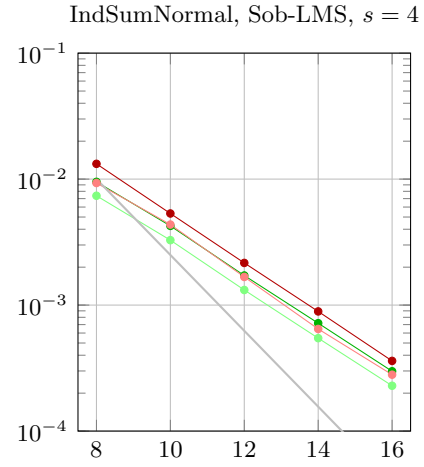
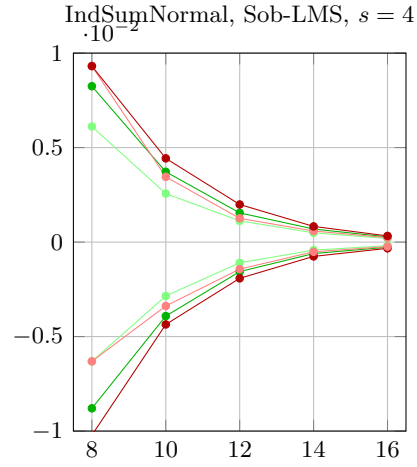


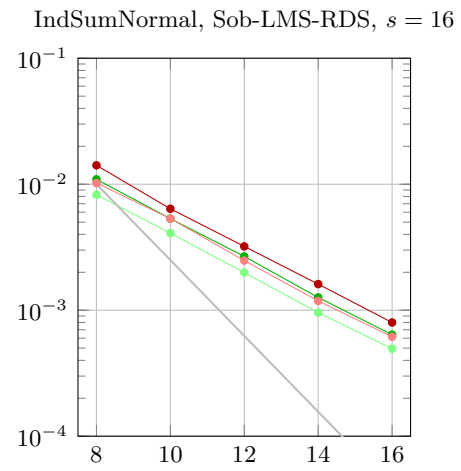
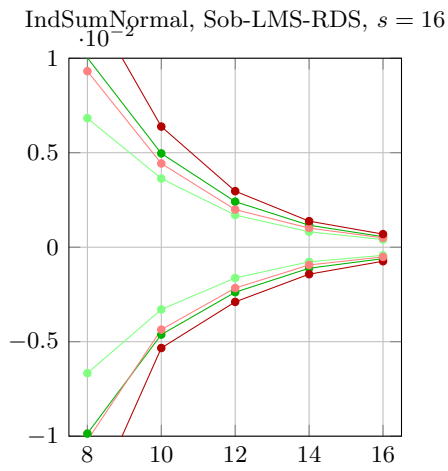
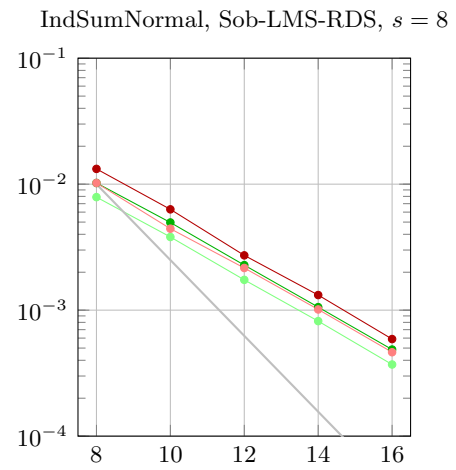
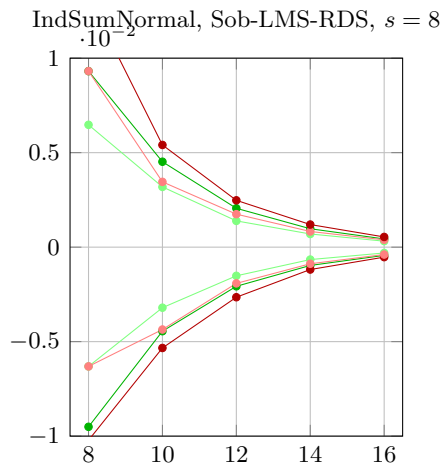
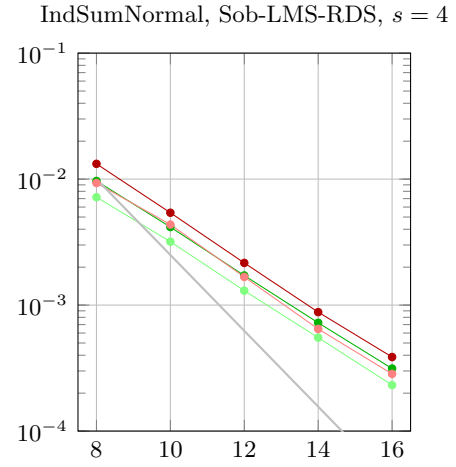
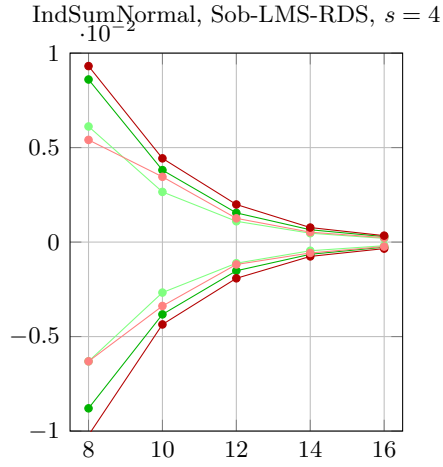












4 More Scatter Plots of Skewness vs Kurtosis, and MSE Ratio

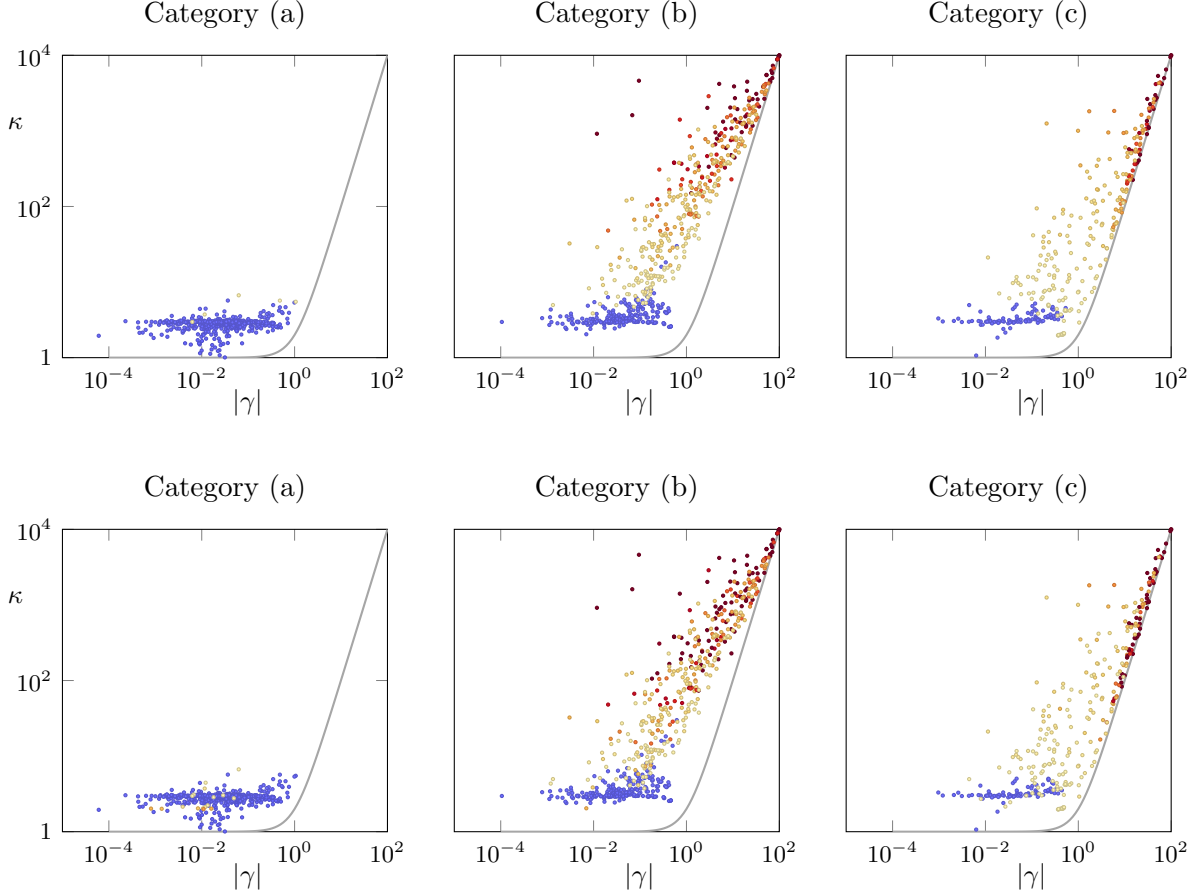


Figure 4: Plots in log scale of the kurtosis κ vs the absolute skewness $|\gamma|$ of X for the three categories of methods: (a) fixed-parameter RQMC (left); (b) random-parameter RQMC (center); (c) random-parameter QMC (right). These plots are restricted to the six models considered in Seljak et al. (2026), in dimensions 2, 4, 8, 16, 32. The upper row is for $r = 5$ and the lower row for $r = 63$. The plots for $r = 11$ are given in Seljak et al. (2026). The color of each point depends on the ratio $\rho = \log(\text{MSE}[A_r]/\text{MSE}[M_r])$. When $\rho > 0$, the color goes from light yellow to dark purple when ρ goes from 0 to 5, and is dark purple above 5. For $\rho < 0$, the color goes from light blue to dark blue when ρ goes from 0 to -0.4 . The points are at the same positions for all values of r , but they may change color. We tend to have slightly more dark purple points when r gets smaller.

Figure 5 shows separate scatter plots for the **PieceLinGauss** and **IndSumNormal**, which are not smooth and represent the types of functions typically encountered in computational finance. The behavior is somewhat similar except that the kurtosis never exceeds about 1000 and the ratio $\text{MSE}[A_r]/\text{MSE}[M_r]$ is never much larger than 1, i.e., using the median is never much better than using the average for these functions. Figure 6 shows histograms of the values of the ratio for each category, also restricted to a single model at a time.

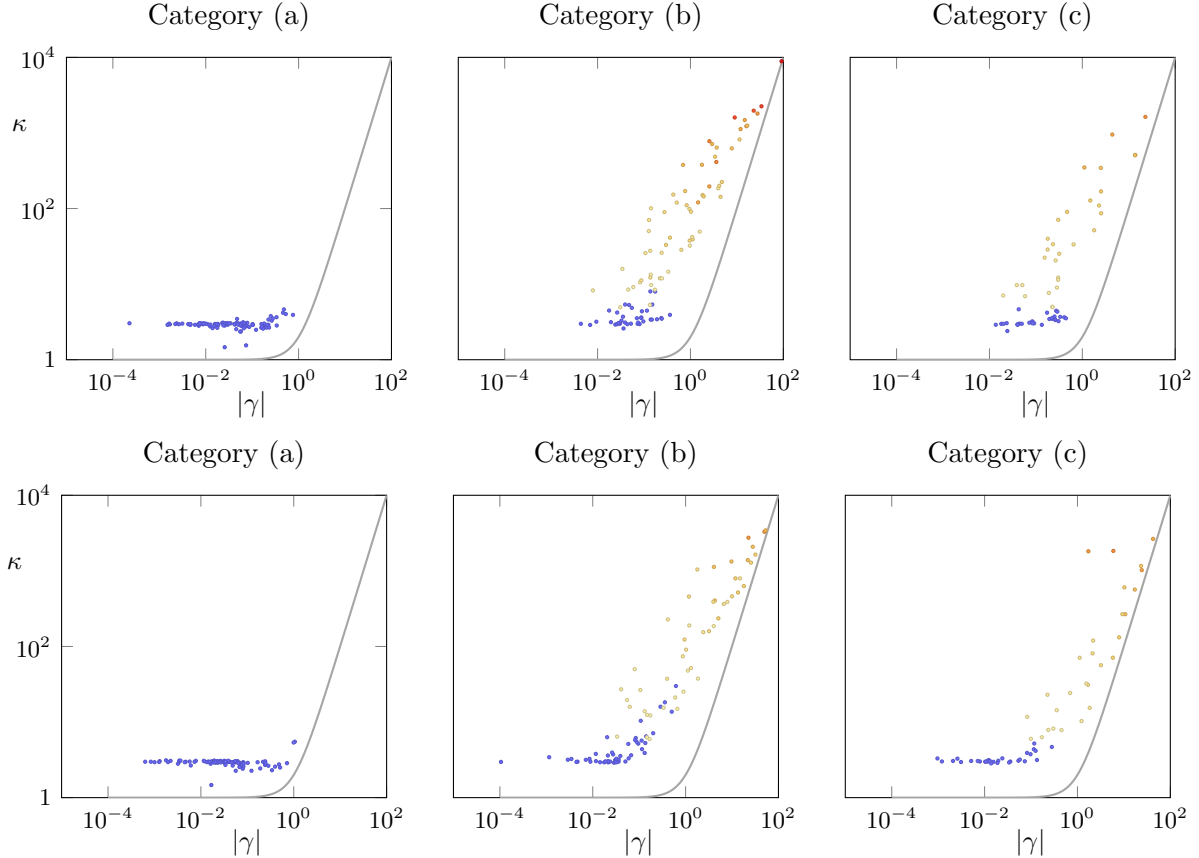


Figure 5: Similar to Figure 4, except that it is restricted to a single model (integrand) at a time, for $r = 11$. The upper row is for **PieceLinGauss** and the lower row is for **IndSumNormal**.

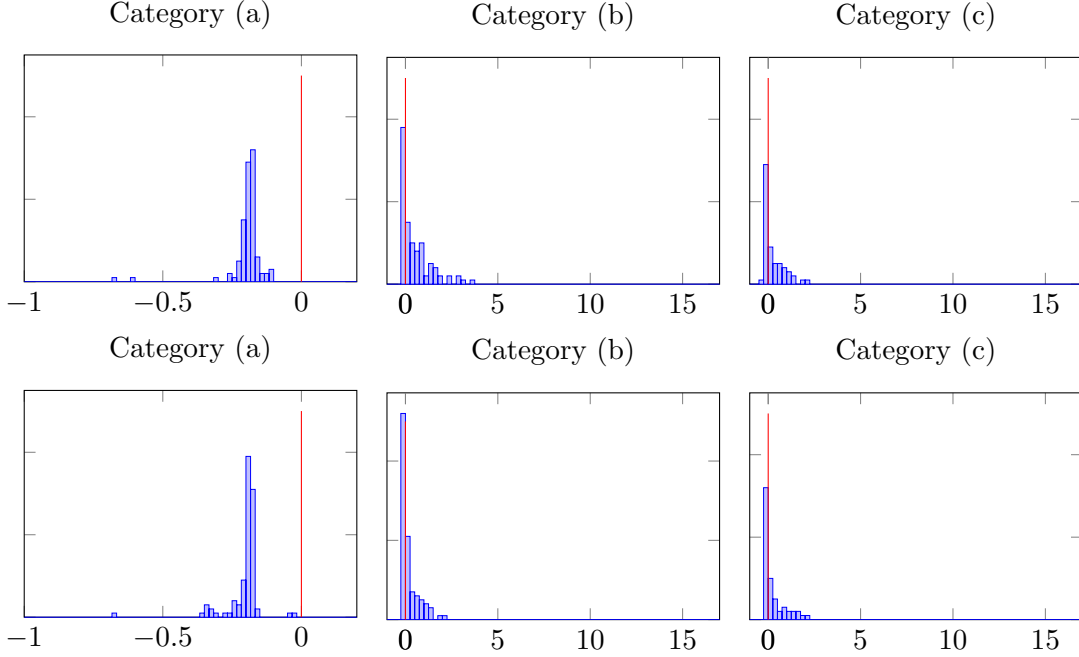


Figure 6: Histograms of the estimated values of $\log_{10}(\text{MSE}[A_r]/\text{MSE}[M_r])$ for $r = 11$ for the three categories as in Figure 4, restricted to a single model at a time. The upper row is for **PieceLinGauss** and the lower row is for **IndSumNormal**. On the left of the vertical red line, A_r wins, while on the right, M_r wins.

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